Gopnik AI

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Read Survey Responses

```
library(ggplot2)
library(gridExtra)
```

```
## Warning: package 'gridExtra' was built under R version 4.4.3
```

```
survey_path <- "227_participants.csv"

survey_raw <- read.csv(
  survey_path,
  stringsAsFactors = FALSE,
  na.strings = c("", "NA"),
  fileEncoding = "UTF-8"
)

survey <- survey_raw
```

Clean Variable Types

```
# Convert true/false fields.
survey$ai_used <- tolower(survey$ai_used) == "true"
survey$assigned_was_top_choice <- tolower(survey$assigned_was_top_choice) == "true"
survey$low_domain_frequency <- tolower(survey$low_domain_frequency) == "true"

# Convert numeric survey and task variables.
numeric_cols <- c(
  "total_duration_ms",
  "consent_page_duration_ms",
  "first_impressions_duration_ms",
  "what_kind_of_thing_is_ai_duration_ms",
  "how_you_use_ai_duration_ms",
  "ai_use_frequency_duration_ms",
  "your_experience_of_ai_duration_ms",
  "about_you_duration_ms",
  "a_short_task_duration_ms",
  "freq_information_guidance",
```

```r
  "freq_writing_communication",
  "freq_coding_math_technical",
  "freq_creative_work",
  "freq_learning_understanding",
  "freq_advice_decisions",
  "freq_summarizing_opinions",
  "freq_automation_repetitive",
  "hours_per_week",
  "exp_ai_does_work_vs_user_decides",
  "exp_independent_vs_reliant",
  "exp_existing_info_vs_creating_new",
  "exp_instrument_vs_point_of_view",
  "age",
  "task_selection_top_frequency",
  "self_reported_freq_in_assigned_category",
  "confidence_level",
  "epistemic_ownership",
  "time_before_ai_ms",
  "ai_visible_duration_ms",
  "response_started_ms",
  "response_submitted_ms",
  "task_duration_ms",
  "jaccard_similarity_to_ai"
)

for (col in intersect(numeric_cols, names(survey))) {
  survey[[col]] <- suppressWarnings(as.numeric(survey[[col]]))
}

# Make key categorical variables easier to work with.
survey$ai_kind_choice <- factor(survey$ai_kind_choice)
survey$assigned_category <- factor(survey$assigned_category)
survey$highest_education <- factor(survey$highest_education)
survey$ai_understanding <- factor(survey$ai_understanding)

survey$ai_use_tenure <- factor(
  survey$ai_use_tenure,
  levels = c(
    "I do not use them, or am just starting out.",
    "Less than 6 months",
    "6 to 12 months",
    "1 to 2 years",
    "More than 2 years"
  ),
  ordered = TRUE
)

survey$ai_reading_depth <- factor(
  survey$ai_reading_depth,
  levels = c("not_read", "glanced", "skimmed", "read_carefully"),
  ordered = TRUE
)
```

Mixed Regression Models

The main research question is whether participants' pre-task beliefs about AI and independence predict cognitive offloading during the task.

Here, "cognitive offloading" is measured in three related ways:

1. **AI access:** whether the participant used the AI assistant at all (`ai_used`).
2. **AI uptake:** how similar the participant's final answer was to the pre-written AI response. I report both lexical overlap (`jaccard_similarity_to_ai`) and, after the embedding section is computed, semantic similarity (`embedding_cosine_similarity_to_ai`).
3. **Epistemic ownership:** whether the participant said the final response reflected mostly AI thinking or mostly their own thinking (`epistemic_ownership`).

The mixed part of the model is `(1 | task_id)`. This means the model allows each task prompt/pre-written AI response to have its own baseline. That matters because some tasks may naturally make people more likely to use or copy AI than others.

```
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.4.3

## Loading required package: Matrix
```

```
# lmerTest adds p-values to the linear mixed models.
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.4.3

##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```
z <- function(x) as.numeric(scale(x))
# z = (person's score - sample mean) / sample standard deviation

model_data <- survey
model_data$task_id <- factor(model_data$task_id)

# Two pre-task independence sliders, kept as separate predictors (r = .20,
# too low to justify combining into a composite).
# Higher values mean the participant described themselves as doing the work /
# staying independent.
model_data$z_user_decides <- z(model_data$exp_ai_does_work_vs_user_decides)
```

```r
model_data$z_independent <- z(-model_data$exp_independent_vs_reliant) # reverse

# Low inter-item correlation; treat as separate predictors rather than a composite.
cor(model_data$z_user_decides, model_data$z_independent, use = "complete.obs")
```

```
## [1] 0.2100162
```

```r
# Control variables.
model_data$z_hours_per_week <- z(model_data$hours_per_week)
model_data$z_instrument_vs_point_of_view <- z(model_data$exp_instrument_vs_point_of_view)
model_data$z_existing_info_vs_creating_new <- z(model_data$exp_existing_info_vs_creating_new)
model_data$z_jaccard_similarity_to_ai <- z(model_data$jaccard_similarity_to_ai)

# Binary recode of epistemic ownership.
# 1-3 = primarily AI's thinking (0); 5-7 = primarily own thinking (1).
# Six participants at the midpoint (4) are excluded from binary ownership models.
model_data$epistemic_ownership_binary <- ifelse(
  model_data$epistemic_ownership <= 3, 0L,
  ifelse(model_data$epistemic_ownership >= 5, 1L, NA_integer_)
)

# Binary reading depth: read_carefully (1) vs. anything less (0).
# Used in place of the ordered 1-4 numeric scale because ~57% of AI users
# reported reading_carefully, leaving few observations in the lower categories.
model_data$read_carefully <- as.integer(
  model_data$ai_reading_depth == "read_carefully"
)

# Use the original epistemic ownership scale.
# Original scale: 1 = mostly AI's thinking, 7 = mostly my thinking.
# Higher values mean more of the participant's own thinking.
ai_users <- subset(model_data, ai_used == TRUE)
```

Model 1: Do Pre-Task Independence Beliefs Predict Whether People Use AI?

This model asks whether participants who described themselves as more independent on either pre-task slider were less likely to consult the AI assistant during the task. The two independence items are entered as separate predictors: `z_user_decides` (higher = participant prefers to do the work themselves) and `z_independent` (higher = participant expects to stay more independent).

Outcome: `ai_used`
Model type: mixed logistic regression
Random intercept: `task_id`

```r
model_ai_used <- glmer(
  ai_used ~ z_user_decides +
    z_independent +
    z_hours_per_week +
    z_instrument_vs_point_of_view +
    (1 | task_id),
  data = model_data,
  family = binomial,
```

```r
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_ai_used)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: ai_used ~ z_user_decides + z_independent + z_hours_per_week +
##     z_instrument_vs_point_of_view + (1 | task_id)
##    Data: model_data
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     262.0     282.6    -125.0     250.0       221
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.7115 -0.9226  0.4116  0.7047  1.3366
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 227, groups:  task_id, 18
##
## Fixed effects:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     0.9816     0.1660   5.912 3.38e-09 ***
## z_user_decides                 -0.7946     0.1989  -3.996 6.45e-05 ***
## z_independent                  -0.1246     0.1600  -0.779   0.4361
## z_hours_per_week                0.3117     0.1807   1.725   0.0846 .
## z_instrument_vs_point_of_view   0.1682     0.1591   1.057   0.2905
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sr_d z_ndpn z_hr__
## z_user_dcds -0.350
## z_indepndnt -0.025 -0.166
## z_hrs_pr_wk  0.119 -0.053  0.032
## z_nstrm____  0.014 -0.020  0.133 -0.224
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
fixed_effects_table(model_ai_used, exponentiate = TRUE)
```

```
##                            term   estimate std_error      p_value odds_ratio
## 1                   (Intercept)  0.9815628 0.1660341 3.383608e-09  2.6686235
## 2                z_user_decides -0.7946191 0.1988651 6.448468e-05  0.4517533
## 3                 z_independent -0.1245733 0.1599696 4.361383e-01  0.8828736
```

```
## 4              z_hours_per_week  0.3116570 0.1807144 8.460266e-02  1.3656862
## 5 z_instrument_vs_point_of_view  0.1682300 0.1591488 2.904837e-01  1.1832087
```

The `user_decides` independence slider significantly predicted lower odds of using the AI assistant, OR = 0.45, p < .001. Participants who more strongly said the user does the work / makes the decisions were less likely to consult AI. The separate `z_independent` slider was not significant, OR = 0.88, p = .436. Hours per week was marginal but not significant, OR = 1.37, p = .085, and instrument-vs-point-of-view beliefs were not significant, OR = 1.18, p = .290.

The mixed model produced a singular fit (task_id random intercept variance = 0). As a robustness check, a fixed-effects logistic regression was fit without the random intercept.

```r
model_ai_used_glm <- glm(
  ai_used ~ z_user_decides +
    z_independent +
    z_hours_per_week +
    z_instrument_vs_point_of_view,
  data = model_data,
  family = binomial
)

summary(model_ai_used_glm)
```

```
##
## Call:
## glm(formula = ai_used ~ z_user_decides + z_independent + z_hours_per_week +
##     z_instrument_vs_point_of_view, family = binomial, data = model_data)
##
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     0.9816     0.1660   5.912 3.38e-09 ***
## z_user_decides                 -0.7946     0.1989  -3.996 6.44e-05 ***
## z_independent                  -0.1246     0.1600  -0.779   0.4361
## z_hours_per_week                0.3117     0.1807   1.725   0.0846 .
## z_instrument_vs_point_of_view   0.1682     0.1591   1.057   0.2905
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 278.84  on 226  degrees of freedom
## Residual deviance: 250.02  on 222  degrees of freedom
## AIC: 260.02
##
## Number of Fisher Scoring iterations: 4
```

```r
fixed_effects_table(model_ai_used_glm, exponentiate = TRUE)
```

```
##                            term   estimate std_error      p_value odds_ratio
## 1                   (Intercept)  0.9815628 0.1660304 3.380870e-09  2.6686235
## 2                z_user_decides -0.7946191 0.1988579 6.444508e-05  0.4517533
## 3                 z_independent -0.1245733 0.1599683 4.361344e-01  0.8828736
## 4              z_hours_per_week  0.3116570 0.1807129 8.460004e-02  1.3656862
## 5 z_instrument_vs_point_of_view  0.1682300 0.1591477 2.904805e-01  1.1832087
```

Model 2: Do Pre-Task Independence Beliefs Predict Text Similarity To AI?

This model only uses participants who consulted the AI assistant. It asks whether people who described themselves as more independent on either slider nevertheless produced answers that were more similar to the AI response.

Note on distribution: Among AI users (n = 158), `jaccard_similarity_to_ai` is severely bimodal. Approximately 35% of AI users (n = 55) had a Jaccard score of exactly 1.0, meaning their response was an exact copy of the pre-written AI response. The remaining participants show a spread of lower values (median approx. 0.31). This bimodality should be kept in mind when interpreting linear models on this outcome. Models are retained as-is, but effects should be interpreted as reflecting this two-group structure.

Outcome: `jaccard_similarity_to_ai`
Model type: linear mixed regression
Random intercept: `task_id`

```r
model_similarity <- lmer(
  jaccard_similarity_to_ai ~ z_user_decides +
    z_independent +
    z_hours_per_week +
    z_instrument_vs_point_of_view +
    (1 | task_id),
  data = ai_users
)

summary(model_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## jaccard_similarity_to_ai ~ z_user_decides + z_independent + z_hours_per_week +
##     z_instrument_vs_point_of_view + (1 | task_id)
##    Data: ai_users
##
## REML criterion at convergence: 191.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6370 -0.8175 -0.2433  0.9534  1.5981
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.02084  0.1444
##  Residual             0.15934  0.3992
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                                Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                     0.50358    0.04748  16.07656  10.607 1.15e-08
## z_user_decides                 -0.04194    0.03304 151.16617  -1.269    0.206
## z_independent                  -0.04106    0.03465 150.52019  -1.185    0.238
## z_hours_per_week                0.01427    0.03207 146.94145   0.445    0.657
## z_instrument_vs_point_of_view   0.01494    0.03418 150.61159   0.437    0.663
##
## (Intercept)                   ***
```

```
## z_user_decides
## z_independent
## z_hours_per_week
## z_instrument_vs_point_of_view
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sr_d z_ndpn z_hr__
## z_user_dcds  0.126
## z_indepndnt  0.041 -0.213
## z_hrs_pr_wk -0.079 -0.087 -0.067
## z_nstrm____ -0.058 -0.177  0.203 -0.124
```

```
fixed_effects_table(model_similarity)
```

```
##                            term    estimate  std_error      p_value
## 1                   (Intercept)  0.50357786 0.04747811 1.146702e-08
## 2                z_user_decides -0.04194347 0.03304321 2.062663e-01
## 3                 z_independent -0.04106184 0.03464951 2.378603e-01
## 4              z_hours_per_week  0.01426732 0.03207427 6.571034e-01
## 5 z_instrument_vs_point_of_view  0.01493647 0.03417693 6.627129e-01
```

Among participants who used the AI assistant, neither pre-task independence predictor significantly predicted Jaccard similarity to the AI response: `z_user_decides`, b = -0.042, p = .206; `z_independent`, b = -0.041, p = .238. Hours per week, b = 0.014, p = .657, and instrument-vs-point-of-view beliefs, b = 0.015, p = .663, were also not significant predictors. This AI-user-only result is weaker than the all-participant correlations reported in the sample-scope check below.

Model 3: Does Objective AI Uptake Predict Epistemic Ownership?

This model asks whether participants whose answers were more similar to the AI response were less likely to feel the response reflected their own thinking. Because `epistemic_ownership` is U-shaped (many 1s, 2s, 6s, 7s; few responses near the midpoint), it is recoded as binary: 0 = primarily AI's thinking (scores 1-3) vs. 1 = primarily own thinking (scores 5-7). The 14 AI users at the midpoint (4) are excluded.

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression
Random intercept: `task_id`

```
model_epistemic_ownership <- glmer(
  epistemic_ownership_binary ~ z_jaccard_similarity_to_ai +
    z_user_decides +
    z_independent +
    z_hours_per_week +
    (1 | task_id),
  data = ai_users,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)

summary(model_epistemic_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## epistemic_ownership_binary ~ z_jaccard_similarity_to_ai + z_user_decides +
##     z_independent + z_hours_per_week + (1 | task_id)
##    Data: ai_users
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     160.8     178.6     -74.4     148.8       138
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9287 -0.6263 -0.3268  0.7832  3.3727
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0.03836  0.1959
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.47898    0.21415  -2.237   0.0253 *
## z_jaccard_similarity_to_ai -1.12868    0.24547  -4.598 4.27e-06 ***
## z_user_decides              0.20344    0.20164   1.009   0.3130
## z_independent              -0.07176    0.20196  -0.355   0.7224
## z_hours_per_week            0.13194    0.19967   0.661   0.5088
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_j___ z_sr_d z_ndpn
## z_jccrd_s__ -0.041
## z_user_dcds  0.151 -0.105
## z_indepndnt  0.044  0.054 -0.164
## z_hrs_pr_wk -0.154 -0.020 -0.152 -0.038
```

```r
fixed_effects_table(model_epistemic_ownership, exponentiate = TRUE)
```

```
##                         term   estimate std_error      p_value odds_ratio
## 1                (Intercept) -0.4789847 0.2141522 2.530887e-02  0.6194119
## 2 z_jaccard_similarity_to_ai -1.1286803 0.2454708 4.265212e-06  0.3234598
## 3             z_user_decides  0.2034378 0.2016415 3.130186e-01  1.2256089
## 4              z_independent -0.0717574 0.2019631 7.223652e-01  0.9307567
## 5           z_hours_per_week  0.1319359 0.1996741 5.087686e-01  1.1410352
```

Greater Jaccard similarity to the AI response significantly predicted lower odds of reporting primarily own-thinking ownership, OR = 0.32, p < .001. Neither independence predictor nor hours per week was significant (`z_user_decides`, OR = 1.23, p = .313; `z_independent`, OR = 0.93, p = .722; hours per week, OR = 1.14, p = .509).

Model 4: Do Pre-Task Independence Beliefs Moderate How People Interpret AI Influence?

For example, imagine two participants both have high similarity to the AI response:

Participant A rated themselves as not very independent. Participant B rated themselves as very independent.

So the interaction asks: does Participant B, the independent person, give a higher epistemic ownership rating than Participant A, although both produced AI-like answers?

Because the two independence sliders are separate predictors, there are two interaction terms:

- `z_user_decides:z_jaccard_similarity_to_ai`
- `z_independent:z_jaccard_similarity_to_ai`

Although sample size has increased to n = 158 AI users, interaction effects require substantial power and should still be interpreted cautiously.

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression with interactions
Random intercept: `task_id`

```
model_contrast <- glmer(
  epistemic_ownership_binary ~
    z_user_decides * z_jaccard_similarity_to_ai +
    z_independent * z_jaccard_similarity_to_ai +
    z_hours_per_week +
    (1 | task_id),
  data = ai_users,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)

summary(model_contrast)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## epistemic_ownership_binary ~ z_user_decides * z_jaccard_similarity_to_ai +
##     z_independent * z_jaccard_similarity_to_ai + z_hours_per_week +
##     (1 | task_id)
##    Data: ai_users
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     163.9     187.7     -74.0     147.9       136
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6938 -0.6330 -0.3097  0.7949  3.9263
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0.03176  0.1782
```

```
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                                 -0.5221     0.2213  -2.359   0.0183
## z_user_decides                               0.2201     0.2115   1.041   0.2981
## z_jaccard_similarity_to_ai                  -1.1775     0.2688  -4.380 1.18e-05
## z_independent                               -0.1264     0.2176  -0.581   0.5613
## z_hours_per_week                             0.1278     0.2001   0.639   0.5229
## z_user_decides:z_jaccard_similarity_to_ai    0.1121     0.2448   0.458   0.6471
## z_jaccard_similarity_to_ai:z_independent    -0.2268     0.2669  -0.850   0.3954
##
## (Intercept)                                *
## z_user_decides
## z_jaccard_similarity_to_ai                 ***
## z_independent
## z_hours_per_week
## z_user_decides:z_jaccard_similarity_to_ai
## z_jaccard_similarity_to_ai:z_independent
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sr_d z_j___ z_ndpn z_hr__ z__:__
## z_user_dcds  0.088
## z_jccrd_s__  0.061 -0.116
## z_indepndnt  0.179 -0.163  0.215
## z_hrs_pr_wk -0.141 -0.170 -0.014 -0.023
## z_sr_d:____ -0.108  0.269  0.020 -0.056 -0.068
## z_jccr___:_  0.254 -0.041  0.353  0.337  0.018 -0.206
```

```r
fixed_effects_table(model_contrast, exponentiate = TRUE)
```

```
##                                        term   estimate std_error      p_value
## 1                               (Intercept) -0.5221174 0.2212833 1.829971e-02
## 2                            z_user_decides  0.2200928 0.2115026 2.980542e-01
## 3                z_jaccard_similarity_to_ai -1.1774767 0.2688091 1.184917e-05
## 4                             z_independent -0.1264309 0.2176280 5.612745e-01
## 5                          z_hours_per_week  0.1278081 0.2000686 5.229391e-01
## 6 z_user_decides:z_jaccard_similarity_to_ai  0.1120546 0.2447612 6.470875e-01
## 7  z_jaccard_similarity_to_ai:z_independent -0.2267967 0.2668848 3.954403e-01
##   odds_ratio
## 1  0.5932630
## 2  1.2461923
## 3  0.3080551
## 4  0.8812351
## 5  1.1363350
## 6  1.1185739
## 7  0.7970828
```

Greater AI similarity again significantly predicted lower odds of reporting primarily own-thinking ownership, OR = 0.31, p < .001. Neither interaction with independence beliefs was significant (`z_user_decides` x similarity: OR = 1.12, p = .647; `z_independent` x similarity: OR = 0.80, p = .395), indicating that the uptake-to-ownership link did not differ reliably by pre-task independence beliefs.

Altogether: Pre-task independence beliefs mattered most for whether participants opened AI, not how much they copied after opening it. **The `user_decides` slider predicted lower AI use, OR = 0.45, p < .001**, but neither independence slider significantly predicted Jaccard similarity among AI users. **Participants' post-task ownership ratings tracked behavioral uptake: higher similarity predicted lower odds of primarily own-thinking ownership in both ownership models, ORs = 0.32 and 0.31, ps < .001.** This suggests that participants were at least partly aware of the AI's contribution. Neither independence predictor moderated this uptake-to-ownership link.

Sample-Scope Check: All Participants vs. AI Users

Several apparent contrasts depend on whether the analysis includes all participants or only people who actually opened the AI assistant. This matters because non-AI users have low or zero AI uptake by design, so all-participant associations partly capture the decision to use AI at all.

Across all participants (N = 227), the composite independence score was negatively related to Jaccard similarity, **Pearson r = -.23, p < .001**; Spearman rho = -.22, p < .001. This matches the broader direction found by the mentor: participants who described themselves as more independent tended to produce less AI-similar final responses overall. However, among AI users only, the individual independence sliders did not significantly predict Jaccard similarity in Model 2. This suggests independence beliefs mainly predicted whether people opened AI, rather than how much they copied after opening it.

Domain-specific AI-use frequency shows the same sample-scope issue. Across all participants, domain frequency was weakly positively related to Jaccard similarity using Pearson correlation, r = .13, p = .043, but not using Spearman correlation, rho = .10, p = .116. In the AI-user-only regression in Analysis 7, domain frequency did not predict Jaccard similarity, b = -0.013, p = .757. **The clearest domain-frequency result is AI access itself: people who reported using AI more often in the assigned task domain were more likely to open AI, OR = 1.82, p < .001.**

The timing data also show that AI use usually happened early. Among AI users, 78 of 158 participants (49%) opened AI within 50 seconds, and 153 of the 157 participants with both timing values started writing after opening AI. This supports the interpretation that most AI users consulted the assistant before drafting their own response.

Note on exploratory analyses (Analyses 1-8): All analyses below are exploratory. They were motivated by theory and by inspection of the data but were not pre-registered. No correction for multiple comparisons is applied across analyses. Within each analysis, Tukey adjustments are used for pairwise comparisons. Results should be interpreted as hypothesis-generating rather than confirmatory, and any significant findings should be treated with appropriate caution pending replication.

Analysis 1: Does Conceptualizing AI Differently Predict Offloading?

This analysis asks whether participants' answer to "What kind of thing is AI?" predicts cognitive offloading during the short task.

This connects most directly to the paper *Large AI models are cultural and social technologies.* That paper argues that large AI models should not only be understood as agents, but also as cultural/social technologies similar to writing, print, libraries, markets, bureaucracies, and other systems for organizing human knowledge.

Because this is exploratory, I do pairwise comparisons. This means I compare every AI conceptualization category with every other category. The models use sum/effects coding for `ai_kind_choice`, so no conceptualization group is treated as the reference group; the pairwise `emmeans()` results are the main comparisons to interpret.

Only 8 participants selected "Writing or print" overall, and only 6 of those participants used the AI assistant. Because that group is too small for stable comparisons, I exclude it from Analysis 1. This means the pairwise comparisons below include the remaining four categories.

```r
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.4.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
concept_data <- subset(model_data, ai_kind_choice != "Writing or print")
concept_data$ai_kind_choice <- droplevels(concept_data$ai_kind_choice)

# Use sum/effects coding so no AI conceptualization category is treated as
# the reference category in the model parameterization.
contrasts(concept_data$ai_kind_choice) <- contr.sum(
  nlevels(concept_data$ai_kind_choice)
)

concept_ai_users <- subset(concept_data, ai_used == TRUE)
concept_ai_users$ai_kind_choice <- droplevels(concept_ai_users$ai_kind_choice)
contrasts(concept_ai_users$ai_kind_choice) <- contr.sum(
  nlevels(concept_ai_users$ai_kind_choice)
)

table(concept_data$ai_kind_choice, useNA = "ifany")
```

```
##
## A calculator or specialized tool           A colleague or advisor
##                               30                               71
##       A library or search engine           A market or aggregator
##                              104                               14
```

```r
table(concept_ai_users$ai_kind_choice, useNA = "ifany")
```

```
##
## A calculator or specialized tool           A colleague or advisor
##                               20                               56
##       A library or search engine           A market or aggregator
##                               67                                9
```

Analysis 1A: AI Conceptualization Predicting Whether Participants Used AI

Outcome: `ai_used`
Model type: mixed logistic regression

Predictor: `ai_kind_choice`
Random intercept: `task_id`

This model asks whether people who conceptualized AI differently were more or less likely to consult the optional AI assistant during the task. The model uses sum/effects coding so no category is treated as the reference category. The `drop1()` result tests whether `ai_kind_choice` as a whole improves the model. The `emmeans()` results show estimated AI-use probabilities for each conceptualization group, and the pairwise comparisons show Tukey-adjusted differences between groups.

```r
model_ai_kind_ai_used <- glmer(
  ai_used ~ ai_kind_choice + (1 | task_id),
  data = concept_data,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)

summary(model_ai_kind_ai_used)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: ai_used ~ ai_kind_choice + (1 | task_id)
##    Data: concept_data
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     275.1     292.0    -132.5     265.1       214
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9513 -1.3413  0.5183  0.7401  0.7495
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0.006044 0.07774
## Number of obs: 219, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)       0.8005     0.2002   3.999 6.37e-05 ***
## ai_kind_choice1  -0.1061     0.3356  -0.316   0.7520
## ai_kind_choice2   0.5198     0.2815   1.846   0.0649 .
## ai_kind_choice3  -0.2066     0.2462  -0.839   0.4013
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_kn_1 a_kn_2
## ai_knd_chc1 -0.009
## ai_knd_chc2 -0.268 -0.235
## ai_knd_chc3 -0.591 -0.120  0.066
```

```r
tests_ai_kind_ai_used <- drop1(model_ai_kind_ai_used, test = "Chisq")
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
tests_ai_kind_ai_used
```

```
## Single term deletions
##
## Model:
## ai_used ~ ai_kind_choice + (1 | task_id)
##                npar    AIC    LRT Pr(Chi)
## <none>              275.05
## ai_kind_choice    3 273.73 4.6708  0.1976
```

```r
emm_ai_kind_ai_used <- emmeans(
  model_ai_kind_ai_used,
  ~ ai_kind_choice,
  type = "response"
)

emm_ai_kind_ai_used
```

```
##  ai_kind_choice                     prob     SE  df asymp.LCL asymp.UCL
##  A calculator or specialized tool 0.667 0.0865 Inf     0.483     0.811
##  A colleague or advisor           0.789 0.0497 Inf     0.676     0.870
##  A library or search engine       0.644 0.0472 Inf     0.547     0.731
##  A market or aggregator           0.644 0.1310 Inf     0.371     0.847
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
pairs_ai_kind_ai_used <- pairs(
  emm_ai_kind_ai_used,
  adjust = "tukey"
)

summary(pairs_ai_kind_ai_used, type = "response")
```

```
##  contrast                                                      odds.ratio    SE
##  A calculator or specialized tool / A colleague or advisor          0.535 0.260
##  A calculator or specialized tool / A library or search engine      1.106 0.486
##  A calculator or specialized tool / A market or aggregator          1.106 0.759
##  A colleague or advisor / A library or search engine                2.068 0.747
##  A colleague or advisor / A market or aggregator                    2.068 1.310
##  A library or search engine / A market or aggregator                1.000 0.607
##   df null z.ratio p.value
##  Inf    1  -1.287  0.5711
##  Inf    1   0.229  0.9958
##  Inf    1   0.147  0.9989
##  Inf    1   2.009  0.1844
##  Inf    1   1.149  0.6588
```

```
##  Inf    1   0.001  1.0000
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log odds ratio scale
```

The overall effect of AI-kind was not significant, chi-squared(3) = 4.67, p = .198. The colleague/advisor category had the highest estimated AI-use probability (.79), followed by calculator/tool (.67), library/search engine (.64), and market/aggregator (.64), but Tukey-adjusted pairwise comparisons were not significant. Overall, AI conceptualization did not reliably predict AI use, though descriptively AI use was highest when AI was seen as a colleague/advisor.

The mixed model produced a singular-fit warning, with the random intercept variance for task_id estimated at zero. This suggests that the model found little evidence of residual variation between task prompts for this outcome, or that the random effect was too weak to estimate reliably. As a robustness check, I fit a simpler logistic regression without (1 | task_id) to test whether the substantive conclusion changed.

```r
model_ai_kind_ai_used_glm <- glm(
  ai_used ~ ai_kind_choice,
  data = concept_data,
  family = binomial
)

summary(model_ai_kind_ai_used_glm)
```

```
##
## Call:
## glm(formula = ai_used ~ ai_kind_choice, family = binomial, data = concept_data)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)       0.7980     0.1916   4.164 3.12e-05 ***
## ai_kind_choice1  -0.1049     0.3343  -0.314   0.7537
## ai_kind_choice2   0.5193     0.2810   1.848   0.0646 .
## ai_kind_choice3  -0.2042     0.2402  -0.850   0.3952
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 269.72  on 218  degrees of freedom
## Residual deviance: 265.06  on 215  degrees of freedom
## AIC: 273.06
##
## Number of Fisher Scoring iterations: 4
```

```r
tests_ai_kind_ai_used_glm <- drop1(
  model_ai_kind_ai_used_glm,
  test = "Chisq"
)

tests_ai_kind_ai_used_glm
```

```
## Single term deletions
```

```
##
## Model:
## ai_used ~ ai_kind_choice
##                Df Deviance    AIC    LRT Pr(>Chi)
## <none>              265.06 273.06
## ai_kind_choice  3   269.73 271.73 4.6687   0.1977
```

```r
emm_ai_kind_ai_used_glm <- emmeans(
  model_ai_kind_ai_used_glm,
  ~ ai_kind_choice,
  type = "response"
)

emm_ai_kind_ai_used_glm
```

```
##  ai_kind_choice                     prob     SE  df asymp.LCL asymp.UCL
##  A calculator or specialized tool 0.667 0.0861 Inf     0.484     0.810
##  A colleague or advisor           0.789 0.0484 Inf     0.679     0.868
##  A library or search engine       0.644 0.0469 Inf     0.548     0.730
##  A market or aggregator           0.643 0.1280 Inf     0.376     0.843
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

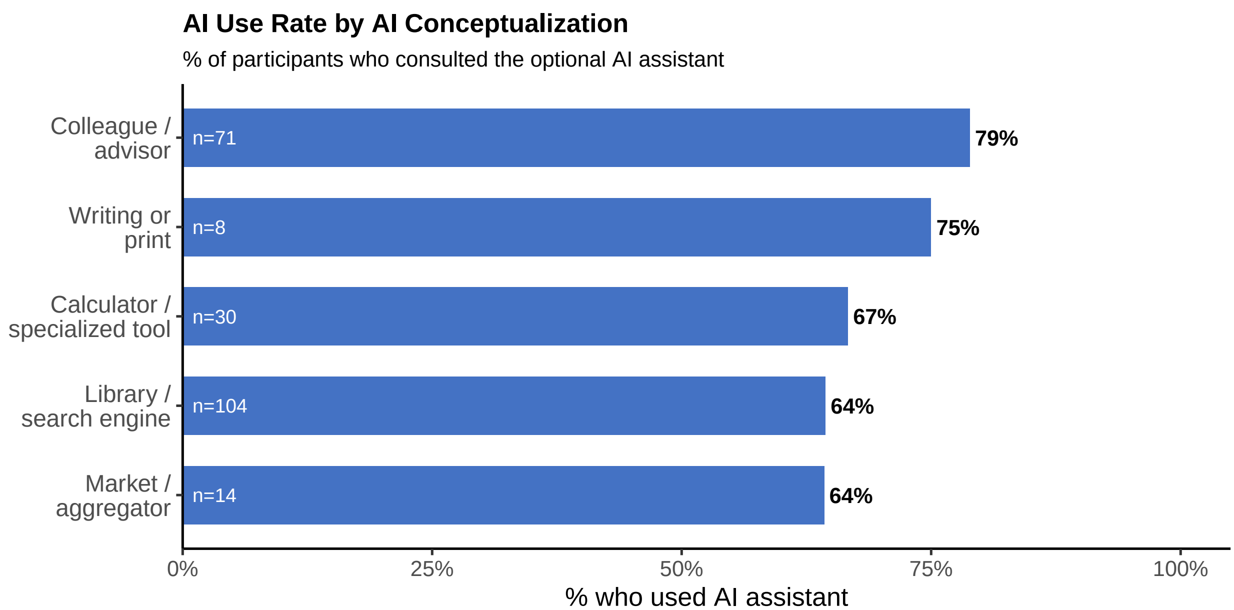
```r
pairs_ai_kind_ai_used_glm <- pairs(
  emm_ai_kind_ai_used_glm,
  adjust = "tukey"
)

summary(pairs_ai_kind_ai_used_glm, type = "response")
```

```
##  contrast                                                     odds.ratio    SE
##  A calculator or specialized tool / A colleague or advisor         0.536 0.259
##  A calculator or specialized tool / A library or search engine     1.104 0.484
##  A calculator or specialized tool / A market or aggregator         1.111 0.755
##  A colleague or advisor / A library or search engine               2.062 0.733
##  A colleague or advisor / A market or aggregator                   2.074 1.300
##  A library or search engine / A market or aggregator               1.006 0.598
##   df null z.ratio p.value
##  Inf    1  -1.289  0.5700
##  Inf    1   0.227  0.9959
##  Inf    1   0.155  0.9987
##  Inf    1   2.034  0.1753
##  Inf    1   1.160  0.6523
##  Inf    1   0.010  1.0000
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log odds ratio scale
```

AI conceptualization did not significantly predict optional assistant use, chi-squared(3) = 4.67, p = .198. Participants who viewed AI as a colleague/advisor showed the highest estimated AI-use probability (.79), followed by calculator/tool (.67), library/search engine (.64), and market/aggregator (.64), but pairwise differences were not statistically reliable after Tukey adjustment.

```r
# AI use rate by conceptualization (all participants including Writing or print)
ai_kind_all <- subset(model_data, !is.na(ai_kind_choice))
ai_kind_use_df <- do.call(rbind, lapply(
  split(ai_kind_all, ai_kind_all$ai_kind_choice),
  function(sub) {
    data.frame(
      ai_kind_choice = as.character(sub$ai_kind_choice[1]),
      pct_used       = mean(sub$ai_used, na.rm = TRUE),
      n              = nrow(sub),
      stringsAsFactors = FALSE
    )
  }
))

kind_lookup2 <- c(
  "A calculator or specialized tool" = "Calculator /\nspecialized tool",
  "A colleague or advisor"           = "Colleague /\nadvisor",
  "A library or search engine"       = "Library /\nsearch engine",
  "A market or aggregator"           = "Market /\naggregator",
  "Writing or print"                 = "Writing or\nprint"
)
ai_kind_use_df$kind_short <- kind_lookup2[ai_kind_use_df$ai_kind_choice]
ai_kind_use_df$kind_short <- factor(
  ai_kind_use_df$kind_short,
  levels = ai_kind_use_df$kind_short[order(ai_kind_use_df$pct_used)]
)

ggplot(ai_kind_use_df, aes(x = kind_short, y = pct_used)) +
  geom_col(fill = "#4472C4", width = 0.65) +
  geom_text(
    aes(label = paste0(round(pct_used * 100), "%")),
    hjust = -0.12, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n), y = 0.01),
    hjust = 0, size = 3, color = "white"
  ) +
  coord_flip() +
  scale_y_continuous(
    limits = c(0, 1.05), expand = c(0, 0),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title    = "AI Use Rate by AI Conceptualization",
    subtitle = "% of participants who consulted the optional AI assistant",
    x        = NULL,
    y        = "% who used AI assistant"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10),
    axis.text.y   = element_text(size = 10.5)
```

```
)
```



Analysis 1B: AI Conceptualization Predicting Similarity To The AI Response

Outcome: `jaccard_similarity_to_ai`
Model type: linear mixed regression
Predictor: `ai_kind_choice`
Random intercept: `task_id`
Sample: only participants who used the AI assistant

This model asks whether people's conceptualization of AI predicted how much their final response resembled the pre-written AI response. The `emmeans()` results show the adjusted average similarity score for each conceptualization group, and the pairwise comparisons test whether any groups differ from one another.

```r
model_ai_kind_similarity <- lmer(
  jaccard_similarity_to_ai ~ ai_kind_choice + (1 | task_id),
  data = concept_ai_users
)

summary(model_ai_kind_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ ai_kind_choice + (1 | task_id)
##    Data: concept_ai_users
##
## REML criterion at convergence: 179.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6964 -0.8675 -0.3148  0.9684  1.5818
##
```

```
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.01529  0.1236
##  Residual             0.16519  0.4064
## Number of obs: 152, groups:  task_id, 18
##
## Fixed effects:
##                  Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)       0.52870    0.05390  33.22106   9.809 2.44e-11 ***
## ai_kind_choice1   0.05266    0.08015 144.22774   0.657    0.512
## ai_kind_choice2   0.02383    0.06073 146.51328   0.392    0.695
## ai_kind_choice3  -0.06049    0.05827 144.76230  -1.038    0.301
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_kn_1 a_kn_2
## ai_knd_chc1  0.031
## ai_knd_chc2 -0.405 -0.175
## ai_knd_chc3 -0.455 -0.163  0.179
```

```r
anova(model_ai_kind_similarity)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                 Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_kind_choice 0.29269 0.097562     3 144.87  0.5906 0.6221
```

```r
emm_ai_kind_similarity <- emmeans(
  model_ai_kind_similarity,
  ~ ai_kind_choice
)
```

```
## Cannot use mode = "kenward-roger" because *pbkrtest* package is not installed
```

```r
emm_ai_kind_similarity
```

```
##  ai_kind_choice                   emmean     SE    df lower.CL upper.CL
##  A calculator or specialized tool  0.581 0.0979 117.7    0.387    0.775
##  A colleague or advisor            0.553 0.0628  50.9    0.427    0.679
##  A library or search engine        0.468 0.0587  41.4    0.350    0.587
##  A market or aggregator            0.513 0.1400 147.3    0.235    0.790
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

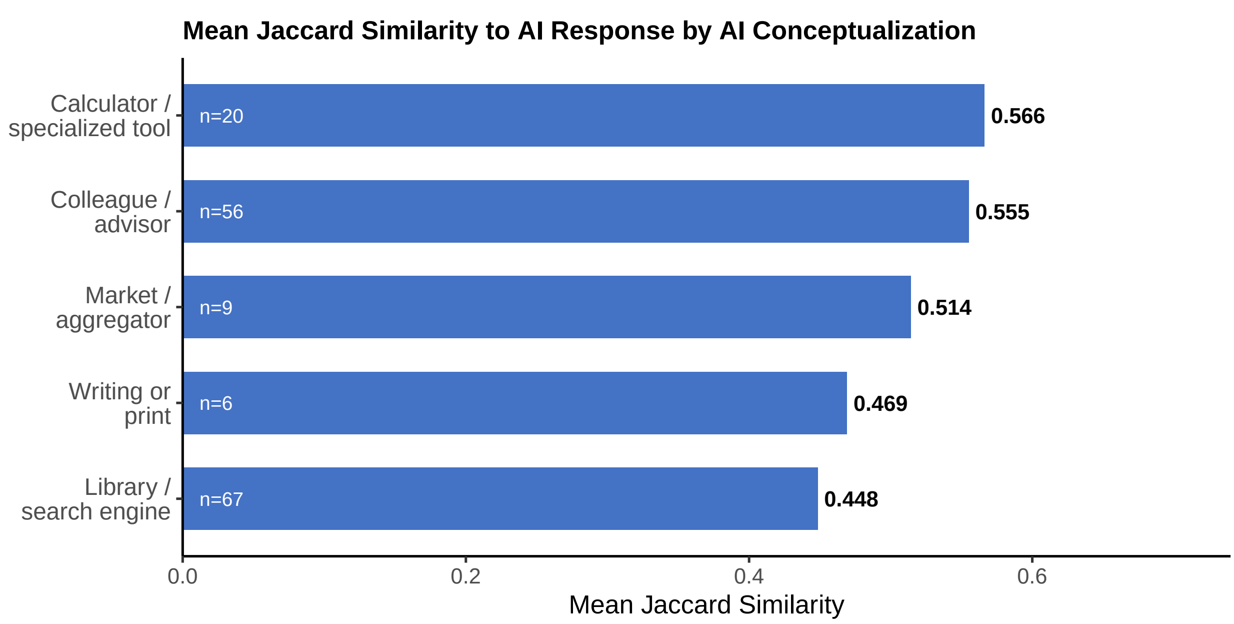
```r
pairs_ai_kind_similarity <- pairs(
  emm_ai_kind_similarity,
  adjust = "tukey"
)

pairs_ai_kind_similarity
```

```
##  contrast                                                  estimate     SE
##  A calculator or specialized tool - A colleague or advisor       0.0288 0.1090
##  A calculator or specialized tool - A library or search engine   0.1131 0.1060
##  A calculator or specialized tool - A market or aggregator       0.0687 0.1660
##  A colleague or advisor - A library or search engine             0.0843 0.0763
##  A colleague or advisor - A market or aggregator                 0.0398 0.1490
##  A library or search engine - A market or aggregator            -0.0445 0.1470
##   df t.ratio p.value
##  145   0.265  0.9934
##  146   1.063  0.7127
##  142   0.413  0.9761
##  148   1.106  0.6866
##  143   0.267  0.9933
##  141  -0.304  0.9902
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 4 estimates
```

AI conceptualization did not significantly predict Jaccard similarity to the AI response (see ANOVA output above). Descriptively, mean Jaccard similarity ranged from .45 (Library/search engine) to .57 (Calculator/tool) across the four main groups, but differences were small and unreliable. Tukey-adjusted pairwise comparisons showed no significant differences between conceptualization groups.

```r
# Include all AI users (including Writing or print) for visualization only.
jaccard_viz_data <- subset(
  ai_users,
  !is.na(ai_kind_choice) & !is.na(jaccard_similarity_to_ai)
)
jaccard_viz_data$ai_kind_choice <- droplevels(jaccard_viz_data$ai_kind_choice)

kind_lookup <- c(
  "A calculator or specialized tool" = "Calculator /\nspecialized tool",
  "A colleague or advisor"           = "Colleague /\nadvisor",
  "A library or search engine"       = "Library /\nsearch engine",
  "A market or aggregator"           = "Market /\naggregator",
  "Writing or print"                 = "Writing or\nprint"
)

library(ggplot2)
# Compute group means and n for bar chart
jac_kind_df <- do.call(rbind, lapply(
  split(jaccard_viz_data, jaccard_viz_data$ai_kind_choice),
  function(sub) {
    data.frame(
      ai_kind_choice = as.character(sub$ai_kind_choice[1]),
      mean_jac       = mean(sub$jaccard_similarity_to_ai, na.rm = TRUE),
      n              = sum(!is.na(sub$jaccard_similarity_to_ai)),
      stringsAsFactors = FALSE
    )
  }
))
jac_kind_df$kind_short <- kind_lookup[jac_kind_df$ai_kind_choice]
jac_kind_df$kind_short <- factor(
  jac_kind_df$kind_short,
```

```r
  levels = jac_kind_df$kind_short[order(jac_kind_df$mean_jac)]
)

ggplot(jac_kind_df, aes(x = kind_short, y = mean_jac)) +
  geom_col(fill = "#4472C4", width = 0.65) +
  geom_text(
    aes(label = sprintf("%.3f", mean_jac)),
    hjust = -0.12, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n), y = 0.012),
    hjust = 0, size = 3, color = "white"
  ) +
  coord_flip() +
  scale_y_continuous(limits = c(0, 0.74), expand = c(0, 0)) +
  labs(
    title    = "Mean Jaccard Similarity to AI Response by AI Conceptualization",
    x        = NULL,
    y        = "Mean Jaccard Similarity"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10),
    axis.text.y   = element_text(size = 10.5)
  )
```



Analysis 1C: AI Conceptualization Predicting Epistemic Ownership

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression
Predictor: `ai_kind_choice`

Random intercept: `task_id`
Sample: only participants who used the AI assistant

This model asks whether people's conceptualization of AI predicted whether they rated the final response as closer to AI thinking or their own thinking. Epistemic ownership is recoded as binary (see Model 3).

```
model_ai_kind_ownership <- glmer(
  epistemic_ownership_binary ~ ai_kind_choice + (1 | task_id),
  data = concept_ai_users,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_ai_kind_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: epistemic_ownership_binary ~ ai_kind_choice + (1 | task_id)
##    Data: concept_ai_users
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     181.8     196.5     -85.9     171.8       133
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.7746 -0.7171 -0.6172  1.3093  1.6202
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 138, groups:  task_id, 18
##
## Fixed effects:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -0.669970   0.241086  -2.779  0.00545 **
## ai_kind_choice1  0.130973   0.413785   0.317  0.75160
## ai_kind_choice2  0.004994   0.316522   0.016  0.98741
## ai_kind_choice3 -0.295111   0.318241  -0.927  0.35376
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_kn_1 a_kn_2
## ai_knd_chc1 -0.016
## ai_knd_chc2 -0.486 -0.149
## ai_knd_chc3 -0.476 -0.152  0.154
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
drop1(model_ai_kind_ownership, test = "Chisq")
```

```
## boundary (singular) fit: see help('isSingular')

## Single term deletions
##
## Model:
## epistemic_ownership_binary ~ ai_kind_choice + (1 | task_id)
##                npar    AIC     LRT Pr(Chi)
## <none>              181.84
## ai_kind_choice    3 176.77 0.93322  0.8174
```

```r
emm_ai_kind_ownership <- emmeans(
  model_ai_kind_ownership,
  ~ ai_kind_choice,
  type = "response"
)

emm_ai_kind_ownership
```

```
##  ai_kind_choice                      prob     SE  df asymp.LCL asymp.UCL
##  A calculator or specialized tool 0.368 0.1110 Inf     0.187     0.597
##  A colleague or advisor           0.340 0.0651 Inf     0.226     0.476
##  A library or search engine       0.276 0.0587 Inf     0.176     0.404
##  A market or aggregator           0.375 0.1710 Inf     0.125     0.715
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
pairs_ai_kind_ownership <- pairs(
  emm_ai_kind_ownership,
  adjust = "tukey"
)

summary(pairs_ai_kind_ownership, type = "response")
```

```
##  contrast                                                      odds.ratio    SE
##  A calculator or specialized tool / A colleague or advisor         1.134 0.632
##  A calculator or specialized tool / A library or search engine     1.531 0.856
##  A calculator or specialized tool / A market or aggregator         0.972 0.847
##  A colleague or advisor / A library or search engine               1.350 0.557
##  A colleague or advisor / A market or aggregator                   0.857 0.674
##  A library or search engine / A market or aggregator               0.635 0.500
##   df null z.ratio p.value
##  Inf    1   0.226  0.9959
##  Inf    1   0.762  0.8714
##  Inf    1  -0.032  1.0000
##  Inf    1   0.727  0.8864
##  Inf    1  -0.196  0.9973
##  Inf    1  -0.577  0.9390
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log odds ratio scale
```

AI conceptualization did not significantly predict binary epistemic ownership, chi-squared(3) = 0.93, p = .817. Estimated probabilities of primarily own-thinking ownership ranged from .28 to .38 across conceptualization groups, and Tukey-adjusted pairwise comparisons showed no significant differences.

Altogether: Conceptualizing AI differently did not significantly predict AI use, similarity to the AI response, or epistemic ownership. Across all three outcomes, the colleague/advisor group showed descriptively higher values, but none of these patterns reached statistical significance. This suggests that participants' explicit mental model of AI was less predictive of cognitive offloading behavior than their actual task-level prompting behavior.

Analysis 2: Learning-Curve Analysis

This analysis is inspired by the Anthropic Economic Index learning-curve report. The idea is that more experienced AI users may use AI differently over time: they may consult it differently, copy it differently, or become more aware of when it has shaped their thinking.

Predictors:

- `ai_use_tenure`: how long the participant has been using AI assistants regularly
- `z_hours_per_week`: approximate weekly AI use (z-scored)
- `ai_understanding`: self-reported understanding of how AI assistants work

Random intercept:

- `task_id`, because participants responded to different short-task prompts

```
learning_curve_data <- subset(
  model_data,
  !is.na(ai_used) &
    !is.na(ai_use_tenure) &
    !is.na(hours_per_week) &
    !is.na(ai_understanding) &
    !is.na(task_id)
)

learning_curve_ai_users <- subset(
  learning_curve_data,
  ai_used == TRUE &
    !is.na(jaccard_similarity_to_ai) &
    !is.na(epistemic_ownership)
)

table(learning_curve_data$ai_use_tenure, useNA = "ifany")
```

```
##
## I do not use them, or am just starting out.
##                                            9
##                           Less than 6 months
##                                           20
##                               6 to 12 months
##                                           49
##                                 1 to 2 years
##                                           89
```

```
##                           More than 2 years
##                                          60
```

```r
table(learning_curve_data$ai_understanding, useNA = "ifany")
```

```
##
##       I have a general sense. It learns from a lot of text and generates responses.
##                                                                                 83
##                              I just use it. I do not think much about how it works.
##                                                                                 14
## I understand it reasonably well. I know about training, prompting, and limitations.
##                                                                                118
##                        I understand it technically. I work with or study AI systems.
##                                                                                 12
```

Analysis 2A: Do More Experienced Users Consult AI Differently?

Outcome: `ai_used`
Model type: mixed logistic regression
Predictors: `ai_use_tenure`, `z_hours_per_week`, `ai_understanding`
Random intercept: `task_id`

This model asks whether more experienced participants were more or less likely to click/use the optional AI assistant during the short task.

```r
model_learning_ai_used <- glmer(
  ai_used ~ ai_use_tenure +
    z_hours_per_week +
    ai_understanding +
    (1 | task_id),
  data = learning_curve_data,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_learning_ai_used)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: ai_used ~ ai_use_tenure + z_hours_per_week + ai_understanding +
##     (1 | task_id)
##    Data: learning_curve_data
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     279.8     314.0    -129.9     259.8       217
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -2.4476 -1.3146  0.5406  0.6718  2.9131
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 227, groups:  task_id, 18
##
## Fixed effects:
##                                                                                                E
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q                                                                                -
## ai_use_tenure.C
## ai_use_tenure^4                                                                                -
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations. -
## ai_understandingI understand it technically. I work with or study AI systems.
##                                                                                                S
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##                                                                                                z
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##                                                                                                P
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L                                                                                *
## ai_use_tenure.Q                                                                                *
## ai_use_tenure.C                                                                                *
## ai_use_tenure^4
```

```
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) a_s_.L a_s_.Q a_s_.C a_s_^4 z_hr__ ajuiIdntmahiw
## ai_us_tnr.L  -0.601
## ai_us_tnr.Q   0.553 -0.711
## ai_us_tnr.C  -0.322  0.529 -0.661
## ai_us_tnr^4   0.128 -0.180  0.327 -0.523
## z_hrs_pr_wk   0.222 -0.163  0.042 -0.067  0.017
## ajuiIdntmahiw -0.316  0.026 -0.083  0.064  0.028 -0.102
## auirwIkatpal  -0.590 -0.018 -0.055  0.095  0.013 -0.125  0.299
## auitIwwosAs   -0.240 -0.038 -0.026  0.032 -0.044 -0.145  0.129
##              auirwIkatpal
## ai_us_tnr.L
## ai_us_tnr.Q
## ai_us_tnr.C
## ai_us_tnr^4
## z_hrs_pr_wk
## ajuiIdntmahiw
## auirwIkatpal
## auitIwwosAs   0.257
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
tests_learning_ai_used <- drop1(
  model_learning_ai_used,
  test = "Chisq"
)
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```r
tests_learning_ai_used
```

```
## Single term deletions
##
## Model:
## ai_used ~ ai_use_tenure + z_hours_per_week + ai_understanding +
##     (1 | task_id)
##                  npar    AIC     LRT  Pr(Chi)
## <none>                279.80
## ai_use_tenure       4 285.74 13.9376 0.007497 **
## z_hours_per_week    1 278.89  1.0930 0.295815
## ai_understanding    3 274.48  0.6795 0.878007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_learning_ai_used_tenure <- emmeans(
  model_learning_ai_used,
  ~ ai_use_tenure,
  type = "response"
)

emm_learning_ai_used_tenure
```

```
##  ai_use_tenure                                      prob     SE  df asymp.LCL
##  I do not use them, or am just starting out. 0.145 0.1350 Inf    0.0196
##  Less than 6 months                          0.741 0.1050 Inf    0.4945
##  6 to 12 months                              0.747 0.0743 Inf    0.5774
##  1 to 2 years                                0.708 0.0656 Inf    0.5657
##  More than 2 years                           0.813 0.0582 Inf    0.6722
##  asymp.UCL
##      0.590
##      0.894
##      0.865
##      0.819
##      0.902
##
## Results are averaged over the levels of: ai_understanding
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

AI-use tenure significantly predicted use of the optional AI assistant, chi-squared(4) = 13.94, p = .0075. Participants who reported not using AI or just starting out had a much lower estimated AI-use probability (.15) than all other tenure groups (.71-.81). This should still be interpreted cautiously because the smallest tenure group produces wide uncertainty. Hours per week, p = .296, and AI understanding, p = .878, were not significant predictors.

```r
model_learning_ai_used_glm <- glm(
  ai_used ~ ai_use_tenure +
    z_hours_per_week +
    ai_understanding,
  data = learning_curve_data,
  family = binomial
)

summary(model_learning_ai_used_glm)
```

```
##
## Call:
## glm(formula = ai_used ~ ai_use_tenure + z_hours_per_week + ai_understanding,
##     family = binomial, data = learning_curve_data)
##
## Coefficients:
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
```

```
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 278.84  on 226  degrees of freedom
## Residual deviance: 259.80  on 218  degrees of freedom
## AIC: 277.8
##
```

```
## Number of Fisher Scoring iterations: 4
```

```r
tests_learning_ai_used_glm <- drop1(
  model_learning_ai_used_glm,
  test = "Chisq"
)

tests_learning_ai_used_glm
```

```
## Single term deletions
##
## Model:
## ai_used ~ ai_use_tenure + z_hours_per_week + ai_understanding
##                  Df Deviance    AIC     LRT Pr(>Chi)
## <none>                259.80 277.80
## ai_use_tenure     4   273.74 283.74 13.9376 0.007497 **
## z_hours_per_week  1   260.89 276.89  1.0930 0.295815
## ai_understanding  3   260.48 272.48  0.6795 0.878007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

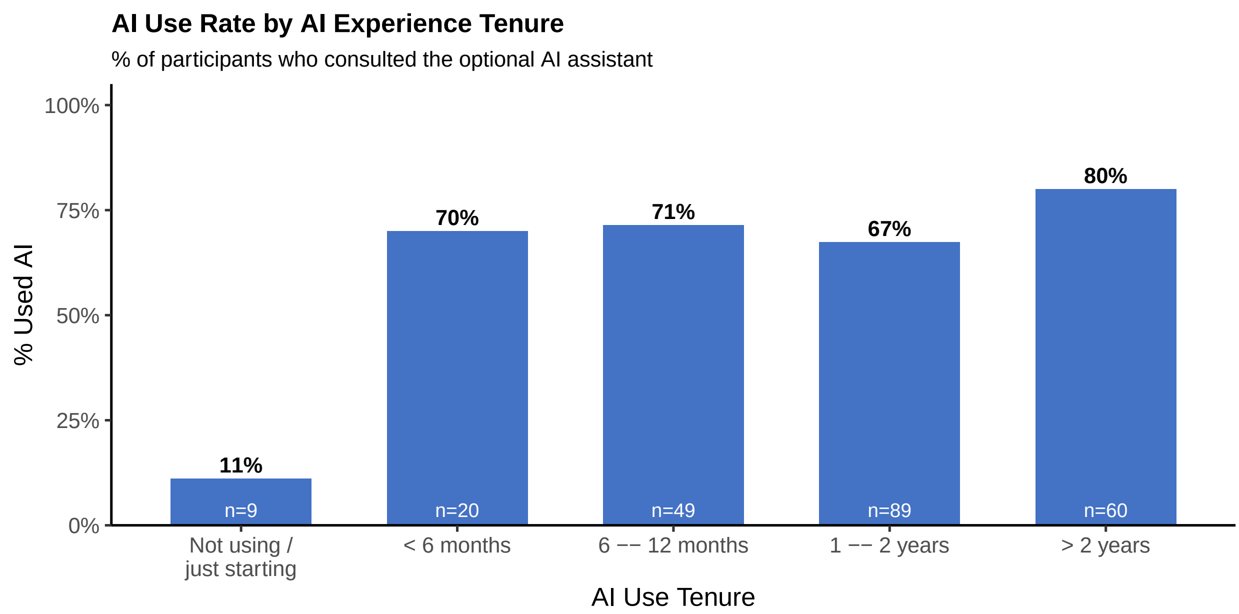
```r
emm_learning_ai_used_tenure_glm <- emmeans(
  model_learning_ai_used_glm,
  ~ ai_use_tenure,
  type = "response"
)

emm_learning_ai_used_tenure_glm
```

```
##  ai_use_tenure                                 prob     SE  df asymp.LCL
##  I do not use them, or am just starting out. 0.145 0.1350 Inf    0.0197
##  Less than 6 months                          0.741 0.1050 Inf    0.4945
##  6 to 12 months                              0.747 0.0743 Inf    0.5774
##  1 to 2 years                                0.708 0.0656 Inf    0.5657
##  More than 2 years                           0.813 0.0582 Inf    0.6722
##  asymp.UCL
##      0.590
##      0.894
##      0.865
##      0.819
##      0.902
##
## Results are averaged over the levels of: ai_understanding
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

Same results but the extreme predicted probabilities and very large standard errors suggest that this result should be interpreted cautiously.

```r
tenure_levels <- c(
  "I do not use them, or am just starting out.",
  "Less than 6 months",
```

```
  "6 to 12 months",
  "1 to 2 years",
  "More than 2 years"
)
tenure_labels_short <- c(
  "Not using /\njust starting",
  "< 6 months",
  "6 -- 12 months",
  "1 -- 2 years",
  "> 2 years"
)

tenure_use_df <- do.call(rbind, lapply(
  seq_along(tenure_levels),
  function(i) {
    sub <- subset(learning_curve_data, ai_use_tenure == tenure_levels[i])
    if (nrow(sub) == 0) return(NULL)
    data.frame(
      tenure   = tenure_labels_short[i],
      pct_used = mean(sub$ai_used, na.rm = TRUE),
      n        = nrow(sub),
      stringsAsFactors = FALSE
    )
  }
))
tenure_use_df$tenure <- factor(tenure_use_df$tenure, levels = tenure_labels_short)

ggplot(tenure_use_df, aes(x = tenure, y = pct_used)) +
  geom_col(fill = "#4472C4", width = 0.65) +
  geom_text(
    aes(label = paste0(round(pct_used * 100), "%")),
    vjust = -0.4, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n), y = 0.02),
    vjust = 0, size = 3, color = "white"
  ) +
  scale_y_continuous(
    limits = c(0, 1.05), expand = c(0, 0),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title    = "AI Use Rate by AI Experience Tenure",
    subtitle = "% of participants who consulted the optional AI assistant",
    x        = "AI Use Tenure",
    y        = "% Used AI"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10),
    axis.text.x   = element_text(size = 10)
  )
```



Analysis 2B: Do More Experienced Users Copy AI Differently?

Outcome: `jaccard_similarity_to_ai`
Model type: linear mixed regression
Predictors: `ai_use_tenure`, `z_hours_per_week`, `ai_understanding`
Random intercept: `task_id`
Sample: only participants who used the AI assistant

This model asks whether more experienced participants produced final responses that were more or less similar to the pre-written AI response.

```
model_learning_similarity <- lmer(
  jaccard_similarity_to_ai ~ ai_use_tenure +
    z_hours_per_week +
    ai_understanding +
    (1 | task_id),
  data = learning_curve_ai_users
)

summary(model_learning_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ ai_use_tenure + z_hours_per_week +
##     ai_understanding + (1 | task_id)
##    Data: learning_curve_ai_users
##
## REML criterion at convergence: 192.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.7644 -0.8611 -0.2303  0.9309  1.6229
##
```

```
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.01908  0.1381
##  Residual             0.16268  0.4033
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
```

```
## ai_understandingI just use it. I do not think much about how it works.                          0
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations. 0
## ai_understandingI understand it technically. I work with or study AI systems.                    0
##
## (Intercept)                                                                                    *
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) a_s_.L a_s_.Q a_s_.C a_s_^4 z_hr__ ajuiIdntmahiw
## ai_us_tnr.L  -0.796
## ai_us_tnr.Q   0.760 -0.908
## ai_us_tnr.C  -0.629  0.780 -0.854
## ai_us_tnr^4   0.365 -0.463  0.538 -0.654
## z_hrs_pr_wk   0.080 -0.094  0.000 -0.063  0.039
## ajuiIdntmahiw -0.205  0.003 -0.022  0.012  0.047 -0.175
## auirwIkatpal -0.463  0.079 -0.118  0.123 -0.019 -0.080  0.306
## auitIwwosAs  -0.207  0.005 -0.038  0.051 -0.075 -0.113  0.169
##              auirwIkatpal
## ai_us_tnr.L
## ai_us_tnr.Q
## ai_us_tnr.C
## ai_us_tnr^4
## z_hrs_pr_wk
## ajuiIdntmahiw
## auirwIkatpal
## auitIwwosAs   0.299
```

```r
tests_learning_similarity <- anova(model_learning_similarity)

tests_learning_similarity
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                   Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_use_tenure    0.68531 0.171328     4 143.31  1.0531 0.3821
## z_hours_per_week 0.00276 0.002759     1 143.19  0.0170 0.8966
## ai_understanding 0.27510 0.091700     3 142.26  0.5637 0.6398
```

```r
emm_learning_similarity_tenure <- emmeans(
  model_learning_similarity,
  ~ ai_use_tenure
)
```

```
## Cannot use mode = "kenward-roger" because *pbkrtest* package is not installed
```

```r
emm_learning_similarity_tenure
```

```
##  ai_use_tenure                                  emmean     SE    df lower.CL
##  I do not use them, or am just starting out.  0.119 0.4220 146.4   -0.714
##  Less than 6 months                           0.527 0.1230 139.6    0.283
##  6 to 12 months                               0.630 0.0863  98.3    0.459
##  1 to 2 years                                 0.493 0.0734  70.4    0.347
##  More than 2 years                            0.600 0.0776  80.8    0.446
##  upper.CL
##     0.952
##     0.770
##     0.801
##     0.639
##     0.755
##
## Results are averaged over the levels of: ai_understanding
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

AI-use tenure did not significantly predict Jaccard similarity to the AI response (see ANOVA output above). Descriptively, observed mean Jaccard ranged from .45 (1–2 years) to .57 (6–12 months) across tenure groups, with no consistent trend with experience level. Hours of AI use per week and self-rated AI understanding were also not significant predictors.

Analysis 2C: Do More Experienced Users Recognize AI Influence Differently?

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression
Predictors: `z_jaccard_similarity_to_ai`, `ai_use_tenure`, `z_hours_per_week`, `ai_understanding`
Interaction: `z_jaccard_similarity_to_ai * ai_use_tenure`
Random intercept: `task_id`
Sample: only participants who used the AI assistant

This model asks whether the link between objective uptake and epistemic ownership differs by experience. Epistemic ownership is recoded as binary (see Model 3). The Jaccard predictor is z-scored for comparability.

```r
model_learning_ownership <- glmer(
  epistemic_ownership_binary ~ z_jaccard_similarity_to_ai *
    ai_use_tenure +
    z_hours_per_week +
    ai_understanding +
    (1 | task_id),
  data = learning_curve_ai_users,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly uniden
##  - Rescale variables?
```

```r
summary(model_learning_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## epistemic_ownership_binary ~ z_jaccard_similarity_to_ai * ai_use_tenure +
##     z_hours_per_week + ai_understanding + (1 | task_id)
##    Data: learning_curve_ai_users
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     173.6     215.2     -72.8     145.6       130
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8062 -0.5751 -0.2845  0.7518  3.5317
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0.02015  0.142
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                                                                                                     E
## (Intercept)
## z_jaccard_similarity_to_ai
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## z_jaccard_similarity_to_ai:ai_use_tenure.L
## z_jaccard_similarity_to_ai:ai_use_tenure.Q
## z_jaccard_similarity_to_ai:ai_use_tenure.C
##                                                                                                     S
## (Intercept)
## z_jaccard_similarity_to_ai
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## z_jaccard_similarity_to_ai:ai_use_tenure.L
## z_jaccard_similarity_to_ai:ai_use_tenure.Q
## z_jaccard_similarity_to_ai:ai_use_tenure.C
##                                                                                                     z
```

```
## (Intercept)
## z_jaccard_similarity_to_ai
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## z_jaccard_similarity_to_ai:ai_use_tenure.L
## z_jaccard_similarity_to_ai:ai_use_tenure.Q
## z_jaccard_similarity_to_ai:ai_use_tenure.C
##
## (Intercept)
## z_jaccard_similarity_to_ai
## ai_use_tenure.L
## ai_use_tenure.Q
## ai_use_tenure.C
## ai_use_tenure^4
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## z_jaccard_similarity_to_ai:ai_use_tenure.L
## z_jaccard_similarity_to_ai:ai_use_tenure.Q
## z_jaccard_similarity_to_ai:ai_use_tenure.C

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(value, correlation=TRUE)  or
##     vcov(value)        if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## optimizer (bobyqa) convergence code: 0 (OK)
## Model is nearly unidentifiable: large eigenvalue ratio
##  - Rescale variables?
```

```r
tests_learning_ownership <- drop1(model_learning_ownership, test = "Chisq")
```

```
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly uniden
##  - Rescale variables?

## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to conve
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
tests_learning_ownership
```

```
## Single term deletions
##
## Model:
## epistemic_ownership_binary ~ z_jaccard_similarity_to_ai * ai_use_tenure +
##     z_hours_per_week + ai_understanding + (1 | task_id)
##                                           npar    AIC     LRT Pr(Chi)
## <none>                                          173.59
## z_hours_per_week                             1 171.96 0.36570  0.5454
## ai_understanding                             3 169.28 1.68867  0.6395
## z_jaccard_similarity_to_ai:ai_use_tenure     3 168.17 0.58377  0.9001
```

```r
ownership_similarity_slopes_by_tenure <- emtrends(
  model_learning_ownership,
  ~ ai_use_tenure,
  var = "z_jaccard_similarity_to_ai"
)

ownership_similarity_slopes_by_tenure
```

```
##  ai_use_tenure                                z_jaccard_similarity_to_ai.trend
##  I do not use them, or am just starting out.                            nonEst
##  Less than 6 months                                                      -1.39
##  6 to 12 months                                                          -1.43
##  1 to 2 years                                                            -1.15
##  More than 2 years                                                       -0.98
##     SE  df asymp.LCL asymp.UCL
##     NA  NA        NA        NA
##  0.852 Inf     -3.06     0.281
##  0.593 Inf     -2.59    -0.262
##  0.391 Inf     -1.92    -0.386
##  0.368 Inf     -1.70    -0.258
##
## Results are averaged over the levels of: ai_understanding
## Confidence level used: 0.95
```

```r
pairs(
  ownership_similarity_slopes_by_tenure,
  adjust = "tukey"
)
```

```
##  contrast                                                         estimate
##  I do not use them, or am just starting out. - Less than 6 months   nonEst
##  I do not use them, or am just starting out. - 6 to 12 months       nonEst
##  I do not use them, or am just starting out. - 1 to 2 years         nonEst
##  I do not use them, or am just starting out. - More than 2 years    nonEst
##  Less than 6 months - 6 to 12 months                                0.0353
##  Less than 6 months - 1 to 2 years                                 -0.2380
##  Less than 6 months - More than 2 years                            -0.4095
```

```
##  6 to 12 months - 1 to 2 years                                      -0.2733
##  6 to 12 months - More than 2 years                                 -0.4448
##  1 to 2 years - More than 2 years                                   -0.1715
##     SE  df z.ratio p.value
##     NA  NA      NA      NA
##     NA  NA      NA      NA
##     NA  NA      NA      NA
##     NA  NA      NA      NA
##  1.020 Inf   0.035  1.0000
##  0.931 Inf  -0.256  0.9941
##  0.921 Inf  -0.444  0.9707
##  0.691 Inf  -0.395  0.9790
##  0.658 Inf  -0.676  0.9063
##  0.526 Inf  -0.326  0.9880
##
## Results are averaged over the levels of: ai_understanding
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Higher Jaccard similarity predicted lower odds of reporting primarily own-thinking epistemic ownership, with negative similarity slopes across tenure groups. The similarity-by-tenure interaction was not significant, chi-squared(3) = 0.58, p = .900. Hours per week, p = .545, and self-reported AI understanding, p = .640, were not significant predictors of epistemic ownership.

Altogether: There is a limited experience effect. **AI-use tenure predicted whether participants consulted the optional assistant** (driven by near-zero use in the no-experience group), but experience did not meaningfully predict how much they copied or converged with the AI response, nor did it change the link between uptake and epistemic ownership.

Analysis 3: Task-Category Analysis

This analysis asks whether different kinds of tasks invite different levels of AI use, uptake, ownership, or confidence.

These models use fixed effects rather than random effects because `assigned_category` is the thing I want to compare directly. This means the models estimate differences between the six task categories, rather than treating task category as background variation. Adding a random intercept for `task_id` within category would likely produce singular fits given the small number of observations per scenario.

```r
library(emmeans)

task_categories <- c(
  "academic",
  "creative",
  "decisions",
  "info",
  "learning",
  "writing"
)

task_category_data <- subset(
  model_data,
  assigned_category %in% task_categories
)
```

```r
task_category_data$assigned_category <- factor(
  task_category_data$assigned_category,
  levels = task_categories
)

task_category_similarity_data <- subset(
  task_category_data,
  ai_used == TRUE & !is.na(jaccard_similarity_to_ai)
)

task_category_ownership_data <- subset(
  task_category_data,
  ai_used == TRUE & !is.na(epistemic_ownership)
)

task_category_confidence_data <- subset(
  task_category_data,
  !is.na(confidence_level)
)

table(task_category_data$assigned_category, useNA = "ifany")
```

```
##
##   academic  creative decisions      info  learning   writing
##         36        35        36        41        42        37
```

Analysis 3A: Does Task Category Predict Whether Participants Used AI?

Outcome: `ai_used`
Model type: logistic regression
Predictor: `assigned_category`

This model asks whether participants were more likely to open/use the optional AI assistant for some task categories than others.

```r
binary_category_summary(task_category_data, "ai_used")
```

```
##   assigned_category  n proportion
## 1          academic 36  0.5833333
## 2          creative 35  0.6571429
## 3         decisions 36  0.6944444
## 4              info 41  0.7073171
## 5          learning 42  0.8571429
## 6           writing 37  0.6486486
```

```r
model_task_ai_used <- glm(
  ai_used ~ assigned_category,
  data = task_category_data,
  family = binomial
)

summary(model_task_ai_used)
```

```
##
## Call:
## glm(formula = ai_used ~ assigned_category, family = binomial,
##     data = task_category_data)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  0.3365     0.3381   0.995  0.31959
## assigned_categorycreative    0.3141     0.4910   0.640  0.52235
## assigned_categorydecisions   0.4845     0.4952   0.978  0.32784
## assigned_categoryinfo        0.5459     0.4818   1.133  0.25715
## assigned_categorylearning    1.4553     0.5556   2.619  0.00882 **
## assigned_categorywriting     0.2766     0.4826   0.573  0.56648
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 278.84  on 226  degrees of freedom
## Residual deviance: 270.22  on 221  degrees of freedom
## AIC: 282.22
##
## Number of Fisher Scoring iterations: 4
```

```
tests_task_ai_used <- drop1(
  model_task_ai_used,
  test = "Chisq"
)

tests_task_ai_used
```

```
## Single term deletions
##
## Model:
## ai_used ~ assigned_category
##                   Df Deviance    AIC    LRT Pr(>Chi)
## <none>                 270.22 282.22
## assigned_category  5   278.84 280.84 8.6245    0.125
```

```
emm_task_ai_used <- emmeans(
  model_task_ai_used,
  ~ assigned_category,
  type = "response"
)

emm_task_ai_used
```

```
##  assigned_category  prob     SE  df asymp.LCL asymp.UCL
##  academic          0.583 0.0822 Inf     0.419     0.731
##  creative          0.657 0.0802 Inf     0.488     0.794
##  decisions         0.694 0.0768 Inf     0.528     0.822
##  info              0.707 0.0711 Inf     0.552     0.826
##  learning          0.857 0.0540 Inf     0.717     0.934
```

```
##  writing           0.649 0.0785 Inf     0.485     0.784
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
pairs_task_ai_used <- pairs(
  emm_task_ai_used,
  adjust = "tukey"
)

summary(pairs_task_ai_used, type = "response")
```

```
##  contrast             odds.ratio    SE  df null z.ratio p.value
##  academic / creative       0.730 0.359 Inf    1  -0.640  0.9880
##  academic / decisions      0.616 0.305 Inf    1  -0.978  0.9249
##  academic / info           0.579 0.279 Inf    1  -1.133  0.8677
##  academic / learning       0.233 0.130 Inf    1  -2.619  0.0926
##  academic / writing        0.758 0.366 Inf    1  -0.573  0.9928
##  creative / decisions      0.843 0.428 Inf    1  -0.336  0.9994
##  creative / info           0.793 0.392 Inf    1  -0.469  0.9972
##  creative / learning       0.319 0.181 Inf    1  -2.013  0.3345
##  creative / writing        1.038 0.514 Inf    1   0.076  1.0000
##  decisions / info          0.940 0.469 Inf    1  -0.123  1.0000
##  decisions / learning      0.379 0.216 Inf    1  -1.702  0.5304
##  decisions / writing       1.231 0.615 Inf    1   0.416  0.9984
##  info / learning           0.403 0.225 Inf    1  -1.627  0.5803
##  info / writing            1.309 0.636 Inf    1   0.554  0.9938
##  learning / writing        3.250 1.820 Inf    1   2.107  0.2837
##
## P value adjustment: tukey method for comparing a family of 6 estimates
## Tests are performed on the log odds ratio scale
```

Assigned task category did not significantly predict whether participants used the AI assistant, chi-squared(5) = 8.62, p = .125. Descriptively, AI use was highest in learning tasks (86%) and lowest in academic tasks (58%), with other categories between 65% and 71%. Tukey-adjusted pairwise comparisons were not significant.

Analysis 3B: Does Task Category Predict Similarity To The AI Response?

Outcome: `jaccard_similarity_to_ai`
Model type: linear regression
Predictor: `assigned_category`
Sample: only participants who used the AI assistant

This model asks whether participants copied, borrowed from, or converged with the AI response more in some task categories than others.

```r
continuous_category_summary(
  task_category_similarity_data,
  "jaccard_similarity_to_ai"
)
```

```
##   assigned_category  n      mean        sd    median
## 1          academic 21 0.5854038 0.4585318 1.0000000
## 2          creative 23 0.7562262 0.4190480 1.0000000
## 3         decisions 25 0.2957821 0.3802970 0.0900000
## 4              info 29 0.4319651 0.4163731 0.1882353
## 5          learning 36 0.3623819 0.3481275 0.2230777
## 6           writing 24 0.7190525 0.3431838 0.9509569
```

```r
model_task_similarity <- lm(
  jaccard_similarity_to_ai ~ assigned_category,
  data = task_category_similarity_data
)

summary(model_task_similarity)
```

```
##
## Call:
## lm(formula = jaccard_similarity_to_ai ~ assigned_category, data = task_category_similarity_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.75623 -0.28297 -0.09218  0.28095  0.70422
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  0.58540    0.08552   6.845 1.76e-10 ***
## assigned_categorycreative    0.17082    0.11828   1.444   0.1507
## assigned_categorydecisions -0.28962    0.11600  -2.497   0.0136 *
## assigned_categoryinfo      -0.15344    0.11229  -1.366   0.1738
## assigned_categorylearning  -0.22302    0.10761  -2.073   0.0399 *
## assigned_categorywriting    0.13365    0.11710   1.141   0.2555
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3919 on 152 degrees of freedom
## Multiple R-squared:  0.1666, Adjusted R-squared:  0.1392
## F-statistic: 6.079 on 5 and 152 DF,  p-value: 3.677e-05
```

```r
tests_task_similarity <- anova(model_task_similarity)

tests_task_similarity
```

```
## Analysis of Variance Table
##
## Response: jaccard_similarity_to_ai
##                    Df  Sum Sq Mean Sq F value    Pr(>F)
## assigned_category   5  4.6682 0.93364  6.0792 3.677e-05 ***
## Residuals         152 23.3441 0.15358
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_task_similarity <- emmeans(
  model_task_similarity,
  ~ assigned_category
)

emm_task_similarity
```

```
##  assigned_category emmean     SE  df lower.CL upper.CL
##  academic           0.585 0.0855 152    0.416    0.754
##  creative           0.756 0.0817 152    0.595    0.918
##  decisions          0.296 0.0784 152    0.141    0.451
##  info               0.432 0.0728 152    0.288    0.576
##  learning           0.362 0.0653 152    0.233    0.491
##  writing            0.719 0.0800 152    0.561    0.877
##
## Confidence level used: 0.95
```

```r
pairs_task_similarity <- pairs(
  emm_task_similarity,
  adjust = "tukey"
)

pairs_task_similarity
```

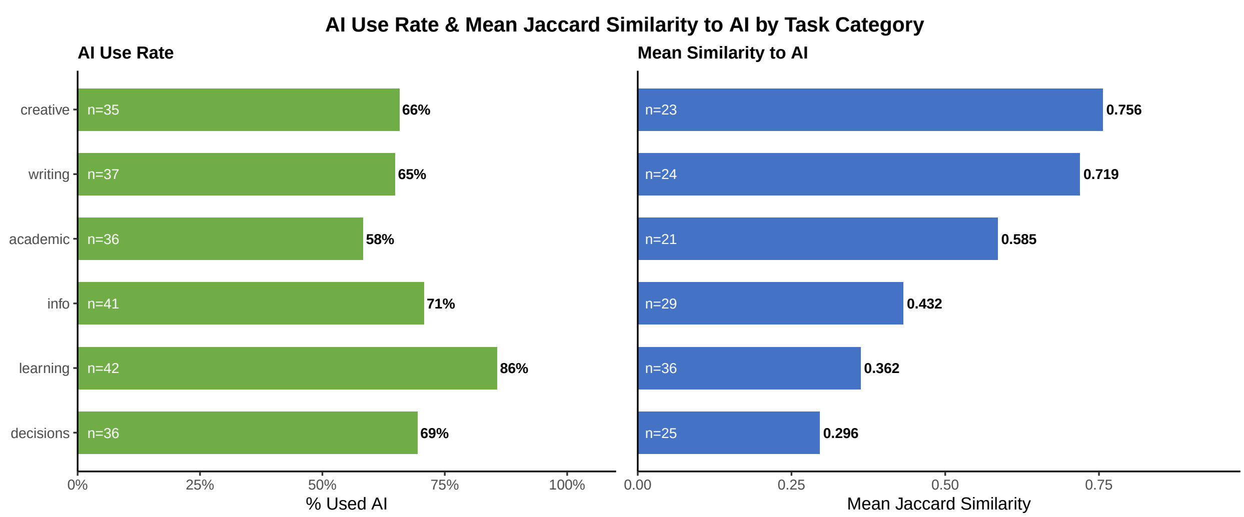
```
##  contrast              estimate     SE  df t.ratio p.value
##  academic - creative    -0.1708 0.1180 152  -1.444  0.7000
##  academic - decisions    0.2896 0.1160 152   2.497  0.1316
##  academic - info         0.1534 0.1120 152   1.366  0.7469
##  academic - learning     0.2230 0.1080 152   2.073  0.3073
##  academic - writing     -0.1336 0.1170 152  -1.141  0.8633
##  creative - decisions    0.4604 0.1130 152   4.067  0.0011
##  creative - info         0.3243 0.1090 152   2.963  0.0405
##  creative - learning     0.3938 0.1050 152   3.765  0.0032
##  creative - writing      0.0372 0.1140 152   0.325  0.9995
##  decisions - info       -0.1362 0.1070 152  -1.273  0.7991
##  decisions - learning   -0.0666 0.1020 152  -0.653  0.9866
##  decisions - writing    -0.4233 0.1120 152  -3.779  0.0030
##  info - learning         0.0696 0.0978 152   0.712  0.9803
##  info - writing         -0.2871 0.1080 152  -2.655  0.0907
##  learning - writing     -0.3567 0.1030 152  -3.454  0.0092
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

Among AI users, assigned task category significantly predicted Jaccard similarity, $F(5, 152) = 6.08$, $p < .001$. Creative (M = .76) and writing (M = .72) tasks showed the highest similarity to the AI response, while decisions (M = .30) and learning (M = .36) showed the lowest. Tukey-adjusted comparisons showed that creative tasks were significantly higher than decisions (p = .001), info (p = .041), and learning (p = .003), and writing tasks were significantly higher than decisions (p = .003) and learning (p = .009).

```r
# Mean Jaccard by category
jac_cat_df <- do.call(rbind, lapply(
  split(task_category_similarity_data,
```

```r
        task_category_similarity_data$assigned_category),
  function(sub) {
    data.frame(
      assigned_category = as.character(sub$assigned_category[1]),
      mean_jac          = mean(sub$jaccard_similarity_to_ai, na.rm = TRUE),
      n                 = sum(!is.na(sub$jaccard_similarity_to_ai)),
      stringsAsFactors  = FALSE
    )
  }
))

jac_cat_df$assigned_category <- factor(
  jac_cat_df$assigned_category,
  levels = jac_cat_df$assigned_category[order(jac_cat_df$mean_jac)]
)

p_jac <- ggplot(jac_cat_df, aes(x = assigned_category, y = mean_jac)) +
  geom_col(fill = "#4472C4", width = 0.65) +
  geom_text(
    aes(label = sprintf("%.3f", mean_jac)),
    hjust = -0.10, size = 3.2, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n), y = 0.012),
    hjust = 0, size = 3, color = "white"
  ) +
  coord_flip() +
  scale_y_continuous(limits = c(0, 0.98), expand = c(0, 0)) +
  labs(title = "Mean Similarity to AI", x = NULL, y = "Mean Jaccard Similarity") +
  theme_classic(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 11),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )

# AI use rate by category, using the same category order
use_cat_df <- do.call(rbind, lapply(
  split(task_category_data, task_category_data$assigned_category),
  function(sub) {
    data.frame(
      assigned_category = as.character(sub$assigned_category[1]),
      pct_used          = mean(sub$ai_used, na.rm = TRUE),
      n                 = nrow(sub),
      stringsAsFactors  = FALSE
    )
  }
))

use_cat_df$assigned_category <- factor(
  use_cat_df$assigned_category,
  levels = levels(jac_cat_df$assigned_category)
)
```

```r
p_use <- ggplot(use_cat_df, aes(x = assigned_category, y = pct_used)) +
  geom_col(fill = "#70AD47", width = 0.65) +
  geom_text(
    aes(label = paste0(round(pct_used * 100), "%")),
    hjust = -0.10, size = 3.2, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n), y = 0.02),
    hjust = 0, size = 3, color = "white"
  ) +
  coord_flip() +
  scale_y_continuous(
    limits = c(0, 1.1), expand = c(0, 0),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(title = "AI Use Rate", x = NULL, y = "% Used AI") +
  theme_classic(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 11)
  )

grid.arrange(
  p_use, p_jac,
  ncol = 2,
  top = grid::textGrob(
    "AI Use Rate & Mean Jaccard Similarity to AI by Task Category",
    gp = grid::gpar(fontface = "bold", fontsize = 13)
  )
)
```



Analysis 3C: Does Task Category Predict Epistemic Ownership?

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: logistic regression
Predictor: `assigned_category`
Sample: only participants who used the AI assistant

This model asks whether epistemic ownership ratings differed across task categories. Epistemic ownership is recoded as binary (see Model 3).

```
binary_category_summary(
  task_category_ownership_data,
  "epistemic_ownership_binary"
)
```

```
##   assigned_category  n proportion
## 1          academic 20  0.4500000
## 2          creative 23  0.3478261
## 3         decisions 24  0.4166667
## 4              info 23  0.2608696
## 5          learning 32  0.2500000
## 6           writing 22  0.3181818
```

```
model_task_ownership <- glm(
  epistemic_ownership_binary ~ assigned_category,
  data = task_category_ownership_data,
  family = binomial
)

summary(model_task_ownership)
```

```
##
## Call:
## glm(formula = epistemic_ownership_binary ~ assigned_category,
##     family = binomial, data = task_category_ownership_data)
##
## Coefficients:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -0.2007     0.4495  -0.446    0.655
## assigned_categorycreative     -0.4279     0.6274  -0.682    0.495
## assigned_categorydecisions    -0.1358     0.6111  -0.222    0.824
## assigned_categoryinfo         -0.8408     0.6538  -1.286    0.198
## assigned_categorylearning     -0.8979     0.6072  -1.479    0.139
## assigned_categorywriting      -0.5615     0.6415  -0.875    0.381
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 183.32  on 143  degrees of freedom
## Residual deviance: 179.76  on 138  degrees of freedom
##   (14 observations deleted due to missingness)
## AIC: 191.76
##
## Number of Fisher Scoring iterations: 4
```

```
tests_task_ownership <- drop1(model_task_ownership, test = "Chisq")

tests_task_ownership
```

```
## Single term deletions
##
```

```
## Model:
## epistemic_ownership_binary ~ assigned_category
##                    Df Deviance    AIC    LRT Pr(>Chi)
## <none>                  179.76 191.76
## assigned_category  5    183.32 185.32 3.5557    0.615
```

```r
emm_task_ownership <- emmeans(
  model_task_ownership,
  ~ assigned_category,
  type = "response"
)

emm_task_ownership
```

```
##  assigned_category  prob     SE  df asymp.LCL asymp.UCL
##  academic          0.450 0.1110 Inf     0.253     0.664
##  creative          0.348 0.0993 Inf     0.184     0.557
##  decisions         0.417 0.1010 Inf     0.241     0.617
##  info              0.261 0.0916 Inf     0.122     0.472
##  learning          0.250 0.0765 Inf     0.130     0.426
##  writing           0.318 0.0993 Inf     0.160     0.534
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
pairs_task_ownership <- pairs(
  emm_task_ownership,
  adjust = "tukey"
)

summary(pairs_task_ownership, type = "response")
```

```
##  contrast              odds.ratio    SE  df null z.ratio p.value
##  academic / creative        1.534 0.963 Inf    1   0.682  0.9840
##  academic / decisions       1.145 0.700 Inf    1   0.222  0.9999
##  academic / info            2.318 1.520 Inf    1   1.286  0.7929
##  academic / learning        2.455 1.490 Inf    1   1.479  0.6779
##  academic / writing         1.753 1.120 Inf    1   0.875  0.9525
##  creative / decisions       0.747 0.450 Inf    1  -0.485  0.9967
##  creative / info            1.511 0.976 Inf    1   0.639  0.9881
##  creative / learning        1.600 0.958 Inf    1   0.785  0.9701
##  creative / writing         1.143 0.724 Inf    1   0.211  0.9999
##  decisions / info           2.024 1.280 Inf    1   1.119  0.8737
##  decisions / learning       2.143 1.250 Inf    1   1.311  0.7793
##  decisions / writing        1.531 0.945 Inf    1   0.690  0.9832
##  info / learning            1.059 0.663 Inf    1   0.091  1.0000
##  info / writing             0.756 0.499 Inf    1  -0.423  0.9983
##  learning / writing         0.714 0.438 Inf    1  -0.549  0.9941
##
## P value adjustment: tukey method for comparing a family of 6 estimates
## Tests are performed on the log odds ratio scale
```

Assigned task category did not significantly predict binary epistemic ownership, chi-squared(5) = 3.56, p = .615. Estimated probabilities of primarily own-thinking ownership varied descriptively from .25 in learning to .45 in academic tasks, but Tukey-adjusted pairwise comparisons were not significant.

Analysis 3D: Does Task Category Predict Confidence?

Outcome: `confidence_level`
Model type: linear regression
Predictor: `assigned_category`

This model asks whether participants felt more confident in their final response for some task categories than others. The confidence scale is treated as approximately continuous.

```r
continuous_category_summary(
  task_category_confidence_data,
  "confidence_level"
)
```

```
##   assigned_category  n     mean        sd median
## 1          academic 36 5.305556 1.3270686      6
## 2          creative 35 5.400000 1.3974767      6
## 3         decisions 36 6.138889 0.8669413      6
## 4              info 41 5.170732 1.4474536      5
## 5          learning 42 4.785714 1.4739314      5
## 6           writing 37 5.864865 1.0317774      6
```

```r
model_task_confidence <- lm(
  confidence_level ~ assigned_category,
  data = task_category_confidence_data
)

summary(model_task_confidence)
```

```
##
## Call:
## lm(formula = confidence_level ~ assigned_category, data = task_category_confidence_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.4000 -0.8649  0.1351  0.8611  2.2143
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 5.30556    0.21435  24.752  < 2e-16 ***
## assigned_categorycreative   0.09444    0.30529   0.309  0.75734
## assigned_categorydecisions  0.83333    0.30314   2.749  0.00647 **
## assigned_categoryinfo      -0.13482    0.29375  -0.459  0.64670
## assigned_categorylearning  -0.51984    0.29211  -1.780  0.07651 .
## assigned_categorywriting    0.55931    0.30108   1.858  0.06455 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.286 on 221 degrees of freedom
```

```
## Multiple R-squared:  0.1115, Adjusted R-squared:  0.09136
## F-statistic: 5.545 on 5 and 221 DF,  p-value: 7.83e-05
```

```r
tests_task_confidence <- anova(model_task_confidence)

tests_task_confidence
```

```
## Analysis of Variance Table
##
## Response: confidence_level
##                      Df Sum Sq Mean Sq F value   Pr(>F)
## assigned_category   5  45.86  9.1712  5.5447 7.83e-05 ***
## Residuals         221 365.55  1.6541
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_task_confidence <- emmeans(
  model_task_confidence,
  ~ assigned_category
)

emm_task_confidence
```

```
##  assigned_category emmean    SE  df lower.CL upper.CL
##  academic            5.31 0.214 221     4.88     5.73
##  creative            5.40 0.217 221     4.97     5.83
##  decisions           6.14 0.214 221     5.72     6.56
##  info                5.17 0.201 221     4.77     5.57
##  learning            4.79 0.198 221     4.39     5.18
##  writing             5.86 0.211 221     5.45     6.28
##
## Confidence level used: 0.95
```

```r
pairs_task_confidence <- pairs(
  emm_task_confidence,
  adjust = "tukey"
)

pairs_task_confidence
```

```
##  contrast              estimate    SE  df t.ratio p.value
##  academic - creative    -0.0944 0.305 221  -0.309  0.9996
##  academic - decisions   -0.8333 0.303 221  -2.749  0.0699
##  academic - info         0.1348 0.294 221   0.459  0.9974
##  academic - learning     0.5198 0.292 221   1.780  0.4811
##  academic - writing     -0.5593 0.301 221  -1.858  0.4312
##  creative - decisions   -0.7389 0.305 221  -2.420  0.1538
##  creative - info         0.2293 0.296 221   0.775  0.9715
##  creative - learning     0.6143 0.294 221   2.087  0.2979
##  creative - writing     -0.4649 0.303 221  -1.533  0.6434
##  decisions - info        0.9682 0.294 221   3.296  0.0144
##  decisions - learning    1.3532 0.292 221   4.632 <0.0001
```

```
##   decisions - writing    0.2740 0.301 221   0.910  0.9436
##   info - learning        0.3850 0.282 221   1.364  0.7487
##   info - writing        -0.6941 0.292 221  -2.380  0.1677
##   learning - writing    -1.0792 0.290 221  -3.722  0.0034
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

Assigned task category significantly predicted confidence, $F(5, 221) = 5.54$, $p < .001$. Confidence was highest for decisions (M = 6.14) and writing (M = 5.86), and lowest for learning (M = 4.79). Tukey-adjusted comparisons showed significantly higher confidence for decisions than info ($p = .014$) and learning ($p < .001$), and higher confidence for writing than learning ($p = .003$).

Altogether: Task category mattered most for uptake and confidence. **Among AI users, creative and writing tasks produced much higher AI similarity than decisions and learning tasks, $F(5, 152) = 6.08$, $p < .001$.** Task category did not significantly predict whether participants opened AI or whether they reported primarily own-thinking ownership, but it did significantly predict confidence, $F(5, 221) = 5.54$, $p < .001$.

Read Intent Codes

```r
intent_path <- "survey_responses_with_ai_intent_codes.csv"

intent_raw <- read.csv(
  intent_path,
  stringsAsFactors = FALSE,
  na.strings = c("", "NA"),
  fileEncoding = "UTF-8"
)

intent <- intent_raw
```

Analysis 4: Asking/Doing/Expressing Intent Coding

This analysis uses the coded `question_for_ai` text, following the ChatGPT paper's Appendix A.2 Asking/Doing/Expressing scheme.

- **Asking**: seeking information or advice to be better informed or make a decision
- **Doing**: requesting that the AI perform a task or produce an output
- **Expressing**: neither asking for information/advice nor requesting a task

This analysis only includes participants who used the AI assistant and had a non-missing `ai_intent_code`. Participants who did not type a prompt are not assigned an intent code because there is no message to classify. Intent codes were assigned by a language model; human validation is a planned next step.

```r
intent$ai_used <- tolower(intent$ai_used) == "true"
intent$jaccard_similarity_to_ai <- suppressWarnings(
  as.numeric(intent$jaccard_similarity_to_ai)
)
intent$epistemic_ownership <- suppressWarnings(
  as.numeric(intent$epistemic_ownership)
)
```

```r
intent$task_id <- factor(intent$task_id)

intent$ai_intent_code <- factor(
  intent$ai_intent_code,
  levels = c("Asking", "Doing", "Expressing")
)

emm_options(lmer.df = "satterthwaite")

# Binary ownership and reading depth recodes (mirrors model-setup chunk).
intent$epistemic_ownership_binary <- ifelse(
  intent$epistemic_ownership <= 3, 0L,
  ifelse(intent$epistemic_ownership >= 5, 1L, NA_integer_)
)
intent$read_carefully <- as.integer(
  intent$ai_reading_depth == "read_carefully"
)

intent_model_data <- subset(
  intent,
  ai_used == TRUE &
    !is.na(ai_intent_code) &
    !is.na(task_id)
)

intent_similarity_data <- subset(
  intent_model_data,
  !is.na(jaccard_similarity_to_ai)
)

table(intent_model_data$ai_intent_code, useNA = "ifany")
```

```
##
##     Asking      Doing Expressing
##         40         86          6
```

```r
table(intent_model_data$task_id, intent_model_data$ai_intent_code)
```

```
##
##                Asking Doing Expressing
##   academic_1        1     3          2
##   academic_2        0     7          0
##   academic_3        1     5          0
##   creative_1        0     4          0
##   creative_2        0     8          0
##   creative_3        0     5          0
##   decisions_1       2     4          1
##   decisions_2       3     4          1
##   decisions_3       3     1          1
##   info_1            4     4          0
##   info_2            3     5          0
##   info_3            4     5          0
```

```
##   learning_1       7     5           0
##   learning_2       6     6           0
##   learning_3       5     4           0
##   writing_1        0     5           0
##   writing_2        0     6           1
##   writing_3        1     5           0
```

Analysis 4A: Does AI Intent Predict Similarity To The AI Response?

Outcome: `jaccard_similarity_to_ai`
Model type: linear mixed regression
Predictor: `ai_intent_code`
Random intercept: `task_id`
Sample: AI users with a coded `question_for_ai`

This model asks whether people who used the AI for Asking, Doing, or Expressing differed in how similar their final response was to the pre-written AI response.

```
model_intent_similarity <- lmer(
  jaccard_similarity_to_ai ~ ai_intent_code + (1 | task_id),
  data = intent_similarity_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_intent_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ ai_intent_code + (1 | task_id)
##    Data: intent_similarity_data
##
## REML criterion at convergence: 98
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.05264 -0.41205  0.03366  0.84441  2.50158
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.0000   0.0000
##  Residual             0.1159   0.3404
## Number of obs: 132, groups:  task_id, 18
##
## Fixed effects:
##                         Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)              0.14850    0.05382 129.00000   2.759  0.00664 **
## ai_intent_codeDoing      0.56408    0.06514 129.00000   8.659 1.67e-14 ***
## ai_intent_codeExpressing  0.24035    0.14902 129.00000   1.613  0.10921
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##             (Intr) a_nt_D
## a_ntnt_cdDn -0.826
## a_ntnt_cdEx -0.361  0.298
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
isSingular(model_intent_similarity)
```

```
## [1] TRUE
```

```r
tests_intent_similarity <- anova(model_intent_similarity)

tests_intent_similarity
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## ai_intent_code 8.8067  4.4034     2   129  38.005 1.056e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_intent_similarity <- emmeans(
  model_intent_similarity,
  ~ ai_intent_code
)

emm_intent_similarity
```

```
##  ai_intent_code emmean     SE  df lower.CL upper.CL
##  Asking          0.148 0.0538 129    0.042    0.255
##  Doing           0.713 0.0367 129    0.640    0.785
##  Expressing      0.389 0.1390 129    0.114    0.664
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

```r
pairs_intent_similarity <- pairs(
  emm_intent_similarity,
  adjust = "tukey"
)

pairs_intent_similarity
```

```
##  contrast            estimate     SE  df t.ratio p.value
##  Asking - Doing        -0.564 0.0651 129  -8.659 <0.0001
##  Asking - Expressing   -0.240 0.1490 129  -1.613  0.2439
##  Doing - Expressing     0.324 0.1440 129   2.252  0.0664
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 3 estimates
```

AI intent significantly predicted Jaccard similarity to the AI response, $F(2, 129) = 38.01$, $p < .001$. Doing prompts produced much higher similarity (M = .713) than Asking prompts (M = .148); Tukey-adjusted Asking - Doing = -0.564, $p < .001$. Expressing was descriptively intermediate (M = .389), but did not differ significantly from Asking ($p = .244$) or Doing ($p = .066$).

Analysis 4B: Does AI Intent Predict Epistemic Ownership?

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression
Predictor: `ai_intent_code`
Random intercept: `task_id`
Sample: AI users with a coded `question_for_ai`

This model asks whether people who used the AI for Asking, Doing, or Expressing differed in how much AI influence they reported in their final answer. Epistemic ownership is recoded as binary (see Model 3).

```r
# Exclude Expressing (n <= 3) -- too sparse for stable binary logistic regression.
intent_ownership_data <- subset(
  intent_model_data,
  !is.na(epistemic_ownership_binary) &
    ai_intent_code %in% c("Asking", "Doing")
)
intent_ownership_data$ai_intent_code <- droplevels(intent_ownership_data$ai_intent_code)

model_intent_ownership <- glmer(
  epistemic_ownership_binary ~ ai_intent_code + (1 | task_id),
  data = intent_ownership_data,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_intent_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: epistemic_ownership_binary ~ ai_intent_code + (1 | task_id)
##    Data: intent_ownership_data
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     132.9     141.1     -63.5     126.9       111
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.7386 -0.5195 -0.5195  0.8856  1.9251
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 114, groups:  task_id, 18
##
## Fixed effects:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -0.6061     0.3589  -1.689   0.0912 .
## ai_intent_codeDoing   -0.7038     0.4511  -1.560   0.1187
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## a_ntnt_cdDn -0.796
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
isSingular(model_intent_ownership)
```

```
## [1] TRUE
```

```
tests_intent_ownership <- drop1(model_intent_ownership, test = "Chisq")
```

```
## boundary (singular) fit: see help('isSingular')
```

```
tests_intent_ownership
```

```
## Single term deletions
##
## Model:
## epistemic_ownership_binary ~ ai_intent_code + (1 | task_id)
##                npar    AIC    LRT Pr(Chi)
## <none>              132.91
## ai_intent_code    1 133.30 2.3904  0.1221
```

```
emm_intent_ownership <- emmeans(
  model_intent_ownership,
  ~ ai_intent_code,
  type = "response"
)

emm_intent_ownership
```

```
##  ai_intent_code  prob     SE  df asymp.LCL asymp.UCL
##  Asking         0.353 0.0820 Inf     0.213     0.524
##  Doing          0.212 0.0457 Inf     0.136     0.316
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```
pairs_intent_ownership <- pairs(
  emm_intent_ownership,
  adjust = "tukey"
)

summary(pairs_intent_ownership, type = "response")
```

```
##  contrast       odds.ratio    SE  df null z.ratio p.value
##  Asking / Doing       2.02 0.912 Inf    1   1.560  0.1187
##
## Tests are performed on the log odds ratio scale
```

The Expressing group (n <= 3) is excluded because it is too sparse for stable binary logistic regression. The Asking vs. Doing comparison showed no significant difference in epistemic ownership.

Altogether: Intent coding showed a clear effect for similarity – Doing prompts led to more AI-like final responses than Asking prompts (Jaccard difference = 0.564, $p < .001$). Evidence for differences in binary epistemic ownership was not significant, chi-squared(1) = 2.39, p = .122.

Analysis 5: AI Reading Depth

This analysis asks whether how carefully participants read the AI response predicted textual similarity and epistemic ownership.

Because 57% of AI users reported reading the AI response carefully (`read_carefully`), the ordered 4-level scale collapses to a binary: `read_carefully` (1) vs. anything less (0). This avoids treating ordinal levels with very few observations as evenly spaced.

Outcome: `jaccard_similarity_to_ai`, `epistemic_ownership_binary`
Model type: linear/logistic mixed regression
Predictor: `read_carefully` (binary)
Random intercept: `task_id`
Sample: AI users with a valid `ai_reading_depth`

```
reading_depth_data <- subset(
  model_data,
  ai_used == TRUE & !is.na(ai_reading_depth) & !is.na(task_id)
)

# read_carefully inherited from model_data
table(reading_depth_data$ai_reading_depth, useNA = "ifany")
```

```
##
##       not_read        glanced        skimmed read_carefully
##              9             14             45             90
```

```
table(reading_depth_data$read_carefully, useNA = "ifany")
```

```
##
##  0  1
## 68 90
```

```
model_reading_similarity <- lmer(
  jaccard_similarity_to_ai ~ read_carefully + (1 | task_id),
  data = subset(reading_depth_data, !is.na(jaccard_similarity_to_ai))
)

summary(model_reading_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ read_carefully + (1 | task_id)
##    Data: subset(reading_depth_data, !is.na(jaccard_similarity_to_ai))
##
```

```
## REML criterion at convergence: 170.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8733 -0.8683 -0.1436  0.9368  1.6588
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.02479  0.1574
##  Residual             0.15011  0.3874
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)      0.62804    0.06096  36.87491  10.303  2.1e-12 ***
## read_carefully  -0.19060    0.06476 152.29432  -2.943  0.00376 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## read_crflly -0.606
```

```
fixed_effects_table(model_reading_similarity)
```

```
##             term   estimate  std_error      p_value
## 1    (Intercept)  0.6280359 0.06095777 2.103428e-12
## 2 read_carefully -0.1906014 0.06475940 3.757009e-03
```

```
model_reading_ownership <- glmer(
  epistemic_ownership_binary ~ read_carefully + (1 | task_id),
  data = subset(reading_depth_data, !is.na(epistemic_ownership_binary)),
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_reading_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: epistemic_ownership_binary ~ read_carefully + (1 | task_id)
##    Data: subset(reading_depth_data, !is.na(epistemic_ownership_binary))
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     187.6     196.5     -90.8     181.6       141
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -0.7796 -0.7796 -0.6146  1.2826  1.6270
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.9734     0.2847  -3.419 0.000628 ***
## read_carefully    0.4756     0.3646   1.305 0.192037
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## read_crflly -0.781
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

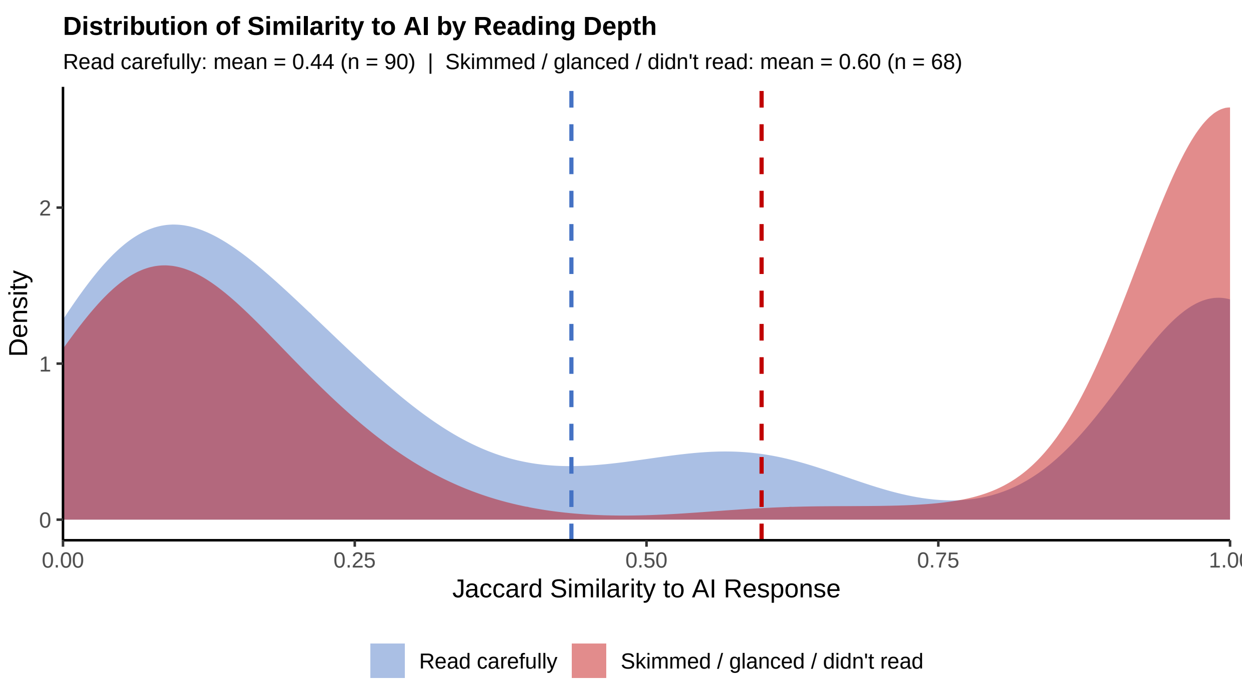
```
fixed_effects_table(model_reading_ownership, exponentiate = TRUE)
```

```
##             term   estimate std_error      p_value odds_ratio
## 1    (Intercept) -0.9734491 0.2846854 0.0006276261  0.3777778
## 2 read_carefully  0.4756107 0.3645705 0.1920365532  1.6089965
```

Reading carefully significantly predicted lower Jaccard similarity, b = -0.191, p = .0038, but did not significantly predict binary epistemic ownership, OR = 1.61, p = .192. Among AI users, 57% reported reading the response carefully, 28% skimmed it, and the remaining 15% glanced or did not read it. The negative similarity result suggests that careful readers may have edited or paraphrased more rather than copying verbatim.

```
# Distribution of Jaccard similarity: read_carefully vs. everyone else
density_data <- subset(
  reading_depth_data,
  !is.na(jaccard_similarity_to_ai)
)
density_data$reading_group <- factor(
  ifelse(density_data$read_carefully == 1, "Read carefully", "Skimmed / glanced / didn't read"),
  levels = c("Read carefully", "Skimmed / glanced / didn't read")
)

group_means <- aggregate(
  jaccard_similarity_to_ai ~ reading_group,
  data = density_data,
  FUN = mean, na.rm = TRUE
)
group_ns <- table(density_data$reading_group)
group_means$n <- as.integer(group_ns[as.character(group_means$reading_group)])
group_means$label <- sprintf(
  "%s: mean = %.2f (n = %d)",
  group_means$reading_group, group_means$jaccard_similarity_to_ai, group_means$n
)
```

```
ggplot(density_data, aes(x = jaccard_similarity_to_ai, fill = reading_group)) +
  geom_density(alpha = 0.45, color = NA, bw = 0.08) +
  geom_vline(
    data = group_means,
    aes(xintercept = jaccard_similarity_to_ai, color = reading_group),
    linetype = "dashed", linewidth = 0.9, show.legend = FALSE
  ) +
  scale_fill_manual(values = c("Read carefully" = "#4472C4",
                               "Skimmed / glanced / didn't read" = "#C00000")) +
  scale_color_manual(values = c("Read carefully" = "#4472C4",
                                "Skimmed / glanced / didn't read" = "#C00000")) +
  scale_x_continuous(limits = c(0, 1), expand = c(0, 0)) +
  labs(
    title    = "Distribution of Similarity to AI by Reading Depth",
    subtitle = paste0(
      group_means$label[group_means$reading_group == "Read carefully"], "  |  ",
      group_means$label[group_means$reading_group == "Skimmed / glanced / didn't read"]
    ),
    x    = "Jaccard Similarity to AI Response",
    y    = "Density",
    fill = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 9.5),
    legend.position = "bottom"
  )
```



Create Local Embedding Cosine Similarity Dataset

This section creates a new dataset in R called `embedding_similarity_data`.

The goal is to measure semantic similarity between each participant's final response and the AI response they were shown. This is different from `jaccard_similarity_to_ai`, which mostly captures exact word overlap. Embedding cosine similarity is better for shared meaning, including synonyms or paraphrased ideas.

Here, we use a free local embedding model `sentence-transformers/all-MiniLM-L6-v2`. Embedding size: 384 dimensions, not 3072 like the OpenAI model.

```r
library(reticulate)

use_virtualenv("r-reticulate-py312", required = TRUE)

py_config()

torch <- import("torch")
sentence_transformers <- import("sentence_transformers")
```

Compute Embedding Cosine Similarity

```r
library(reticulate)
```

```
## Warning: package 'reticulate' was built under R version 4.4.3
```

```r
use_virtualenv("r-reticulate-py312", required = TRUE)
py_config()
```

```
## python:         C:/Users/lengo/Documents/.virtualenvs/r-reticulate-py312/Scripts/python.exe
## libpython:      C:/Users/lengo/AppData/Local/r-reticulate/r-reticulate/pyenv/pyenv-win/versions/3.12
## pythonhome:     C:/Users/lengo/Documents/.virtualenvs/r-reticulate-py312
## version:        3.12.8 (tags/v3.12.8:2dc476b, Dec  3 2024, 19:30:04) [MSC v.1942 64 bit (AMD64)]
## Architecture:   64bit
## numpy:          C:/Users/lengo/Documents/.virtualenvs/r-reticulate-py312/Lib/site-packages/numpy
## numpy_version:  2.4.6
##
## NOTE: Python version was forced by use_python() function
```

```r
sentence_transformers <- tryCatch(
  import("sentence_transformers", delay_load = FALSE),
  error = function(e) {
    python_config <- paste(capture.output(py_config()), collapse = "\n")
    stop(
      "R could not import the Python module 'sentence_transformers'.\n\n",
      "Python configuration being used by reticulate:\n",
      python_config,
      "\n\nOriginal Python import error:\n",
      conditionMessage(e),
      "\n\nTry restarting R, running the one-time setup chunk, ",
      "and then running this chunk again.",
```

```r
      call. = FALSE
    )
  }
)

embedding_model_name <- "sentence-transformers/all-MiniLM-L6-v2"
embedding_model <- sentence_transformers$SentenceTransformer(
  embedding_model_name
)

embedding_similarity_data <- intent

embedding_similarity_data$embedding_cosine_similarity_to_ai <- NA_real_
embedding_similarity_data$embedding_model_name <- embedding_model_name

embedding_rows <- which(
  embedding_similarity_data$ai_used == TRUE &
    !is.na(embedding_similarity_data$participant_response) &
    !is.na(embedding_similarity_data$ai_response_shown) &
    trimws(embedding_similarity_data$participant_response) != "" &
    trimws(embedding_similarity_data$ai_response_shown) != ""
)

participant_texts <- as.character(
  embedding_similarity_data$participant_response[embedding_rows]
)

ai_texts <- as.character(
  embedding_similarity_data$ai_response_shown[embedding_rows]
)

texts_to_embed <- c(participant_texts, ai_texts)

# normalize_embeddings = TRUE makes cosine similarity equal to the dot product.
embeddings <- embedding_model$encode(
  texts_to_embed,
  convert_to_numpy = TRUE,
  normalize_embeddings = TRUE,
  show_progress_bar = TRUE
)

embeddings <- py_to_r(embeddings)

n_pairs <- length(embedding_rows)

participant_embeddings <- embeddings[seq_len(n_pairs), , drop = FALSE]
ai_embeddings <- embeddings[n_pairs + seq_len(n_pairs), , drop = FALSE]

embedding_cosine_similarity <- rowSums(
  participant_embeddings * ai_embeddings
)

embedding_similarity_data$embedding_cosine_similarity_to_ai[
```

```r
  embedding_rows
] <- embedding_cosine_similarity

summary(embedding_similarity_data$embedding_cosine_similarity_to_ai)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
## 0.05209 0.58189 0.86238 0.77955 1.00000 1.00000      70
```

Embedding Similarity Robustness Checks

Because Jaccard similarity and embedding cosine similarity were strongly correlated, the embedding measure is best used as a robustness check and semantic extension. These models ask whether the main findings still hold when similarity means shared meaning rather than exact word overlap.

```r
embedding_analysis_data <- subset(
  embedding_similarity_data,
  ai_used == TRUE &
    !is.na(embedding_cosine_similarity_to_ai) &
    !is.na(task_id)
)

embedding_analysis_data$task_id <- factor(embedding_analysis_data$task_id)

embedding_analysis_data$ai_intent_code <- factor(
  embedding_analysis_data$ai_intent_code,
  levels = c("Asking", "Doing", "Expressing")
)

embedding_analysis_data$ai_kind_choice <- factor(
  embedding_analysis_data$ai_kind_choice
)

embedding_analysis_data$assigned_category <- factor(
  embedding_analysis_data$assigned_category
)

embedding_analysis_data$z_user_decides <- z(
  embedding_analysis_data$exp_ai_does_work_vs_user_decides
)
embedding_analysis_data$z_independent <- z(
  -embedding_analysis_data$exp_independent_vs_reliant
)

embedding_analysis_data$z_hours_per_week <- z(
  embedding_analysis_data$hours_per_week
)

embedding_analysis_data$z_instrument_vs_point_of_view <- z(
  embedding_analysis_data$exp_instrument_vs_point_of_view
)

embedding_analysis_data$z_existing_info_vs_creating_new <- z(
```

```
  embedding_analysis_data$exp_existing_info_vs_creating_new
)

embedding_analysis_data$z_embedding_cosine_similarity_to_ai <- z(
  embedding_analysis_data$embedding_cosine_similarity_to_ai
)
table(embedding_analysis_data$ai_intent_code, useNA = "ifany")
```

```
##
##     Asking      Doing Expressing       <NA>
##         40         86          6         25
```

```
summary(embedding_analysis_data$embedding_cosine_similarity_to_ai)
```

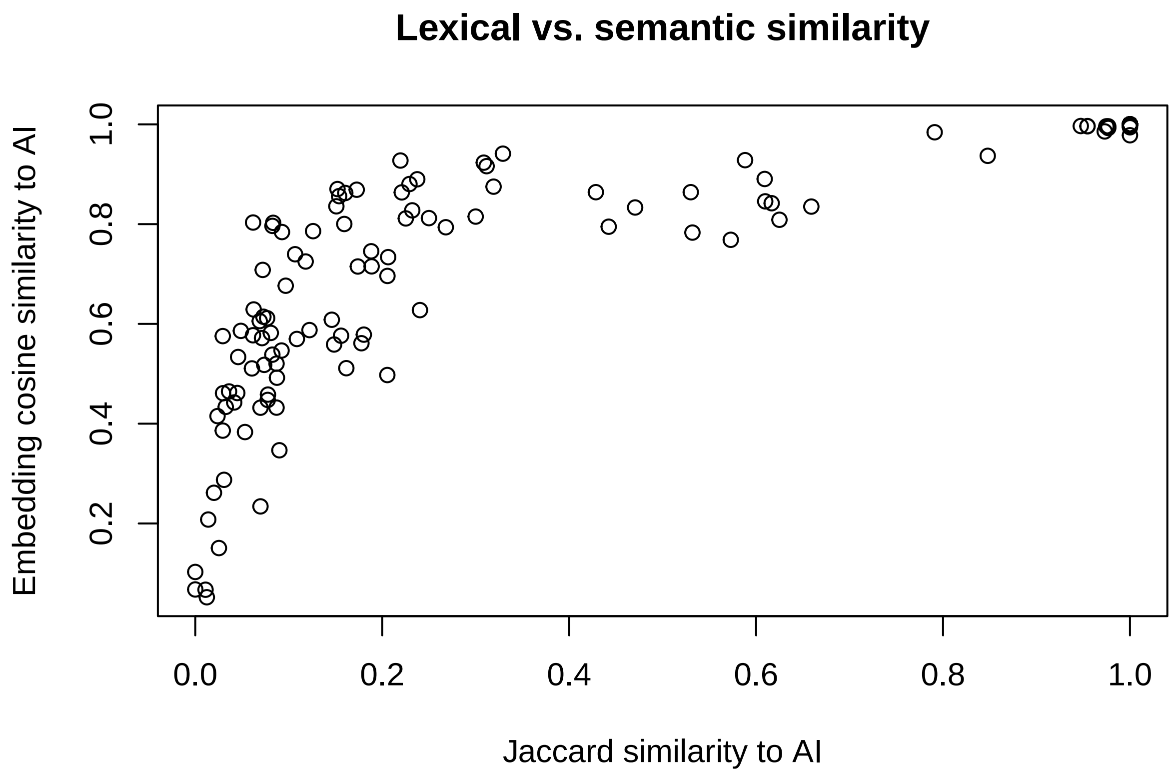
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.05209 0.58189 0.86238 0.77955 1.00000 1.00000
```

Embedding Robustness 1: Jaccard vs. Embedding Similarity

```
similarity_check_data <- subset(
  embedding_analysis_data,
  !is.na(jaccard_similarity_to_ai) &
    !is.na(embedding_cosine_similarity_to_ai)
)

embedding_jaccard_correlation <- cor(
  similarity_check_data$jaccard_similarity_to_ai,
  similarity_check_data$embedding_cosine_similarity_to_ai,
  use = "complete.obs"
)

embedding_jaccard_correlation
```

```
## [1] 0.8283619
```

```
plot(
  similarity_check_data$jaccard_similarity_to_ai,
  similarity_check_data$embedding_cosine_similarity_to_ai,
  xlab = "Jaccard similarity to AI",
  ylab = "Embedding cosine similarity to AI",
  main = "Lexical vs. semantic similarity"
)
```



Jaccard similarity to the AI response was strongly positively associated with embedding cosine similarity to the AI response, Pearson r = .83. This suggests that the lexical overlap measure captured meaningful semantic convergence with the AI response, not only exact-word copying.

Embedding Robustness 2: Does AI Intent Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictor: `ai_intent_code`
Random intercept: `task_id`
Sample: AI users with embedding similarity scores

This is the semantic version of Analysis 4A. It asks whether Doing prompts led to responses that were closer in meaning to the AI response, not just closer in wording.

```
embedding_intent_data <- subset(
  embedding_analysis_data,
  !is.na(ai_intent_code)
)

model_embedding_intent <- lmer(
  embedding_cosine_similarity_to_ai ~ ai_intent_code + (1 | task_id),
  data = embedding_intent_data
)

summary(model_embedding_intent)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: embedding_cosine_similarity_to_ai ~ ai_intent_code + (1 | task_id)
##    Data: embedding_intent_data
##
## REML criterion at convergence: -26.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1151 -0.3742  0.4133  0.5825  1.8866
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.002806 0.05297
##  Residual             0.041897 0.20469
## Number of obs: 132, groups:  task_id, 18
##
## Fixed effects:
##                           Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                0.59546    0.03618  51.53334  16.456  < 2e-16 ***
## ai_intent_codeDoing        0.28883    0.04085 126.46581   7.071 9.26e-11 ***
## ai_intent_codeExpressing   0.02322    0.09232 128.99347   0.251    0.802
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_nt_D
## a_ntnt_cdDn -0.786
## a_ntnt_cdEx -0.352  0.306
```

```r
isSingular(model_embedding_intent)
```

```
## [1] FALSE
```

```r
tests_embedding_intent <- anova(model_embedding_intent)

tests_embedding_intent
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## ai_intent_code 2.2633  1.1317     2 128.05  27.011 1.633e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_embedding_intent <- emmeans(
  model_embedding_intent,
  ~ ai_intent_code
)

emm_embedding_intent
```

```
##  ai_intent_code emmean     SE    df lower.CL upper.CL
```

```
##  Asking          0.595 0.0362  51.5    0.523    0.668
##  Doing           0.884 0.0256  27.8    0.832    0.937
##  Expressing      0.619 0.0865 127.2    0.448    0.790
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

```
pairs_embedding_intent <- pairs(
  emm_embedding_intent,
  adjust = "tukey"
)

pairs_embedding_intent
```

```
##  contrast            estimate     SE  df t.ratio p.value
##  Asking - Doing       -0.2888 0.0408 126  -7.071 <0.0001
##  Asking - Expressing  -0.0232 0.0923 129  -0.251  0.9657
##  Doing - Expressing    0.2656 0.0888 129   2.991  0.0093
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 3 estimates
```

AI intent significantly predicted embedding-based semantic similarity to the AI response, $F(2, 128.05) = 27.01$, $p < .001$. Doing prompts were more semantically similar to the AI response (M = .884) than Asking prompts (M = .595), Tukey-adjusted Asking - Doing = -0.289, $p < .001$. Doing was also higher than Expressing (M = .619), Doing - Expressing = 0.266, $p = .009$, while Asking and Expressing did not differ ($p = .966$).

Embedding Robustness 3: Do Pre-Task Independence Beliefs Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictors: `z_user_decides`, `z_independent`
Random intercept: `task_id`
Sample: AI users with embedding similarity scores

This is the semantic version of the earlier stated-independence similarity model. It asks whether participants who described themselves as more independent on either slider produced responses that were less semantically similar to the AI response.

```
model_embedding_independence <- lmer(
  embedding_cosine_similarity_to_ai ~ z_user_decides +
    z_independent +
    z_hours_per_week +
    z_instrument_vs_point_of_view +
    (1 | task_id),
  data = embedding_analysis_data
)

summary(model_embedding_independence)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
## Formula: embedding_cosine_similarity_to_ai ~ z_user_decides + z_independent +
##     z_hours_per_week + z_instrument_vs_point_of_view + (1 | task_id)
##    Data: embedding_analysis_data
##
## REML criterion at convergence: 31.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0158 -0.7210  0.2456  0.7842  1.6869
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.008454 0.09194
##  Residual             0.055897 0.23643
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##                                Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                    0.778696   0.028912  14.653318  26.933 6.85e-14
## z_user_decides                -0.028370   0.020553 148.722718  -1.380    0.170
## z_independent                 -0.007937   0.020655 149.186153  -0.384    0.701
## z_hours_per_week               0.005288   0.019841 145.197136   0.267    0.790
## z_instrument_vs_point_of_view  0.008407   0.020611 148.265054   0.408    0.684
##
## (Intercept)                   ***
## z_user_decides
## z_independent
## z_hours_per_week
## z_instrument_vs_point_of_view
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sr_d z_ndpn z_hr__
## z_user_dcds -0.005
## z_indepndnt  0.014 -0.228
## z_hrs_pr_wk -0.015 -0.080 -0.078
## z_nstrm____ -0.006 -0.186  0.216 -0.127
```

```r
isSingular(model_embedding_independence)
```

```
## [1] FALSE
```

```r
fixed_effects_table(model_embedding_independence)
```

```
##                            term     estimate  std_error      p_value
## 1                   (Intercept)  0.778695623 0.02891202 6.849850e-14
## 2                z_user_decides -0.028369873 0.02055262 1.695492e-01
## 3                 z_independent -0.007937422 0.02065540 7.013194e-01
## 4              z_hours_per_week  0.005287936 0.01984099 7.902202e-01
## 5 z_instrument_vs_point_of_view  0.008407357 0.02061130 6.839352e-01
```

Neither independence predictor significantly predicted embedding-based semantic similarity to the AI response: `z_user_decides`, b = -0.028, p = .170; `z_independent`, b = -0.008, p = .701. Hours per week (b = 0.005, p = .790) and instrument-vs-point-of-view beliefs (b = 0.008, p = .684) were also not significant. Participants who described themselves as more independent did not reliably produce responses that were less semantically similar to the AI response.

Embedding Robustness 4: Does AI Conceptualization Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictor: `ai_kind_choice`
Random intercept: `task_id`
Sample: AI users with embedding similarity scores, excluding `Writing or print` because that group had only 3 participants overall

This is the semantic version of Analysis 1B. It asks whether people's conceptualization of AI predicted meaning-level convergence with the AI response.

```r
embedding_concept_data <- subset(
  embedding_analysis_data,
  !is.na(ai_kind_choice) & ai_kind_choice != "Writing or print"
)

embedding_concept_data$ai_kind_choice <- droplevels(
  embedding_concept_data$ai_kind_choice
)

contrasts(embedding_concept_data$ai_kind_choice) <- contr.sum(
  nlevels(embedding_concept_data$ai_kind_choice)
)

model_embedding_ai_kind_similarity <- lmer(
  embedding_cosine_similarity_to_ai ~ ai_kind_choice + (1 | task_id),
  data = embedding_concept_data
)

summary(model_embedding_ai_kind_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: embedding_cosine_similarity_to_ai ~ ai_kind_choice + (1 | task_id)
##    Data: embedding_concept_data
##
## REML criterion at convergence: 27.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9971 -0.7779  0.2943  0.7361  1.4732
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.005923 0.07696
##  Residual             0.058927 0.24275
```

```
## Number of obs: 152, groups:  task_id, 18
##
## Fixed effects:
##                    Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)        0.780751   0.032608  30.048925  23.944   <2e-16 ***
## ai_kind_choice1    0.037248   0.047923 143.480556   0.777    0.438
## ai_kind_choice2   -0.001736   0.036323 146.089782  -0.048    0.962
## ai_kind_choice3   -0.017423   0.034843 144.089550  -0.500    0.618
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_kn_1 a_kn_2
## ai_knd_chc1  0.031
## ai_knd_chc2 -0.401 -0.175
## ai_knd_chc3 -0.450 -0.163  0.179
```

```
anova(model_embedding_ai_kind_similarity)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_kind_choice 0.045363 0.015121     3 144.24  0.2566 0.8565
```

```
emm_embedding_ai_kind_similarity <- emmeans(
  model_embedding_ai_kind_similarity,
  ~ ai_kind_choice
)

emm_embedding_ai_kind_similarity
```

```
##  ai_kind_choice                     emmean     SE    df lower.CL upper.CL
##  A calculator or specialized tool  0.818 0.0588 114.2    0.702    0.934
##  A colleague or advisor            0.779 0.0379  46.5    0.703    0.855
##  A library or search engine        0.763 0.0354  37.7    0.692    0.835
##  A market or aggregator            0.763 0.0840 147.1    0.597    0.929
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

```
pairs_embedding_ai_kind_similarity <- pairs(
  emm_embedding_ai_kind_similarity,
  adjust = "tukey"
)

pairs_embedding_ai_kind_similarity
```

```
##  contrast                                                    estimate     SE
##  A calculator or specialized tool - A colleague or advisor     0.038984 0.0650
##  A calculator or specialized tool - A library or search engine 0.054670 0.0637
##  A calculator or specialized tool - A market or aggregator     0.055336 0.0993
##  A colleague or advisor - A library or search engine           0.015686 0.0456
##  A colleague or advisor - A market or aggregator               0.016352 0.0890
```

```
##  A library or search engine - A market or aggregator          0.000666 0.0876
##   df t.ratio p.value
##  145   0.600  0.9320
##  145   0.859  0.8261
##  141   0.557  0.9444
##  147   0.344  0.9860
##  142   0.184  0.9978
##  140   0.008  1.0000
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 4 estimates
```

AI conceptualization did not significantly predict semantic similarity to the AI response, F(3, 144.24) = 0.26, p = .857. Tukey-adjusted pairwise comparisons showed no significant differences between conceptualization groups, all p >= .826, consistent with the Jaccard-based result in Analysis 1B.

Embedding Robustness 5: Do Learning-Curve Variables Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictors: `ai_use_tenure`, `z_hours_per_week`, `ai_understanding`
Random intercept: `task_id`
Sample: AI users with embedding similarity scores

This is the semantic version of Analysis 2B. It asks whether more experienced participants produced responses that were closer in meaning to the AI response.

```r
embedding_learning_data <- subset(
  embedding_analysis_data,
  !is.na(ai_use_tenure) &
    !is.na(z_hours_per_week) &
    !is.na(ai_understanding)
)

model_embedding_learning_similarity <- lmer(
  embedding_cosine_similarity_to_ai ~ ai_use_tenure +
    z_hours_per_week +
    ai_understanding +
    (1 | task_id),
  data = embedding_learning_data
)

summary(model_embedding_learning_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## embedding_cosine_similarity_to_ai ~ ai_use_tenure + z_hours_per_week +
##     ai_understanding + (1 | task_id)
##    Data: embedding_learning_data
##
## REML criterion at convergence: 36
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9325 -0.6558  0.2366  0.7327  1.4598
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.007623 0.08731
##  Residual             0.057196 0.23916
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##
## (Intercept)
## ai_use_tenure6 to 12 months
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure6 to 12 months
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure6 to 12 months
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure6 to 12 months
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure6 to 12 months
```

```
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
##
## (Intercept)
## ai_use_tenure6 to 12 months
## ai_use_tenureI do not use them, or am just starting out.
## ai_use_tenureLess than 6 months
## ai_use_tenureMore than 2 years
## z_hours_per_week
## ai_understandingI just use it. I do not think much about how it works.
## ai_understandingI understand it reasonably well. I know about training, prompting, and limitations.
## ai_understandingI understand it technically. I work with or study AI systems.
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##               (Intr) a__t1m adnutoajso a__t6m a__t2y z_hr__ ajuiIdntmahiw
## a_s_tn6t12m   -0.470
## a_dnutoajso   -0.045  0.092
## a_s_tnrLt6m   -0.317  0.264  0.061
## a_s_tnrMt2y   -0.447  0.382  0.085      0.251
## z_hrs_pr_wk    0.096  0.069  0.054      0.026 -0.186
## ajuiIdntmahiw -0.303  0.084 -0.004      0.007  0.007 -0.174
## auirwIkatpal  -0.588  0.114 -0.090      0.063  0.021 -0.078  0.307
## auitIwwosAs   -0.267 -0.016 -0.025      0.068 -0.043 -0.112  0.169
##               auirwIkatpal
## a_s_tn6t12m
## a_dnutoajso
## a_s_tnrLt6m
## a_s_tnrMt2y
## z_hrs_pr_wk
## ajuiIdntmahiw
## auirwIkatpal
## auitIwwosAs    0.299
```

```r
anova(model_embedding_learning_similarity)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                    Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_use_tenure    0.172450 0.043113     4 141.87  0.7538 0.5571
## z_hours_per_week 0.004093 0.004093     1 141.38  0.0716 0.7895
## ai_understanding 0.059564 0.019855     3 140.40  0.3471 0.7913
```

```r
emm_embedding_learning_tenure <- emmeans(
  model_embedding_learning_similarity,
  ~ ai_use_tenure
)

emm_embedding_learning_tenure
```

```
##  ai_use_tenure                                      emmean     SE    df lower.CL
##  1 to 2 years                                        0.759 0.0441  65.8    0.671
##  6 to 12 months                                      0.825 0.0517  93.2    0.722
##  I do not use them, or am just starting out.         0.802 0.2510 144.9    0.307
##  Less than 6 months                                  0.787 0.0759 137.3    0.637
##  More than 2 years                                   0.835 0.0466  75.8    0.742
##  upper.CL
##     0.847
##     0.927
##     1.297
##     0.938
##     0.928
##
## Results are averaged over the levels of: ai_understanding
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

AI-use tenure, F(4, 141.87) = 0.75, p = .557, hours per week, F(1, 141.38) = 0.07, p = .790, and AI understanding, F(3, 140.40) = 0.35, p = .791, did not significantly predict embedding-based semantic similarity to the AI response. This pattern is consistent with the Jaccard-based result in Analysis 2B.

Embedding Robustness 6: Does Task Category Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear regression
Predictor: `assigned_category`
Sample: AI users with embedding similarity scores

This is the semantic version of Analysis 3B. It uses fixed effects for task category, matching the task-category analysis above.

```r
embedding_task_category_data <- subset(
  embedding_analysis_data,
  assigned_category %in% task_categories &
    !is.na(embedding_cosine_similarity_to_ai)
)

embedding_task_category_data$assigned_category <- factor(
  embedding_task_category_data$assigned_category,
  levels = task_categories
)

continuous_category_summary(
  embedding_task_category_data,
  "embedding_cosine_similarity_to_ai"
)
```

```
##   assigned_category  n      mean        sd    median
## 1          academic 21 0.8014962 0.2390487 0.9981539
## 2          creative 23 0.8479434 0.2936878 0.9999999
## 3         decisions 24 0.6022662 0.2684879 0.5655401
```

```
## 4              info 29 0.7316817 0.2624763 0.7841010
## 5          learning 36 0.8105107 0.1952642 0.8629885
## 6           writing 24 0.8835004 0.1494671 0.9892157
```

```r
model_embedding_task_category_similarity <- lm(
  embedding_cosine_similarity_to_ai ~ assigned_category,
  data = embedding_task_category_data
)

summary(model_embedding_task_category_similarity)
```

```
##
## Call:
## lm(formula = embedding_cosine_similarity_to_ai ~ assigned_category,
##     data = embedding_task_category_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.78024 -0.14966  0.05841  0.15206  0.39773
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 0.801496   0.051700  15.503  < 2e-16 ***
## assigned_categorycreative   0.046447   0.071508   0.650  0.51698
## assigned_categorydecisions -0.199230   0.070793  -2.814  0.00554 **
## assigned_categoryinfo      -0.069814   0.067885  -1.028  0.30540
## assigned_categorylearning   0.009015   0.065054   0.139  0.88998
## assigned_categorywriting    0.082004   0.070793   1.158  0.24854
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2369 on 151 degrees of freedom
## Multiple R-squared:  0.1269, Adjusted R-squared:  0.09803
## F-statistic: 4.391 on 5 and 151 DF,  p-value: 0.0009236
```

```r
anova(model_embedding_task_category_similarity)
```

```
## Analysis of Variance Table
##
## Response: embedding_cosine_similarity_to_ai
##                    Df Sum Sq  Mean Sq F value    Pr(>F)
## assigned_category   5 1.2323 0.246461  4.3908 0.0009236 ***
## Residuals         151 8.4758 0.056131
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emm_embedding_task_category_similarity <- emmeans(
  model_embedding_task_category_similarity,
  ~ assigned_category
)

emm_embedding_task_category_similarity
```

```
## assigned_category emmean     SE  df lower.CL upper.CL
## academic           0.801 0.0517 151    0.699    0.904
## creative           0.848 0.0494 151    0.750    0.946
## decisions          0.602 0.0484 151    0.507    0.698
## info               0.732 0.0440 151    0.645    0.819
## learning           0.811 0.0395 151    0.732    0.889
## writing            0.884 0.0484 151    0.788    0.979
##
## Confidence level used: 0.95
```

```r
pairs_embedding_task_category_similarity <- pairs(
  emm_embedding_task_category_similarity,
  adjust = "tukey"
)

pairs_embedding_task_category_similarity
```

```
## contrast               estimate     SE  df t.ratio p.value
## academic - creative    -0.04645 0.0715 151  -0.650  0.9869
## academic - decisions    0.19923 0.0708 151   2.814  0.0606
## academic - info         0.06981 0.0679 151   1.028  0.9077
## academic - learning    -0.00901 0.0651 151  -0.139  1.0000
## academic - writing     -0.08200 0.0708 151  -1.158  0.8557
## creative - decisions    0.24568 0.0691 151   3.554  0.0066
## creative - info         0.11626 0.0662 151   1.758  0.4963
## creative - learning     0.03743 0.0632 151   0.592  0.9914
## creative - writing     -0.03556 0.0691 151  -0.514  0.9956
## decisions - info       -0.12942 0.0654 151  -1.979  0.3590
## decisions - learning   -0.20824 0.0624 151  -3.335  0.0134
## decisions - writing    -0.28123 0.0684 151  -4.112  0.0009
## info - learning        -0.07883 0.0591 151  -1.333  0.7660
## info - writing         -0.15182 0.0654 151  -2.322  0.1917
## learning - writing     -0.07299 0.0624 151  -1.169  0.8508
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

Task category significantly predicted embedding-based semantic similarity, $F(5, 151) = 4.39$, $p < .001$. Writing tasks had the highest estimated semantic similarity ($M = .884$) and decisions tasks had the lowest ($M = .602$). Tukey-adjusted pairwise comparisons showed that decisions were significantly lower than creative tasks (difference = -0.246, $p = .0066$), learning tasks (difference = -0.208, $p = .013$), and writing tasks (difference = -0.281, $p < .001$).

Embedding Robustness 7: Does Reading Depth Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictor: `read_carefully` (binary)
Random intercept: `task_id`
Sample: AI users with embedding similarity scores and a valid `ai_reading_depth`

This is the semantic version of the reading-depth similarity model in Analysis 5.

```r
embedding_reading_depth_data <- subset(
  embedding_analysis_data,
  !is.na(ai_reading_depth)
)

embedding_reading_depth_data$read_carefully <- as.integer(
  embedding_reading_depth_data$ai_reading_depth == "read_carefully"
)

model_embedding_reading_similarity <- lmer(
  embedding_cosine_similarity_to_ai ~ read_carefully + (1 | task_id),
  data = embedding_reading_depth_data
)

summary(model_embedding_reading_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: embedding_cosine_similarity_to_ai ~ read_carefully + (1 | task_id)
##    Data: embedding_reading_depth_data
##
## REML criterion at convergence: 10
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3647 -0.6968  0.3560  0.7494  1.5124
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.008758 0.09359
##  Residual             0.053663 0.23165
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)      0.82938    0.03637  34.88666  22.807   <2e-16 ***
## read_carefully  -0.08902    0.03881 151.06051  -2.294   0.0232 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## read_crflly -0.605
```

```r
fixed_effects_table(model_embedding_reading_similarity)
```

```
##             term    estimate  std_error      p_value
## 1    (Intercept)  0.82938155 0.03636554 1.512924e-22
## 2 read_carefully -0.08902257 0.03880571 2.316537e-02
```

Reading carefully significantly predicted lower embedding-based semantic similarity, b = -0.089, p = .023. This is consistent with Analysis 5: careful readers showed lower lexical and semantic similarity, suggesting more selective editing or paraphrasing rather than direct copying.

Cases Where Lexical And Semantic Similarity Diverge

This table helps identify cases where participants did not share many exact words with the AI response but still had high semantic similarity.

```r
embedding_analysis_data$jaccard_z <- z(
  embedding_analysis_data$jaccard_similarity_to_ai
)

embedding_analysis_data$embedding_similarity_z <- z(
  embedding_analysis_data$embedding_cosine_similarity_to_ai
)

embedding_analysis_data$semantic_minus_lexical <-
  embedding_analysis_data$embedding_similarity_z -
  embedding_analysis_data$jaccard_z

embedding_divergence_cases <- embedding_analysis_data[
  order(-embedding_analysis_data$semantic_minus_lexical),
  c(
    "submission_id",
    "assigned_category",
    "ai_intent_code",
    "jaccard_similarity_to_ai",
    "embedding_cosine_similarity_to_ai",
    "semantic_minus_lexical",
    "participant_response",
    "ai_response_shown"
  )
]

embedding_divergence_cases
```

```
##                                submission_id assigned_category ai_intent_code
## 200 ai-d07f7fe1-634e-4b7a-8715-38695fe51985          learning         Asking
## 48  ai-fb25bda4-b9d2-4548-a930-4367a7620940          learning          Doing
## 183 ai-0194fcbf-b43b-48ee-909a-0d5b523e377e          learning         Asking
## 168 ai-00113ca9-1216-4a5f-a4c7-9107e42f0948          learning           <NA>
## 187 ai-4684f7ed-a24f-4e26-8d48-b636360f7b63              info         Asking
## 13  ai-802fec76-fc09-429b-b4f5-74f6643a19ec          learning          Doing
## 203 ai-8bbfa3d6-4e68-4150-afd0-91febbcd78f0          learning         Asking
## 68  ai-e3007e63-495b-4e2a-9c90-7b613d0bba23          learning         Asking
## 55  ai-f32f8340-4cd5-44ff-a028-1cc1a787d7b7          learning         Asking
## 75  ai-7de8404f-ff3d-4cac-ab7f-121868326d22          learning          Doing
## 95  ai-658b6665-1e89-43ac-8293-8912d94f181c          learning         Asking
## 185 ai-375b84d5-de93-44e5-85f6-ee8b0fbace0c              info          Doing
## 34  ai-25a61e97-2444-4095-b07a-7f763e4b440b          learning          Doing
## 197 ai-d7cc626b-1951-4ec8-b50c-3412cb5f981a          learning          Doing
## 11  ai-c8b66e6b-7ac7-4889-995b-1fa326f42f26          learning         Asking
## 105 ai-c82af804-e29b-4c2e-bc99-678ad2a59c94              info         Asking
## 35  ai-cfc1376a-921a-4e11-a818-35a4038be455          academic          Doing
## 212 ai-8676c9ec-ae2e-4fe3-8384-513dc06313b6          learning         Asking
## 14  ai-480b82e9-6a2b-4b09-bd0b-58e20601ca0c          creative           <NA>
## 88  ai-41df940e-deea-43c9-8a35-3f8a65228a5e          learning         Asking
```

```
## 63  ai-57f22ce2-b49d-48db-8c44-0e619737f951          learning          <NA>
## 9   ai-f41b68bb-6e9e-46e5-8f45-9aa7bae58ab9          academic         Doing
## 166 ai-043a2017-3c73-41de-85eb-2824ad332b38              info         Doing
## 132 ai-7fc47df5-97e6-4e5f-b31a-44aea4ce424c          learning         Doing
## 146 ai-a4d0a785-bea7-4b5b-8f90-9cc3a5216012          learning        Asking
## 4   ai-2c713b0c-b1a6-46a7-92ca-d737c06830bf         decisions         Doing
## 80  ai-3de92f98-3c76-4079-9e0c-74bdb130a8fc          learning          <NA>
## 19  ai-a19188e1-a1de-4a59-acb3-52e0153f56eb              info        Asking
## 40  ai-7f22fa45-3ff8-4e9f-9e48-ed938199c411          creative          <NA>
## 70  ai-74efcb75-a131-45b9-9ddd-2be2a3e26c70              info        Asking
## 69  ai-506e63d2-9d4d-4049-a9c0-621d5d0306b4           writing          <NA>
## 62  ai-fc01ab3a-1580-4992-83c5-c1161cd2f3c1          learning         Doing
## 72  ai-fa6af708-08a0-43bc-b150-b0c473526855           writing          <NA>
## 142 ai-036b9943-41ef-4153-837e-97f59cef9f2f           writing         Doing
## 90  ai-d3fffa42-0196-4a02-ac79-145cf281b5d5              info         Doing
## 79  ai-fad2aa5b-4252-4a19-808e-597378908dd7         decisions          <NA>
## 184 ai-a2410e08-e9d0-4261-9265-ccb057246e6b          academic         Doing
## 123 ai-98a9f8b0-a38c-437c-a1d3-0ef8b4556755          learning        Asking
## 33  ai-c47c4c3e-71e5-4427-97ec-d799ea4ade11          academic         Doing
## 83             ai-1780384949407-83z90pyml8g         decisions        Asking
## 53  ai-e72053f5-bc7a-49be-b10f-26816f6e0ead          academic         Doing
## 121 ai-84d1e4a7-b9f2-4141-ade7-0dc4eb2dcd63           writing         Doing
## 165 ai-f214ca17-23e9-43b6-86bd-7d30fc3c8f3b              info         Doing
## 213 ai-79050dc5-cdf7-4c59-b6cf-eb67420ab638         decisions        Asking
## 51  ai-2dea9244-b173-4d30-a2ee-43fb7f38e4fc          learning        Asking
## 57             ai-1780382223533-1e3jjyghs5d          learning         Doing
## 52  ai-db819a6a-0bfa-48fd-9041-dc47bf3fbcb7              info         Doing
## 170 ai-7a02838d-03c5-467f-9c28-31a9ec31bd58          learning        Asking
## 32  ai-53d14de3-84d0-4a54-9094-537b083a4ae2              info         Doing
## 135 ai-4fc82e3c-a8f9-40df-9f64-12289cf675bd          learning        Asking
## 41             ai-1780382055100-pq94eggrdj9         decisions          <NA>
## 179 ai-80b80b6b-e15a-4c1b-8d01-f64def255968          learning        Asking
## 61  ai-7327a28a-cfdd-4295-be40-d82bde104f0a         decisions        Asking
## 25  ai-380c80db-8894-40a5-b665-de02e34a8dca           writing          <NA>
## 156 ai-2bc96f0e-2c97-4d8e-bd85-358ecf9951d7         decisions    Expressing
## 8   ai-63ff3c65-a7d6-480a-b00e-5dac58c41dc8              info          <NA>
## 18  ai-d2bf71f2-9eff-46a0-a8a8-53e578fb760d           writing          <NA>
## 161 ai-5ceb0401-c7f6-454e-9738-eb5dbabf72da          learning        Asking
## 155 ai-3639d520-8bb0-4c0b-a552-b68b3d5a4bfd         decisions         Doing
## 130 ai-3d2ab6d2-5f92-4eca-89f1-81176f1dd244              info        Asking
## 49  ai-2434a9cb-1da0-49cd-aabe-c3180a6aaafd          academic        Asking
## 133 ai-644b27be-a736-455d-888b-fa4973460df7           writing        Asking
## 6   ai-4719caa3-7a55-4c33-bd88-c3e639489708              info        Asking
## 225 ai-6e3d867c-f4df-4470-8205-bd2d154b069f         decisions        Asking
## 150 ai-5788b73b-bc5c-4bc9-a01b-eee88aa034f3           writing         Doing
## 58  ai-15cf7924-5bd5-42c1-8e33-c1321d94f46c         decisions        Asking
## 125 ai-541e2136-5a21-4ae8-b7de-d35fbc31c86d           writing         Doing
## 16  ai-05b7871a-ef13-48b4-a0de-388551d4749d              info          <NA>
## 169 ai-f6f48ce2-5f6e-4491-847e-9f74a7c591d2         decisions         Doing
## 137 ai-04f36787-468d-4c09-9452-4d63906e467b              info        Asking
## 117 ai-894cf2d9-1671-4771-9dc1-77f10ef0478f           writing         Doing
## 93              ai-1780386396588-tqlpk93nsc           writing          <NA>
## 159 ai-31c8400c-a2d3-4fdd-a6af-c29726676721           writing         Doing
## 60  ai-a04ac4f3-8a89-4928-9f55-dc334773e45f          creative         Doing
```

```
## 100 ai-5a46312c-0103-4c62-b370-2d2b718063d7      writing       Doing
## 17  ai-50e57cb2-6cfb-4bc0-92a6-48169a4453cd      writing       Doing
## 78  ai-64101525-589f-4b2a-b2d0-716a0958b87f     academic        <NA>
## 84  ai-b3e0b5a5-3721-47e5-86af-d92e787d8eb2     creative       Doing
## 59  ai-b559fa87-7123-43e3-9a85-c63cbb8471ec      writing       Doing
## 77             ai-1780383992866-v3cn2v08wgi     academic       Doing
## 54  ai-1c27e5ae-866e-4349-9db4-7ec42752d43d     academic  Expressing
## 199 ai-df1fb0c8-3356-4f21-a0ac-643fcec9f0ac      writing  Expressing
## 45  ai-c0a9d318-d72f-43e5-8bf2-71a9b2fc7647         info        <NA>
## 139 ai-1668fdcd-a675-488e-bf06-77b879fee727         info      Asking
## 50  ai-f0740d95-0b49-49f3-92a6-31168e14e10a      writing       Doing
## 24  ai-cce5b570-8aee-4ffb-bbc6-7a4d9337ccd7         info       Doing
## 145 ai-e2feda33-be10-408b-9202-59190975df89     learning       Doing
## 147 ai-667e3853-b8b2-4cf6-ac50-fea11083b967     learning       Doing
## 23  ai-85186445-5b06-41b0-a628-22c6e0927db4     academic       Doing
## 108 ai-4a158d13-7a0b-4d89-b9c7-22122eca67e0     learning       Doing
## 226 ai-2e2f1669-1c7a-40ef-a382-6a184af445f7     learning       Doing
## 92  ai-5865e427-b54f-46eb-acd1-0a606f5b4e31         info      Asking
## 107 ai-f487f5b5-1e43-4c50-a1a4-43b83313aca4         info       Doing
## 224 ai-7ecfef37-065b-42a7-8136-f8f5501add5d         info       Doing
## 46  ai-4676ece5-b290-422b-916c-18427b38f067    decisions       Doing
## 149 ai-9898e418-d2bf-4042-830f-f4e4b1e747f2    decisions       Doing
## 37  ai-76be743e-4c44-4477-826d-b3f1784b95f6     academic       Doing
## 91  ai-25a62fbe-faa4-4d2c-a117-8db9b579db77     academic       Doing
## 208 ai-ef69d085-7c3b-41e2-991f-9c117a6f5883     academic        <NA>
## 36  ai-ef7f768d-4b70-457b-a2f4-02537384463a         info       Doing
## 148 ai-57fc84ce-fea3-42a3-b26e-27b262f862fd         info       Doing
## 201 ai-702a40b4-475a-4a70-9f40-ff834a8aa076         info       Doing
## 1   ai-05e682ab-248a-4f40-9b3a-b74f583a72c8     learning       Doing
## 5   ai-b412fe12-44c2-4976-9a4e-aed83f18f7cf     learning       Doing
## 220 ai-9987ba30-607e-4a41-836d-9c56b0141daa     learning       Doing
## 98  ai-cbeda50f-64f7-4185-962b-9d155e1be1be     academic       Doing
## 167 ai-94345d83-3eb9-4ed4-9745-351c45165fa8     academic  Expressing
## 172 ai-3bc4dc10-e3b0-4274-bf09-22cf680b8a41     academic       Doing
## 2   ai-ab784340-49a7-4104-92ca-d547be96dd4f     creative       Doing
## 30  ai-f7e06690-c0e7-43ff-b639-9ec468aa5799     creative       Doing
## 103 ai-faccebf0-68d5-44cf-a96c-bc07c48dee7b     creative       Doing
## 113 ai-257878db-84a4-4695-979c-de2e1b7f6dbe     creative       Doing
## 128 ai-2deb3418-5c3c-45f0-986a-e7948e00c5dc     creative       Doing
## 163 ai-ddaca5e9-35e3-4267-aa27-44326fa6b702     creative       Doing
## 109 ai-391cad87-9a13-4343-b940-e5d560df149c     creative       Doing
## 190 ai-4b327991-4786-435c-9ed2-fdc881b355e5     creative       Doing
## 195 ai-311527bb-108a-438d-b60e-d05ad53db747     creative       Doing
## 222 ai-1e2c1956-950f-4567-98be-717561906681     creative       Doing
## 74  ai-4790c4bd-f7a1-4555-a54e-37ff1653afa1      writing       Doing
## 86  ai-bb078926-7e23-4be3-942c-f53bb4c3e699      writing       Doing
## 140 ai-d72a192b-c0e6-4916-aa2b-f1d33cb95d4f      writing        <NA>
## 198 ai-acd9a310-9901-44f0-866f-68d00e582abf      writing       Doing
## 206 ai-45d87509-95df-4ba3-8684-bf4c47114357      writing       Doing
## 110 ai-75a1f5ad-5a10-4338-b298-1c27400165f2    decisions       Doing
## 28  ai-c40c3831-5a60-4414-9aaa-ce98e0436784         info        <NA>
## 116 ai-9c92e553-a702-47dd-850f-7c30b013b9cf         info       Doing
## 27  ai-7fb8b321-79aa-4e6a-907d-11427d683586      writing       Doing
## 144 ai-38ee6a8e-b11b-4d64-8b0c-ffae3e9d810b    decisions       Doing
```

```
## 66  ai-dc688d91-f689-4178-aee9-dfd03ae8d0b3          academic          Doing
## 87  ai-cb4d9ae9-eebf-415f-aaff-5488c1680abc          academic          Doing
## 209 ai-55542a1e-5319-410a-bfe1-21e9adbfdab4          academic          Doing
## 81  ai-e412922d-abea-41e5-891f-8db0a799fc1d          creative           <NA>
## 82            ai-1780384840980-qjcy7t5l4lk          creative          Doing
## 174 ai-513a1235-0fa4-4d1c-9b33-98640ebb5206          creative          Doing
## 211 ai-31ec013c-4843-41d9-8de2-eae2bcbd19ea          creative           <NA>
## 215 ai-d5022e49-dc2d-4f15-b18b-c348087fc1e7          creative           <NA>
## 106 ai-df9e2071-c4b3-4686-ba0a-6ec41505e2f3          academic          Doing
## 44  ai-3951d5d6-f3d8-4baf-a22b-e511fe50633d         decisions          Doing
## 21  ai-ccb3f336-7ef2-4a73-8c1c-67cad746792f           writing          Doing
## 71  ai-99d0e015-dce5-4c20-93b0-f6b8fa6201af              info          Doing
## 221 ai-950ef311-6489-4b30-93f7-f4ebaf0e9bfe         decisions           <NA>
## 219 ai-f1d710a4-c334-4aa5-a196-f629b8740b8a         decisions         Asking
## 194 ai-580297a3-a980-491c-98fd-3ffce776fefa          creative          Doing
## 120 ai-17ff6899-7836-41d6-8eae-34a95814dffc         decisions     Expressing
## 56  ai-162f6f01-48ee-4777-a486-ae23dcc85f88         decisions          Doing
## 173 ai-88a80b3e-f515-4568-942a-5e5b8877b0d7          learning         Asking
## 15  ai-d654fb95-6b07-4461-9e67-b9559ee8cd55         decisions         Asking
## 96  ai-b768b11f-81a2-4bc3-9226-527d4db25c37         decisions           <NA>
## 216 ai-e55ed7e9-0128-4486-bbbe-76bb1c519a36          creative          Doing
## 43  ai-e06bfee9-6ad4-4045-8b89-1be0929e2d6a          academic         Asking
## 162 ai-28cf02f6-7f90-47ad-9afe-685556e6baf5          creative          Doing
## 136 ai-47a74470-e9db-45fc-864a-62c8cac63c36              info         Asking
## 127 ai-26bcd679-30ca-4067-814d-0da75d2f6434         decisions     Expressing
## 112 ai-74a5dc89-0f9d-437c-844d-7710520ab12e          learning         Asking
## 65  ai-ca973dc9-9ed3-4e89-ab3e-d29a66f54760          creative           <NA>
## 204 ai-82458211-0cb5-4c13-a29a-cf67131c4304              info         Asking
## 126 ai-002afe42-6bae-47f0-a433-654c09bf2d9b         decisions         Asking
##     jaccard_similarity_to_ai embedding_cosine_similarity_to_ai
## 200               0.21951219                        0.92747154
## 48                0.15217391                        0.87027097
## 183               0.16049383                        0.86238122
## 168               0.17272727                        0.86892432
## 187               0.06185567                        0.80321242
## 13                0.15384615                        0.85613012
## 203               0.08333333                        0.80262313
## 68                0.23750000                        0.88992872
## 55                0.08256881                        0.79644042
## 75                0.32911392                        0.94123846
## 95                0.15094340                        0.83572366
## 185               0.22916667                        0.88032279
## 34                0.30864198                        0.92320698
## 197               0.22093023                        0.86359574
## 11                0.31168831                        0.91618313
## 105               0.09278351                        0.78410096
## 35                0.12605042                        0.78592362
## 212               0.15942029                        0.80028588
## 14                0.23214286                        0.82759778
## 88                0.31914894                        0.87511480
## 63                0.22522523                        0.81141141
## 9                 0.10679612                        0.73965127
## 166               0.07216495                        0.70820976
## 132               0.25000000                        0.81205038
```

```
## 146                 0.11811024                        0.72505046
## 4                   0.30000000                        0.81514832
## 80                  0.26804124                        0.79375868
## 19                  0.18823529                        0.74557059
## 40                  0.09677419                        0.67650601
## 70                  0.17391304                        0.71505866
## 69                  0.20634921                        0.73383604
## 62                  0.42857143                        0.86404002
## 72                  0.18867924                        0.71530161
## 142                 0.06250000                        0.62896121
## 90                  0.58823529                        0.92822698
## 79                  0.20560748                        0.69615750
## 184                 0.07291667                        0.61436780
## 123                 0.07692308                        0.61142447
## 33                  0.06896552                        0.60565974
## 83                  0.02941176                        0.57556314
## 53                  0.04878049                        0.58604164
## 121                 0.47058824                        0.83333394
## 165                 0.53012048                        0.86401546
## 213                 0.06172839                        0.57719992
## 51                  0.08080808                        0.58188596
## 57                  0.44230769                        0.79480718
## 52                  0.60919540                        0.89048543
## 170                 0.07142857                        0.57155796
## 32                  0.14606742                        0.60833183
## 135                 0.79104478                        0.98414635
## 41                  0.12222222                        0.58762524
## 179                 0.04587156                        0.53361043
## 61                  0.10869565                        0.56979391
## 25                  0.09230769                        0.54662596
## 156                 0.08247423                        0.53835117
## 8                   0.24038461                        0.62767349
## 18                  0.60975610                        0.84562434
## 161                 0.15596330                        0.57629191
## 155                 0.61666667                        0.84191849
## 130                 0.06060606                        0.51074529
## 49                  0.07377049                        0.51786934
## 133                 0.18032787                        0.57847486
## 6                   0.14851485                        0.55871269
## 225                 0.08695652                        0.52030264
## 150                 0.53191489                        0.78305764
## 58                  0.17777778                        0.56128624
## 125                 0.65909091                        0.83516472
## 16                  0.02970297                        0.46104758
## 169                 0.03614458                        0.46458904
## 137                 0.08737864                        0.49214858
## 117                 0.62500000                        0.80871253
## 93                  0.94736842                        0.99663703
## 159                 0.84782609                        0.93687419
## 60                  0.04494382                        0.46146694
## 100                 0.95454545                        0.99651358
## 17                  0.95454545                        0.99624594
## 78                  0.57291667                        0.76856086
## 84                  0.97468354                        0.99598547
```

```
## 59                    0.97674419                        0.99608053
## 77                    0.04166667                        0.44250570
## 54                    0.16161616                        0.51126033
## 199                   0.97674419                        0.99265851
## 45                    0.03260870                        0.43369648
## 139                   0.07777778                        0.45829505
## 50                    0.97297297                        0.98577294
## 24                    1.00000000                        1.00000014
## 145                   1.00000000                        1.00000014
## 147                   1.00000000                        1.00000014
## 23                    1.00000000                        1.00000013
## 108                   1.00000000                        1.00000011
## 226                   1.00000000                        1.00000010
## 92                    1.00000000                        1.00000009
## 107                   1.00000000                        1.00000009
## 224                   1.00000000                        1.00000009
## 46                    1.00000000                        1.00000009
## 149                   1.00000000                        1.00000009
## 37                    1.00000000                        1.00000009
## 91                    1.00000000                        1.00000009
## 208                   1.00000000                        1.00000009
## 36                    1.00000000                        1.00000008
## 148                   1.00000000                        1.00000008
## 201                   1.00000000                        1.00000008
## 1                     1.00000000                        1.00000006
## 5                     1.00000000                        1.00000006
## 220                   1.00000000                        1.00000006
## 98                    1.00000000                        1.00000005
## 167                   1.00000000                        1.00000005
## 172                   1.00000000                        1.00000005
## 2                     1.00000000                        1.00000005
## 30                    1.00000000                        1.00000005
## 103                   1.00000000                        1.00000005
## 113                   1.00000000                        1.00000005
## 128                   1.00000000                        1.00000005
## 163                   1.00000000                        1.00000005
## 109                   1.00000000                        1.00000004
## 190                   1.00000000                        1.00000004
## 195                   1.00000000                        1.00000004
## 222                   1.00000000                        1.00000004
## 74                    1.00000000                        1.00000004
## 86                    1.00000000                        1.00000004
## 140                   1.00000000                        1.00000004
## 198                   1.00000000                        1.00000004
## 206                   1.00000000                        1.00000004
## 110                   1.00000000                        1.00000003
## 28                    1.00000000                        1.00000003
## 116                   1.00000000                        1.00000003
## 27                    1.00000000                        0.99999999
## 144                   1.00000000                        0.99999999
## 66                    1.00000000                        0.99999993
## 87                    1.00000000                        0.99999993
## 209                   1.00000000                        0.99999993
## 81                    1.00000000                        0.99999990
```

```
## 82              1.00000000                                0.99999990
## 174             1.00000000                                0.99999990
## 211             1.00000000                                0.99999990
## 215             1.00000000                                0.99999990
## 106             1.00000000                                0.99815390
## 44              1.00000000                                0.99692608
## 21              1.00000000                                0.99413293
## 71              0.07751938                                0.44758251
## 221             0.02380952                                0.41516187
## 219             0.06976744                                0.43202401
## 194             1.00000000                                0.97817635
## 120             0.08695652                                0.43216898
## 56              0.20547945                                0.49751966
## 173             0.02941176                                0.38607977
## 15              0.05319149                                0.38319880
## 96              0.09000000                                0.34668298
## 216             0.03076923                                0.28742891
## 43              0.02000000                                0.26142506
## 162             0.01388889                                0.20783077
## 136             0.06976744                                0.23416447
## 127             0.02531646                                0.15068111
## 112             0.00000000                                0.10275160
## 65              0.00000000                                0.06770594
## 204             0.01098901                                0.06716727
## 126             0.01234568                                0.05208959
##     semantic_minus_lexical
## 200           1.279242418
## 48            1.209591204
## 183           1.158239353
## 168           1.155465303
## 187           1.154903619
## 13            1.148941226
## 203           1.101622355
## 68            1.086101775
## 55            1.078650669
## 75            1.074586429
## 95            1.074021056
## 185           1.067351718
## 34            1.050839517
## 197           1.019825972
## 11            1.015461261
## 105           1.004969452
## 35            0.933407009
## 212           0.911867104
## 14            0.848940670
## 88            0.833145905
## 63            0.800455696
## 9             0.793566089
## 166           0.749631819
## 132           0.744281362
## 146           0.708213419
## 4             0.638160409
## 80            0.628184710
## 19            0.624219396
```

```
## 40              0.564199844
## 70              0.535863339
## 69              0.534235639
## 62              0.529333587
## 72              0.501829673
## 142             0.454867101
## 90              0.408106267
## 79              0.384954760
## 184             0.371671791
## 123             0.350374689
## 33              0.346131704
## 83              0.319259100
## 53              0.315344421
## 121             0.306631284
## 165             0.288483825
## 213             0.249204462
## 51              0.222755161
## 57              0.219238855
## 52              0.207122283
## 170             0.203590867
## 32              0.174050835
## 135             0.151448050
## 41              0.147577655
## 179             0.112063434
## 61              0.108167069
## 25              0.054147680
## 156             0.044290139
## 8               0.027978311
## 18              0.025961431
## 161             0.022153541
## 155            -0.005277479
## 130            -0.014526868
## 49             -0.017179257
## 133            -0.026859021
## 6              -0.030656370
## 225            -0.038686326
## 150            -0.040300434
## 58             -0.089716147
## 125            -0.132929741
## 16             -0.140481986
## 169            -0.141557277
## 137            -0.152546336
## 117            -0.158144511
## 93             -0.169091670
## 159            -0.172665088
## 60             -0.174933779
## 100            -0.186601768
## 17             -0.187674632
## 78             -0.195619194
## 84             -0.236461926
## 59             -0.240966194
## 77             -0.243172971
## 54             -0.251935824
## 199            -0.254683819
```

```
## 45              -0.257011342
## 139             -0.265491088
## 50              -0.273344791
## 24              -0.280388538
## 145             -0.280388564
## 147             -0.280388564
## 23              -0.280388580
## 108             -0.280388683
## 226             -0.280388692
## 92              -0.280388746
## 107             -0.280388746
## 224             -0.280388746
## 46              -0.280388747
## 149             -0.280388747
## 37              -0.280388766
## 91              -0.280388766
## 208             -0.280388766
## 36              -0.280388773
## 148             -0.280388773
## 201             -0.280388773
## 1               -0.280388858
## 5               -0.280388858
## 220             -0.280388858
## 98              -0.280388898
## 167             -0.280388898
## 172             -0.280388898
## 2               -0.280388900
## 30              -0.280388900
## 103             -0.280388900
## 113             -0.280388900
## 128             -0.280388900
## 163             -0.280388900
## 109             -0.280388951
## 190             -0.280388951
## 195             -0.280388951
## 222             -0.280388951
## 74              -0.280388952
## 86              -0.280388952
## 140             -0.280388952
## 198             -0.280388952
## 206             -0.280388952
## 110             -0.280388998
## 28              -0.280389011
## 116             -0.280389011
## 27              -0.280389145
## 144             -0.280389149
## 66              -0.280389396
## 87              -0.280389396
## 209             -0.280389396
## 81              -0.280389530
## 82              -0.280389530
## 174             -0.280389530
## 211             -0.280389530
## 215             -0.280389530
```

```
## 106          -0.287789450
## 44           -0.292711319
## 21           -0.303908034
## 71           -0.307821072
## 221          -0.310448792
## 219          -0.351811190
## 194          -0.367872077
## 120          -0.391981733
## 56           -0.411007701
## 173          -0.440309953
## 15           -0.508235374
## 96           -0.741879066
## 216          -0.838983197
## 43           -0.917691486
## 162          -1.118043111
## 136          -1.144957218
## 127          -1.374227383
## 112          -1.506339195
## 65           -1.646824340
## 204          -1.675036297
## 126          -1.738693526
##
## 200
## 48
## 183
## 168
## 187
## 13
## 203
## 68
## 55
## 75
## 95
## 185
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## 146
## 4                                                                                Having eight months of s
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```

```
## 72
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## 8
## 18
## 161
## 155
## 130
## 49
## 133
## 6
## 225
## 150
## 58
## 125
## 16
## 169
## 137
## 117
## 93
## 159
## 60
## 100
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```

```
## 145                                                 General anesthesia has been used in surgery sinc
## 147                                                 General anesthesia has been used in surgery sinc
## 23                            The Norwegian study's within-family design is its most important methodolog
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## 224
## 46
## 149
## 37                                                                                                 The co
## 91                                                                                                   The
## 208                                                                                                  The
## 36
## 148
## 201
## 1   Research in epigenetics has shown that certain experiences - particularly stress, starvation, or
## 5     Research in epigenetics has shown that certain experiences - particularly stress, starvation,
## 220   Research in epigenetics has shown that certain experiences - particularly stress, starvation,
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## 172
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## 222
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## 86
## 140
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## 206
## 110
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## 66                                              The Norwegian study's within-family design is its
## 87                                              The Norwegian study's within-family design is its
## 209                                             The Norwegian study's within-family design is its
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```

```
## 221
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## 62                                               General anesthesia has been used in surgery since
## 72
## 142
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```

```
## 184
## 123                                              General anesthesia has been used in surgery since
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## 53                     The Norwegian study's within-family design is its most important methodologica
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## 57  Research in epigenetics has shown that certain experiences - particularly stress, starvation, or
## 52
## 170                                              General anesthesia has been used in surgery since
## 32
## 135
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## 179 Research in epigenetics has shown that certain experiences - particularly stress, starvation, or
## 61
## 25
## 156
## 8
## 18
## 161 Research in epigenetics has shown that certain experiences - particularly stress, starvation, or
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## 145                                              General anesthesia has been used in surgery since
## 147                                              General anesthesia has been used in surgery since
## 23                     The Norwegian study's within-family design is its most important methodologica
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## 28
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## 144
## 66                                            The Norwegian study's within-family design is its mo
## 87                                            The Norwegian study's within-family design is its mo
## 209                                           The Norwegian study's within-family design is its mo
## 81
## 82
## 174
## 211
## 215
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## 44
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```

```
## 56
## 173 Research in epigenetics has shown that certain experiences - particularly stress, starvation, or
## 15
## 96
## 216
## 43                    The Norwegian study's within-family design is its most important methodologica
## 162
## 136
## 127
## 112
## 65
## 204
## 126
```

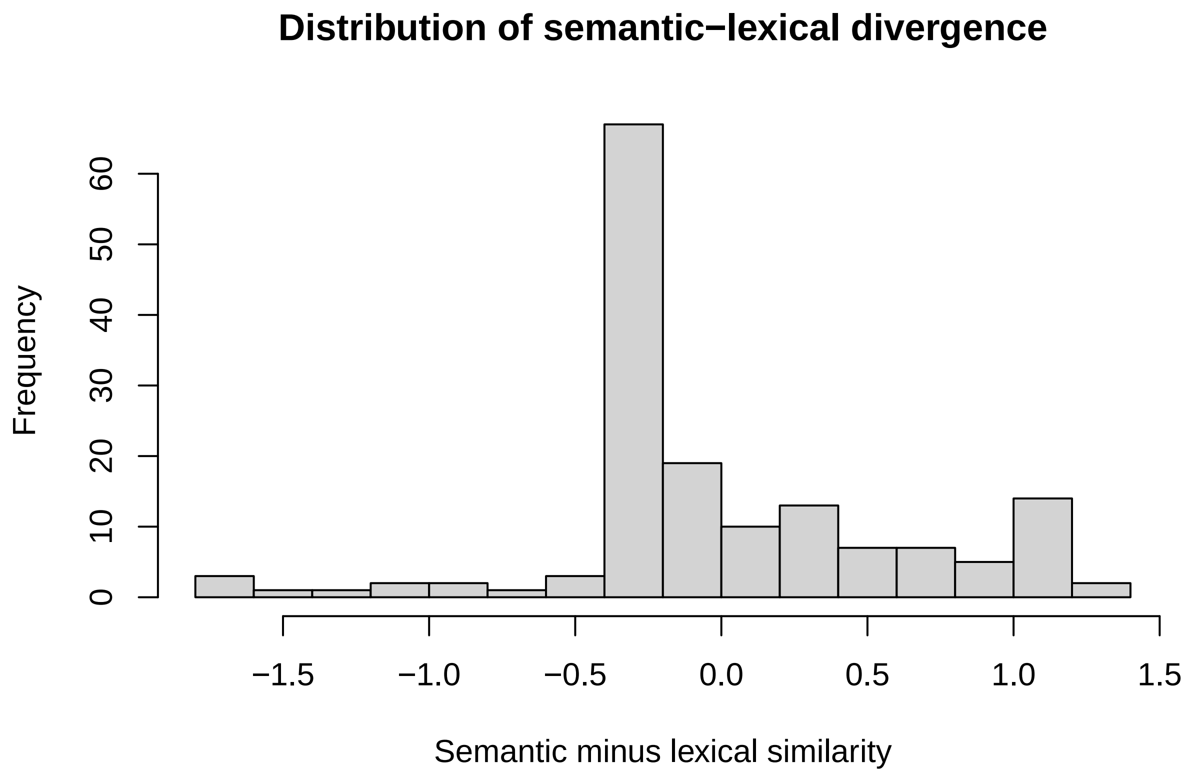
```r
summary(embedding_divergence_cases)
```

```
##  submission_id      assigned_category    ai_intent_code
##  Length:157         academic :21      Asking    :40
##  Class :character   creative :23      Doing     :86
##  Mode  :character   decisions:24      Expressing: 6
##                     info     :29      NA's      :25
##                     learning :36
##                     writing  :24
##  jaccard_similarity_to_ai embedding_cosine_similarity_to_ai
##  Min.   :0.00000          Min.   :0.05209
##  1st Qu.:0.08738          1st Qu.:0.58189
##  Median :0.31169          Median :0.86238
##  Mean   :0.50899          Mean   :0.77955
##  3rd Qu.:1.00000          3rd Qu.:1.00000
##  Max.   :1.00000          Max.   :1.00000
##  semantic_minus_lexical participant_response ai_response_shown
##  Min.   :-1.7387        Length:157           Length:157
##  1st Qu.:-0.2804        Class :character     Class :character
##  Median :-0.2410        Mode  :character     Mode  :character
##  Mean   : 0.0000
##  3rd Qu.: 0.3193
##  Max.   : 1.2792
```

High positive cases mean relatively high semantic similarity but lower lexical overlap so possible paraphrasing / same ideas in different words.

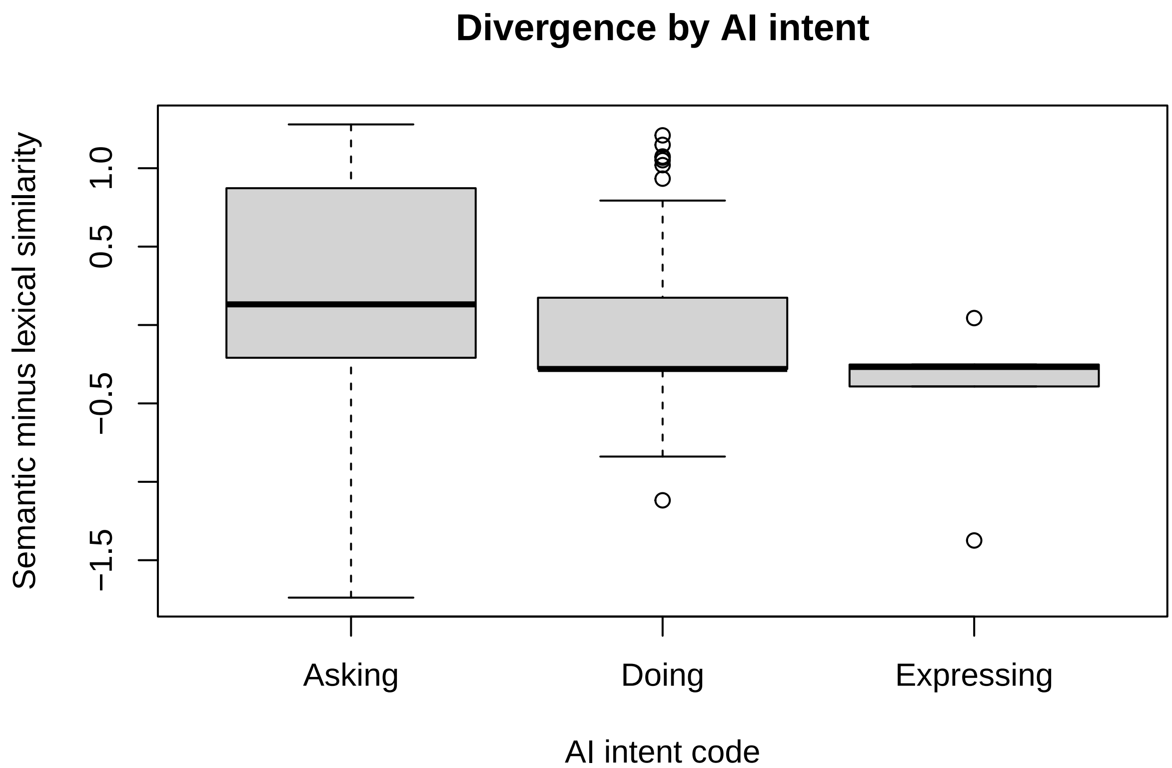
High negative cases mean relatively high lexical overlap but lower semantic similarity so possible shared keywords, copied phrases, or task-specific vocabulary without the whole response being as semantically close.

```r
hist(
  embedding_analysis_data$semantic_minus_lexical,
  breaks = 15,
  xlab = "Semantic minus lexical similarity",
  main = "Distribution of semantic-lexical divergence"
)
```



The divergence distribution was centered slightly below zero, suggesting that lexical and semantic similarity generally aligned, with a small tendency for lexical overlap to be relatively stronger. However, there were more moderately positive divergence cases than strongly negative ones, indicating several responses that were semantically close to the AI response despite lower exact word overlap.

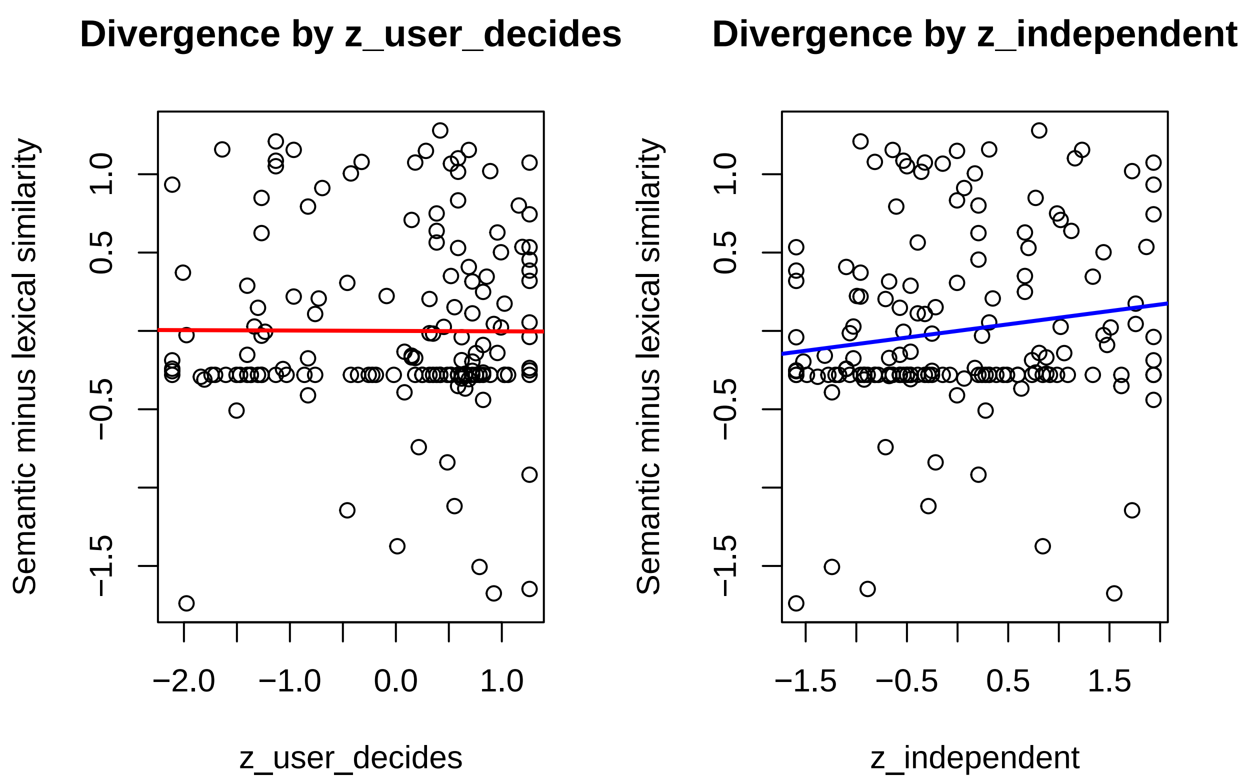
```
boxplot(
  semantic_minus_lexical ~ ai_intent_code,
  data = embedding_analysis_data,
  xlab = "AI intent code",
  ylab = "Semantic minus lexical similarity",
  main = "Divergence by AI intent"
)
```



Asking has the highest and most variable semantic_minus_lexical values. Many Asking cases are positive or near positive, meaning people who asked for information/advice may have taken up the AI's ideas while using their own wording. Doing is centered lower, around negative values. That suggests Doing prompts tended to produce responses where lexical overlap was relatively strong compared with semantic-only divergence.

```r
par(mfrow = c(1, 2))

plot(
  embedding_analysis_data$z_user_decides,
  embedding_analysis_data$semantic_minus_lexical,
  xlab = "z_user_decides",
  ylab = "Semantic minus lexical similarity",
  main = "Divergence by z_user_decides"
)
abline(
  lm(
    semantic_minus_lexical ~ z_user_decides,
    data = embedding_analysis_data
  ),
  col = "red",
  lwd = 2
)

plot(
  embedding_analysis_data$z_independent,
  embedding_analysis_data$semantic_minus_lexical,
```

```r
  xlab = "z_independent",
  ylab = "Semantic minus lexical similarity",
  main = "Divergence by z_independent"
)
abline(
  lm(
    semantic_minus_lexical ~ z_independent,
    data = embedding_analysis_data
  ),
  col = "blue",
  lwd = 2
)
```




```r
par(mfrow = c(1, 1))

summary(lm(
  semantic_minus_lexical ~ z_user_decides + z_independent,
  data = embedding_analysis_data
))
```

```
##
## Call:
## lm(formula = semantic_minus_lexical ~ z_user_decides + z_independent,
##     data = embedding_analysis_data)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7931 -0.2804 -0.1841  0.3090  1.2711
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.828e-16  4.654e-02   0.000   1.0000
## z_user_decides -2.058e-02  4.769e-02  -0.432   0.6666
## z_independent   8.858e-02  4.769e-02   1.857   0.0652 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5832 on 154 degrees of freedom
## Multiple R-squared:  0.02193,    Adjusted R-squared:  0.009227
## F-statistic: 1.726 on 2 and 154 DF,  p-value: 0.1813
```

Neither independence predictor significantly predicted semantic-lexical divergence. Participants who described themselves as more independent on either slider were not more likely to show idea-level uptake without word-level overlap.

Altogether: Embedding similarity analyses suggest that participants' overlap with the AI response also reflected meaningful semantic convergence besides shared words. Asking prompts appear to involve using AI as an informational or advisory resource, where participants may incorporate AI-generated ideas while still rephrasing or integrating them into their own response. Doing prompts, by contrast, more directly delegate the task of producing the response to the AI. This was reflected in higher lexical and semantic similarity to the AI response among Doing prompts, suggesting a stronger form of cognitive offloading. However, neither pre-task independence predictor reliably predicted lower semantic uptake or greater paraphrasing. Actual prompting behavior predicted AI uptake more clearly than prior independence beliefs.

Exploratory Clustering: AI Use Profiles

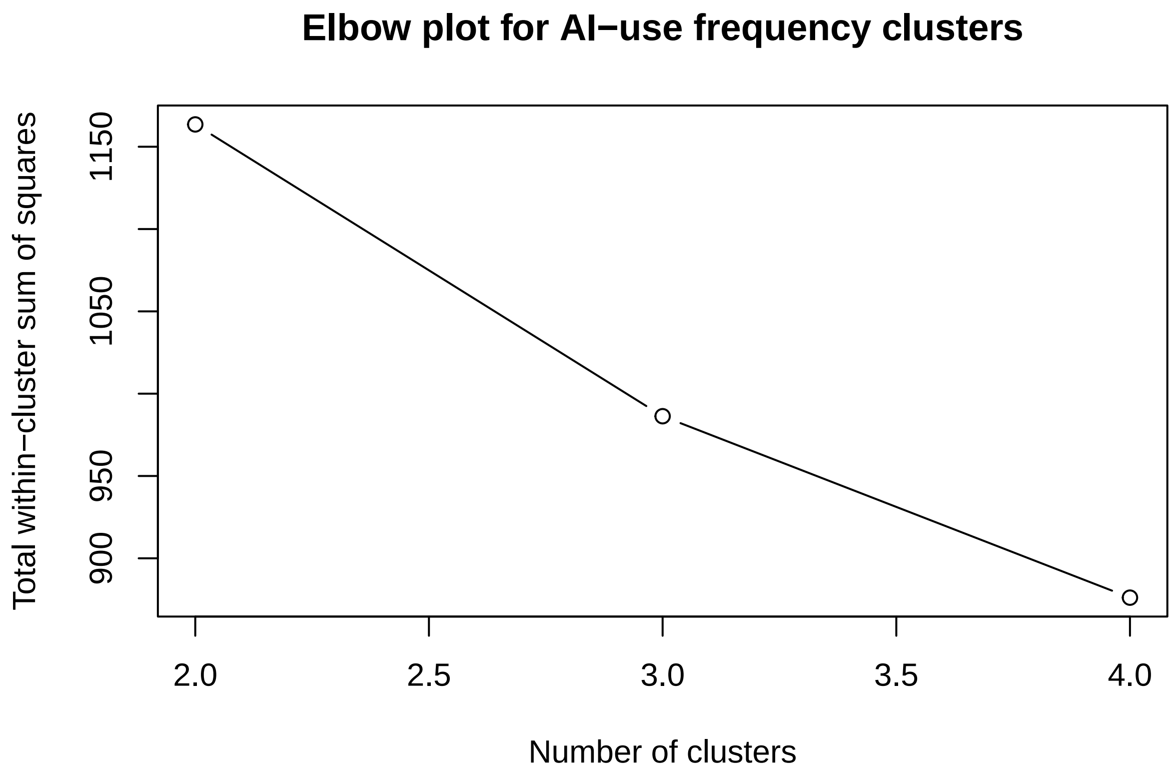
This exploratory analysis asks whether participants fall into AI-use profiles based on how often they use AI for different purposes. Because the eight frequency items are on the same 1-7 scale but may vary in spread, I standardize them before clustering. The cluster profiles are then described using the original 1-7 means so they remain interpretable.

```r
cluster_frequency_items <- c(
  "freq_information_guidance",
  "freq_writing_communication",
  "freq_coding_math_technical",
  "freq_creative_work",
  "freq_learning_understanding",
  "freq_advice_decisions",
  "freq_summarizing_opinions",
  "freq_automation_repetitive"
)

cluster_data <- embedding_similarity_data

for (item in cluster_frequency_items) {
  cluster_data[[item]] <- suppressWarnings(as.numeric(cluster_data[[item]]))
}
```

```r
cluster_complete_data <- cluster_data[
  complete.cases(cluster_data[, cluster_frequency_items]),
]

cluster_matrix <- scale(cluster_complete_data[, cluster_frequency_items])

set.seed(123)

cluster_fit_table <- data.frame()
cluster_models <- list()

for (k in 2:4) {
  fit <- kmeans(
    cluster_matrix,
    centers = k,
    nstart = 50
  )

  cluster_models[[as.character(k)]] <- fit

  cluster_fit_table <- rbind(
    cluster_fit_table,
    data.frame(
      n_clusters = k,
      total_withinss = fit$tot.withinss,
      between_total_ratio = fit$betweenss / fit$totss
    )
  )
}

cluster_fit_table
```

```
##   n_clusters total_withinss between_total_ratio
## 1          2      1163.5507           0.3564432
## 2          3       986.3183           0.4544700
## 3          4       876.1107           0.5154255
```

```r
plot(
  cluster_fit_table$n_clusters,
  cluster_fit_table$total_withinss,
  type = "b",
  xlab = "Number of clusters",
  ylab = "Total within-cluster sum of squares",
  main = "Elbow plot for AI-use frequency clusters"
)
```




```r
# Choose the number of clusters after inspecting the elbow plot and profiles.
# Three clusters is the default because it is interpretable and comparable to
# the earlier exploratory LCA result.
best_n_clusters <- 3
best_cluster_model <- cluster_models[[as.character(best_n_clusters)]]

cluster_complete_data$ai_use_cluster <- factor(
  best_cluster_model$cluster
)

table(cluster_complete_data$ai_use_cluster)
```

```
##
##  1  2  3
## 51 87 89
```

```r
cluster_profiles <- aggregate(
  cluster_complete_data[, cluster_frequency_items],
  by = list(ai_use_cluster = cluster_complete_data$ai_use_cluster),
  FUN = mean,
  na.rm = TRUE
)

cluster_profiles
```

```
##   ai_use_cluster freq_information_guidance freq_writing_communication
## 1              1                  3.352941                   2.862745
## 2              2                  5.977011                   4.160920
## 3              3                  6.078652                   5.505618
##   freq_coding_math_technical freq_creative_work freq_learning_understanding
## 1                   2.843137           2.352941                    2.862745
## 2                   2.195402           3.793103                    5.310345
## 3                   5.123596           5.629213                    5.898876
##   freq_advice_decisions freq_summarizing_opinions freq_automation_repetitive
## 1              1.647059                  2.411765                   2.176471
## 2              4.379310                  4.505747                   2.816092
## 3              5.280899                  5.786517                   5.550562
```

The elbow plot suggested that a 3-cluster solution provided a reasonable balance between fit and interpretability. It explained approximately 47% of the variation in AI-use frequency patterns.

The clusters reflected distinct AI-use profiles.

Cluster 1: Low-frequency AI users, with relatively low use across all purposes.

Cluster 2: Broad high-frequency AI users, with high use across information, writing, coding, creative work, learning, advice, summarizing, and automation.

Cluster 3: Information/learning/advice users, with relatively high use for information, learning, advice, and summarizing, but lower use for coding, creative work, and automation.

```r
model_cluster_ai_used <- glm(
  ai_used ~ ai_use_cluster,
  data = cluster_complete_data,
  family = binomial
)

summary(model_cluster_ai_used)
```

```
##
## Call:
## glm(formula = ai_used ~ ai_use_cluster, family = binomial, data = cluster_complete_data)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.1967     0.2814  -0.699 0.484544
## ai_use_cluster2   1.4057     0.3796   3.703 0.000213 ***
## ai_use_cluster3   1.3717     0.3762   3.646 0.000266 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 278.84  on 226  degrees of freedom
## Residual deviance: 261.27  on 224  degrees of freedom
## AIC: 267.27
##
## Number of Fisher Scoring iterations: 4
```

```r
drop1(model_cluster_ai_used, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## ai_used ~ ai_use_cluster
##                Df Deviance    AIC    LRT  Pr(>Chi)
## <none>              261.27 267.27
## ai_use_cluster  2   278.84 280.84 17.566 0.0001533 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emmeans(model_cluster_ai_used, ~ ai_use_cluster, type = "response")
```

```
##  ai_use_cluster  prob     SE  df asymp.LCL asymp.UCL
##  1              0.451 0.0697 Inf     0.321     0.588
##  2              0.770 0.0451 Inf     0.670     0.847
##  3              0.764 0.0450 Inf     0.665     0.841
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
cluster_ai_users <- subset(
  cluster_complete_data,
  ai_used == TRUE & !is.na(task_id)
)

cluster_ai_users$task_id <- factor(cluster_ai_users$task_id)

model_cluster_jaccard <- lmer(
  jaccard_similarity_to_ai ~ ai_use_cluster + (1 | task_id),
  data = subset(cluster_ai_users, !is.na(jaccard_similarity_to_ai))
)

summary(model_cluster_jaccard)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ ai_use_cluster + (1 | task_id)
##    Data: subset(cluster_ai_users, !is.na(jaccard_similarity_to_ai))
##
## REML criterion at convergence: 180.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6785 -0.8369 -0.3121  0.9796  1.6087
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.02013  0.1419
##  Residual             0.16098  0.4012
```

```
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                   Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)        0.44048    0.09266 109.28109   4.754 6.13e-06 ***
## ai_use_cluster2    0.08937    0.09915 148.21167   0.901    0.369
## ai_use_cluster3    0.09142    0.10192 154.96608   0.897    0.371
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_s_c2
## ai_s_clstr2 -0.800
## ai_s_clstr3 -0.811  0.734
```

```r
anova(model_cluster_jaccard)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                 Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_use_cluster 0.15008 0.075041     2 154.06  0.4662 0.6283
```

```r
emmeans(model_cluster_jaccard, pairwise ~ ai_use_cluster, adjust = "tukey")
```

```
## $emmeans
##  ai_use_cluster emmean     SE    df lower.CL upper.CL
##  1               0.440 0.0927 109.3    0.257    0.624
##  2               0.530 0.0609  39.6    0.407    0.653
##  3               0.532 0.0605  38.6    0.409    0.654
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
##
## $contrasts
##  contrast                          estimate     SE  df t.ratio p.value
##  ai_use_cluster1 - ai_use_cluster2 -0.08937 0.0991 148  -0.901  0.6403
##  ai_use_cluster1 - ai_use_cluster3 -0.09142 0.1020 155  -0.897  0.6431
##  ai_use_cluster2 - ai_use_cluster3 -0.00205 0.0733 155  -0.028  0.9996
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
model_cluster_embedding <- lmer(
  embedding_cosine_similarity_to_ai ~ ai_use_cluster + (1 | task_id),
  data = subset(cluster_ai_users, !is.na(embedding_cosine_similarity_to_ai))
)

summary(model_cluster_embedding)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: embedding_cosine_similarity_to_ai ~ ai_use_cluster + (1 | task_id)
##    Data: subset(cluster_ai_users, !is.na(embedding_cosine_similarity_to_ai))
```

```
##
## REML criterion at convergence: 18.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0747 -0.7351  0.2317  0.7835  1.5862
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.007559 0.08694
##  Residual             0.056248 0.23717
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##                  Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)       0.74446    0.05510 104.01351  13.511   <2e-16 ***
## ai_use_cluster2   0.03418    0.05866 146.26727   0.583    0.561
## ai_use_cluster3   0.04610    0.06044 153.84268   0.763    0.447
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_s_c2
## ai_s_clstr2 -0.796
## ai_s_clstr3 -0.805  0.732
```

```r
anova(model_cluster_embedding)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_use_cluster 0.032791 0.016396     2 152.76  0.2915 0.7476
```

```r
emmeans(model_cluster_embedding, pairwise ~ ai_use_cluster, adjust = "tukey")
```

```
## $emmeans
##  ai_use_cluster emmean     SE    df lower.CL upper.CL
##  1               0.744 0.0551 104.0    0.635    0.854
##  2               0.779 0.0365  36.1    0.705    0.853
##  3               0.791 0.0364  35.8    0.717    0.864
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
##
## $contrasts
##  contrast                          estimate     SE  df t.ratio p.value
##  ai_use_cluster1 - ai_use_cluster2  -0.0342 0.0587 146  -0.583  0.8296
##  ai_use_cluster1 - ai_use_cluster3  -0.0461 0.0604 154  -0.763  0.7264
##  ai_use_cluster2 - ai_use_cluster3  -0.0119 0.0436 154  -0.273  0.9596
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
cluster_ai_users$epistemic_ownership_binary <- ifelse(
  cluster_ai_users$epistemic_ownership <= 3, 0L,
  ifelse(cluster_ai_users$epistemic_ownership >= 5, 1L, NA_integer_)
)

model_cluster_ownership <- glmer(
  epistemic_ownership_binary ~ ai_use_cluster + (1 | task_id),
  data = subset(cluster_ai_users, !is.na(epistemic_ownership_binary)),
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_cluster_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: epistemic_ownership_binary ~ ai_use_cluster + (1 | task_id)
##    Data: subset(cluster_ai_users, !is.na(epistemic_ownership_binary))
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     188.9     200.8     -90.4     180.9       140
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.8165 -0.8165 -0.6100  1.2247  1.6394
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.8473     0.4880  -1.736   0.0825 .
## ai_use_cluster2  -0.1413     0.5691  -0.248   0.8039
## ai_use_cluster3   0.4418     0.5497   0.804   0.4216
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) a_s_c2
## ai_s_clstr2 -0.857
## ai_s_clstr3 -0.888  0.761
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
drop1(model_cluster_ownership, test = "Chisq")
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Single term deletions
##
## Model:
## epistemic_ownership_binary ~ ai_use_cluster + (1 | task_id)
##                npar    AIC    LRT Pr(Chi)
## <none>              188.89
## ai_use_cluster    2 187.32 2.4266  0.2972
```

```
emmeans(model_cluster_ownership, pairwise ~ ai_use_cluster, adjust = "tukey", type = "response")
```

```
## $emmeans
##  ai_use_cluster  prob     SE  df asymp.LCL asymp.UCL
##  1              0.300 0.1020 Inf     0.141     0.527
##  2              0.271 0.0579 Inf     0.173     0.398
##  3              0.400 0.0608 Inf     0.289     0.523
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
##
## $contrasts
##  contrast                          odds.ratio    SE  df null z.ratio p.value
##  ai_use_cluster1 / ai_use_cluster2      1.152 0.655 Inf    1   0.248  0.9666
##  ai_use_cluster1 / ai_use_cluster3      0.643 0.353 Inf    1  -0.804  0.7007
##  ai_use_cluster2 / ai_use_cluster3      0.558 0.216 Inf    1  -1.506  0.2879
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log odds ratio scale
```

Cluster membership significantly predicted whether participants used the optional AI assistant, chi-squared(2) = 17.57, p < .001. Participants in Cluster 1, the low-frequency AI-use group, had an estimated AI-use probability of .45. Participants in Cluster 2, the broad high-frequency AI-use group, had a higher estimated probability of AI use (.77) and were significantly more likely to use AI than Cluster 1, b = 1.41, SE = 0.38, z = 3.70, p < .001. Participants in Cluster 3, the information/learning/advice group, also had a higher estimated probability of AI use (.76) and were significantly more likely to use AI than Cluster 1, b = 1.37, SE = 0.38, z = 3.65, p < .001.

However, among participants who used AI, cluster membership did not significantly predict lexical similarity to the AI response, F(2, 154.06) = 0.47, p = .628. Estimated Jaccard similarity was similar across clusters: Cluster 1, M = .440; Cluster 2, M = .530; Cluster 3, M = .532. Tukey-adjusted pairwise comparisons were all non-significant, all p >= .640.

Cluster membership also did not significantly predict embedding-based semantic similarity to the AI response, F(2, 152.76) = 0.29, p = .748. Estimated semantic similarity was similar across clusters: Cluster 1, M = .744; Cluster 2, M = .779; Cluster 3, M = .791. Tukey-adjusted pairwise comparisons were again non-significant, all p >= .726.

Finally, cluster membership did not significantly predict binary epistemic ownership, chi-squared(2) = 2.43, p = .297. Estimated probabilities of primarily own-thinking ownership were .30 for Cluster 1, .27 for Cluster 2, and .40 for Cluster 3. Tukey-adjusted pairwise comparisons were not significant, all p >= .288.

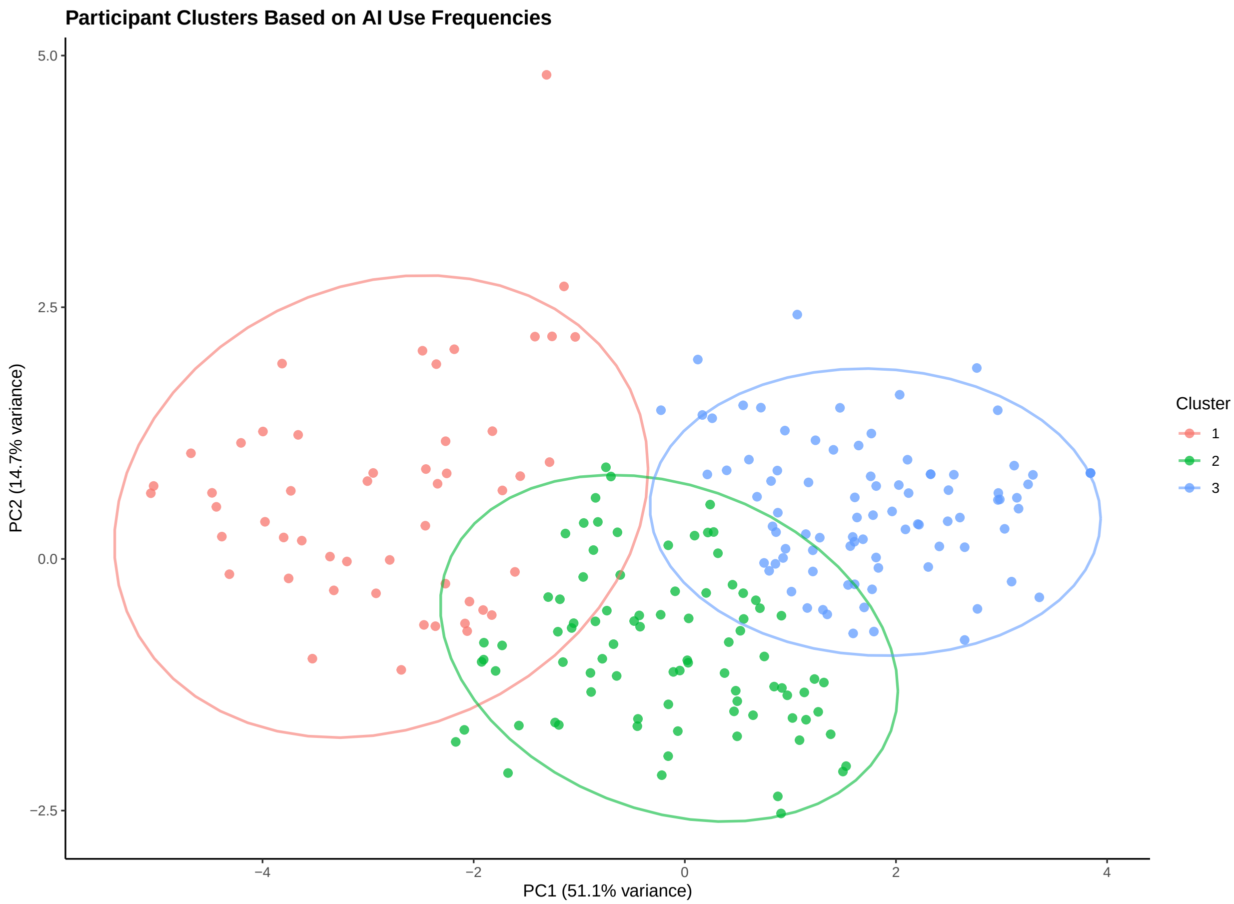
```
# -----------------------------
# 1. Cluster profile heatmap
# -----------------------------
```

```r
cluster_profile_long <- do.call(rbind, lapply(
  seq_len(nrow(cluster_profiles)),
  function(i) {
    data.frame(
      ai_use_cluster = cluster_profiles$ai_use_cluster[i],
      frequency_item = cluster_frequency_items,
      mean_frequency = as.numeric(cluster_profiles[i, cluster_frequency_items]),
      stringsAsFactors = FALSE
    )
  }
))

cluster_profile_long$frequency_label <- factor(
  cluster_profile_long$frequency_item,
  levels = cluster_frequency_items,
  labels = c(
    "Info / guidance",
    "Writing / communication",
    "Coding / technical",
    "Creative work",
    "Learning",
    "Advice / decisions",
    "Summarizing",
    "Automation"
  )
)

ggplot(cluster_profile_long, aes(
  x = frequency_label,
  y = ai_use_cluster,
  fill = mean_frequency
)) +
  geom_tile(color = "white", linewidth = 0.6) +
  geom_text(
    aes(label = sprintf("%.1f", mean_frequency)),
    size = 3.4,
    fontface = "bold"
  ) +
  scale_fill_gradient(
    low = "#f2f2f2",
    high = "#4472C4",
    limits = c(1, 7)
  ) +
  labs(
    title = "AI Use Frequency Profiles by Cluster",
    x = NULL,
    y = "AI use cluster",
    fill = "Mean\nfrequency"
  ) +
  theme_classic(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 13),
    axis.text.x = element_text(angle = 35, hjust = 1)
```

```
)
```


```
# ------------------------------
# 2. PCA visualization of clusters
# ------------------------------

cluster_pca <- prcomp(cluster_matrix, center = FALSE, scale. = FALSE)

cluster_pca_data <- data.frame(
  pc1 = cluster_pca$x[, 1],
  pc2 = cluster_pca$x[, 2],
  ai_use_cluster = cluster_complete_data$ai_use_cluster
)

pca_var <- round(100 * summary(cluster_pca)$importance[2, 1:2], 1)

ggplot(cluster_pca_data, aes(
  x = pc1,
  y = pc2,
  color = ai_use_cluster
)) +
  geom_point(alpha = 0.75, size = 2.2) +
  stat_ellipse(linewidth = 0.8, alpha = 0.6) +
  labs(
```

```r
    title = "Participant Clusters Based on AI Use Frequencies",
    x = paste0("PC1 (", pca_var[1], "% variance)"),
    y = paste0("PC2 (", pca_var[2], "% variance)"),
    color = "Cluster"
  ) +
  theme_classic(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 13)
  )
```

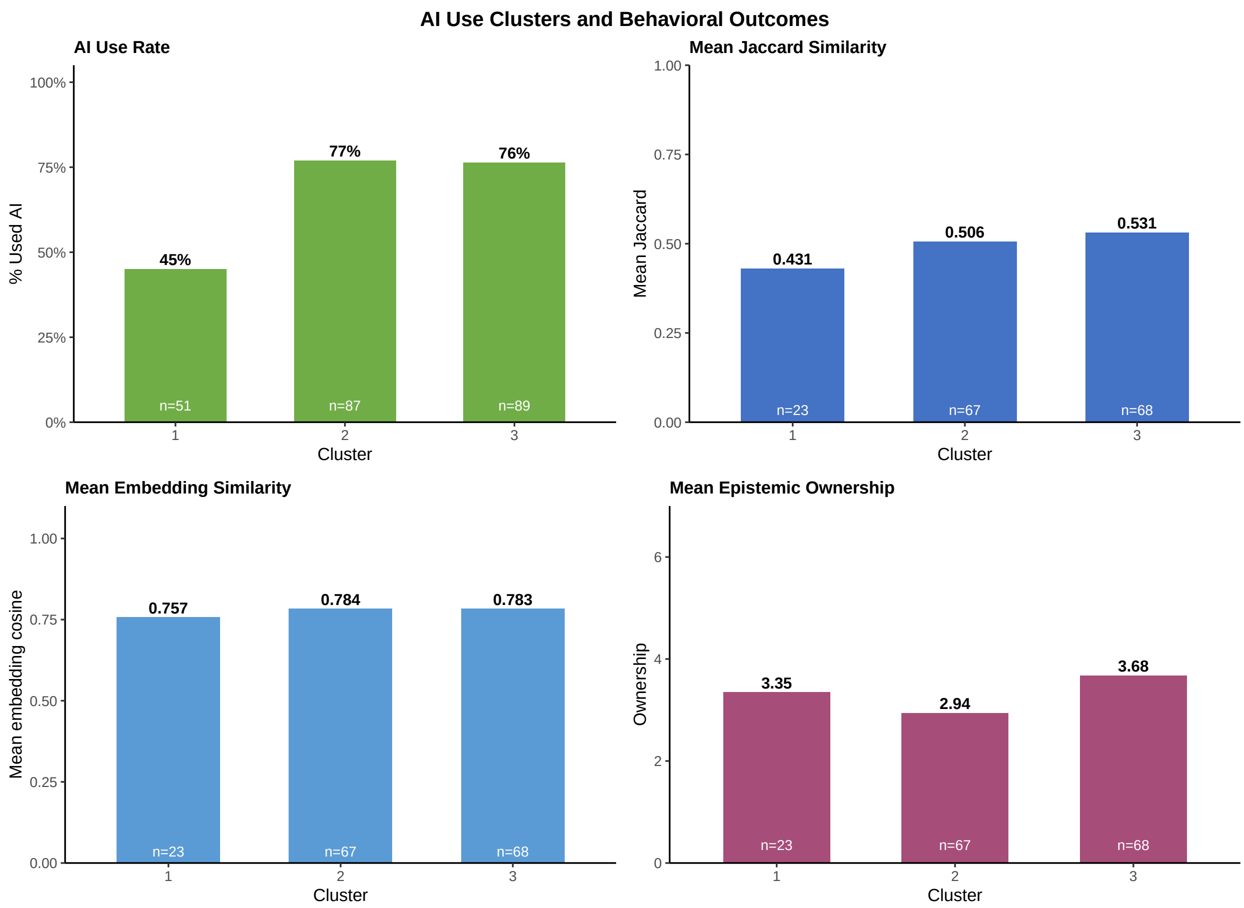



```r
# ------------------------------
# 3. Cluster relationships with outcomes
# ------------------------------

cluster_outcome_summary <- do.call(rbind, lapply(
  split(cluster_complete_data, cluster_complete_data$ai_use_cluster),
  function(sub) {
    data.frame(
      ai_use_cluster = as.character(sub$ai_use_cluster[1]),
      ai_use_rate = mean(sub$ai_used, na.rm = TRUE),
      mean_jaccard = mean(
        sub$jaccard_similarity_to_ai[sub$ai_used == TRUE],
        na.rm = TRUE
      ),
```

```
      mean_embedding = mean(
        sub$embedding_cosine_similarity_to_ai[sub$ai_used == TRUE],
        na.rm = TRUE
      ),
      mean_epistemic_ownership = mean(
        sub$epistemic_ownership[sub$ai_used == TRUE],
        na.rm = TRUE
      ),
      n_all = nrow(sub),
      n_ai_users = sum(sub$ai_used == TRUE, na.rm = TRUE),
      stringsAsFactors = FALSE
    )
  }
))

cluster_outcome_summary$ai_use_cluster <- factor(
  cluster_outcome_summary$ai_use_cluster,
  levels = sort(unique(cluster_outcome_summary$ai_use_cluster))
)

p_cluster_use <- ggplot(
  cluster_outcome_summary,
  aes(x = ai_use_cluster, y = ai_use_rate)
) +
  geom_col(fill = "#70AD47", width = 0.6) +
  geom_text(
    aes(label = paste0(round(ai_use_rate * 100), "%")),
    vjust = -0.35, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n_all), y = 0.05),
    color = "white", size = 3.1
  ) +
  scale_y_continuous(
    limits = c(0, 1.05),
    expand = c(0, 0),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title = "AI Use Rate",
    x = "Cluster",
    y = "% Used AI"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 11))

p_cluster_jaccard <- ggplot(
  cluster_outcome_summary,
  aes(x = ai_use_cluster, y = mean_jaccard)
) +
  geom_col(fill = "#4472C4", width = 0.6) +
  geom_text(
    aes(label = sprintf("%.3f", mean_jaccard)),
```

```
    vjust = -0.35, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n_ai_users), y = 0.035),
    color = "white", size = 3.1
  ) +
  scale_y_continuous(limits = c(0, 1), expand = c(0, 0)) +
  labs(
    title = "Mean Jaccard Similarity",
    x = "Cluster",
    y = "Mean Jaccard"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 11))

p_cluster_embedding <- ggplot(
  cluster_outcome_summary,
  aes(x = ai_use_cluster, y = mean_embedding)
) +
  geom_col(fill = "#5B9BD5", width = 0.6) +
  geom_text(
    aes(label = sprintf("%.3f", mean_embedding)),
    vjust = -0.35, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n_ai_users), y = 0.035),
    color = "white", size = 3.1
  ) +
  scale_y_continuous(limits = c(0, 1.1), expand = c(0, 0)) +
  labs(
    title = "Mean Embedding Similarity",
    x = "Cluster",
    y = "Mean embedding cosine"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 11))

p_cluster_ownership <- ggplot(
  cluster_outcome_summary,
  aes(x = ai_use_cluster, y = mean_epistemic_ownership)
) +
  geom_col(fill = "#A64D79", width = 0.6) +
  geom_text(
    aes(label = sprintf("%.2f", mean_epistemic_ownership)),
    vjust = -0.35, size = 3.5, fontface = "bold"
  ) +
  geom_text(
    aes(label = paste0("n=", n_ai_users), y = 0.35),
    color = "white", size = 3.1
  ) +
  scale_y_continuous(limits = c(0, 7), expand = c(0, 0)) +
  labs(
    title = "Mean Epistemic Ownership",
```

```
    x = "Cluster",
    y = "Ownership"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 11))

grid.arrange(
  p_cluster_use,
  p_cluster_jaccard,
  p_cluster_embedding,
  p_cluster_ownership,
  ncol = 2,
  top = grid::textGrob(
    "AI Use Clusters and Behavioral Outcomes",
    gp = grid::gpar(fontface = "bold", fontsize = 13)
  )
)
```



Altogether: Participants' general AI-use profiles predicted whether they chose to consult the AI assistant, chi-squared(2) = 17.57, p < .001, but not how much they incorporated the AI response once they used it, whether uptake was measured lexically, semantically, or through binary epistemic ownership.

Analysis 7: Domain-Specific AI Use Frequency vs. Task Behavior

This analysis asks whether self-reported AI use frequency in the specific domain of the assigned task predicts actual AI use and uptake in that task. For example, does someone who says they frequently use AI for writing tasks actually consult AI and copy more in a writing task?

Rather than constructing a category-to-item mapping, this analysis uses `self_reported_freq_in_assigned_category`, which directly records how often each participant reported using AI for the type of task they were assigned. This is a more direct measure and avoids the assumption that a single frequency item maps cleanly to each task category.

```
domain_freq_data <- model_data

domain_freq_data$z_self_reported_freq <- z(
  domain_freq_data$self_reported_freq_in_assigned_category
)

summary(domain_freq_data$self_reported_freq_in_assigned_category)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.000   6.000   6.000   6.084   7.000   7.000
```

```
table(domain_freq_data$assigned_category, useNA = "ifany")
```

```
##
##  academic  creative decisions      info  learning   writing
##        36        35        36        41        42        37
```

```
model_dfreq_ai_used <- glmer(
  ai_used ~ z_self_reported_freq +
    z_hours_per_week +
    (1 | task_id),
  data = subset(domain_freq_data, !is.na(z_self_reported_freq)),
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_dfreq_ai_used)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: ai_used ~ z_self_reported_freq + z_hours_per_week + (1 | task_id)
##    Data: subset(domain_freq_data, !is.na(z_self_reported_freq))
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     267.0     280.7    -129.5     259.0       223
##
## Scaled residuals:
```

```
##      Min       1Q   Median       3Q      Max 
## -2.4987 -0.8759  0.5059  0.6595  1.5757 
## 
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0       
## Number of obs: 227, groups:  task_id, 18
## 
## Fixed effects:
##                     Estimate Std. Error z value Pr(>|z|)    
## (Intercept)           0.8861     0.1530   5.792 6.97e-09 ***
## z_self_reported_freq   0.5998     0.1584   3.785 0.000153 ***
## z_hours_per_week       0.1333     0.1771   0.752 0.451842    
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Correlation of Fixed Effects:
##             (Intr) z_sl__
## z_slf_rprt_  0.103       
## z_hrs_pr_wk  0.073 -0.268
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
fixed_effects_table(model_dfreq_ai_used, exponentiate = TRUE)
```

```
##                   term  estimate std_error      p_value odds_ratio
## 1          (Intercept) 0.8860543 0.1529866 6.967258e-09   2.425540
## 2 z_self_reported_freq 0.5997989 0.1584489 1.534364e-04   1.821752
## 3     z_hours_per_week 0.1332590 0.1771243 4.518422e-01   1.142546
```

Self-reported domain-specific AI use frequency significantly predicted whether participants used the AI assistant during the task, OR = 1.82, p < .001. A one-standard-deviation increase in reported AI use for the assigned task domain was associated with higher odds of consulting the optional assistant, controlling for hours per week. Hours per week was not significant, OR = 1.14, p = .452.

```r
domain_freq_ai_users <- subset(
  domain_freq_data,
  ai_used == TRUE & !is.na(z_self_reported_freq) & !is.na(jaccard_similarity_to_ai)
)

model_dfreq_similarity <- lmer(
  jaccard_similarity_to_ai ~ z_self_reported_freq +
    z_hours_per_week +
    (1 | task_id),
  data = domain_freq_ai_users
)

summary(model_dfreq_similarity)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ z_self_reported_freq + z_hours_per_week +
```

```
##     (1 | task_id)
##    Data: domain_freq_ai_users
##
## REML criterion at convergence: 184.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6443 -0.8696 -0.2867  0.9693  1.6106
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.01895  0.1377
##  Residual             0.16237  0.4030
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)            0.51906    0.04673  16.89077  11.108 3.48e-09 ***
## z_self_reported_freq  -0.01318    0.04250 154.22434  -0.310    0.757
## z_hours_per_week       0.01125    0.03238 149.82863   0.347    0.729
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sl__
## z_slf_rprt_ -0.176
## z_hrs_pr_wk -0.036 -0.186
```

```r
fixed_effects_table(model_dfreq_similarity)
```

```
##                   term    estimate  std_error      p_value
## 1          (Intercept)  0.51905897 0.04672640 3.483237e-09
## 2 z_self_reported_freq -0.01317750 0.04250093 7.569400e-01
## 3     z_hours_per_week  0.01125076 0.03238021 7.287349e-01
```

Among participants who used the assistant, domain-specific AI use frequency did not predict lexical similarity to the AI response, b = -0.013, p = .757. Hours per week was also not significant, b = 0.011, p = .729. This is an AI-user-only model, so it asks whether domain frequency predicts copying after participants have already chosen to use AI.

```r
# Use self_reported_freq_in_assigned_category on embedding_analysis_data.
embedding_analysis_data$z_self_reported_freq <- z(
  embedding_analysis_data$self_reported_freq_in_assigned_category
)

domain_freq_embedding_data <- subset(
  embedding_analysis_data,
  !is.na(z_self_reported_freq) & !is.na(embedding_cosine_similarity_to_ai)
)

model_dfreq_embedding <- lmer(
  embedding_cosine_similarity_to_ai ~ z_self_reported_freq +
    z_hours_per_week +
```

```r
    (1 | task_id),
  data = domain_freq_embedding_data
)

summary(model_dfreq_embedding)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## embedding_cosine_similarity_to_ai ~ z_self_reported_freq + z_hours_per_week +
##     (1 | task_id)
##    Data: domain_freq_embedding_data
##
## REML criterion at convergence: 22.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0282 -0.7537  0.2385  0.7331  1.5901
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.006661 0.08162
##  Residual             0.056757 0.23824
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)           0.779022   0.027215  14.802363  28.625 2.26e-14 ***
## z_self_reported_freq -0.012016   0.020262 152.960807  -0.593    0.554
## z_hours_per_week      0.004629   0.019954 148.418030   0.232    0.817
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_sl__
## z_slf_rprt_ -0.010
## z_hrs_pr_wk -0.013 -0.188
```

```r
fixed_effects_table(model_dfreq_embedding)
```

```
##                   term     estimate  std_error      p_value
## 1          (Intercept)  0.779021717 0.02721507 2.259196e-14
## 2 z_self_reported_freq -0.012016468 0.02026217 5.540239e-01
## 3     z_hours_per_week  0.004628941 0.01995416 8.168740e-01
```

```r
domain_freq_embedding_data$z_embedding_cosine_similarity_to_ai <- z(
  domain_freq_embedding_data$embedding_cosine_similarity_to_ai
)
```

Domain-specific AI use frequency also did not predict embedding-based semantic similarity to the AI response among AI users, b = -0.012, p = .554. Hours per week was not significant, b = 0.005, p = .817.

```r
domain_freq_data$epistemic_ownership_binary <- ifelse(
  domain_freq_data$epistemic_ownership <= 3, 0L,
  ifelse(domain_freq_data$epistemic_ownership >= 5, 1L, NA_integer_)
)

domain_freq_ownership_data <- subset(
  domain_freq_data,
  ai_used == TRUE & !is.na(z_self_reported_freq) & !is.na(epistemic_ownership_binary)
)

model_dfreq_ownership <- glmer(
  epistemic_ownership_binary ~ z_jaccard_similarity_to_ai +
    z_self_reported_freq +
    z_hours_per_week +
    (1 | task_id),
  data = domain_freq_ownership_data,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_dfreq_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## epistemic_ownership_binary ~ z_jaccard_similarity_to_ai + z_self_reported_freq +
##     z_hours_per_week + (1 | task_id)
##    Data: domain_freq_ownership_data
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     159.3     174.1     -74.6     149.3       139
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.7142 -0.6034 -0.3321  0.8239  3.5480
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 -0.5557     0.2092  -2.656  0.00791 **
## z_jaccard_similarity_to_ai  -1.1036     0.2189  -5.042 4.62e-07 ***
## z_self_reported_freq         0.1909     0.2462   0.775  0.43825
## z_hours_per_week             0.1364     0.1944   0.702  0.48289
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##             (Intr) z_j___ z_sl__
## z_jccrd_s__  0.058
## z_slf_rprt_ -0.272 -0.008
## z_hrs_pr_wk -0.045 -0.133 -0.173
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
fixed_effects_table(model_dfreq_ownership, exponentiate = TRUE)
```

```
##                         term   estimate std_error      p_value odds_ratio
## 1                (Intercept) -0.5556580 0.2092207 7.910960e-03  0.5736947
## 2 z_jaccard_similarity_to_ai -1.1035684 0.2188948 4.617809e-07  0.3316854
## 3       z_self_reported_freq  0.1908550 0.2462140 4.382456e-01  1.2102840
## 4           z_hours_per_week  0.1364096 0.1944096 4.828907e-01  1.1461512
```

```
domain_freq_embedding_data$epistemic_ownership_binary <- ifelse(
  domain_freq_embedding_data$epistemic_ownership <= 3, 0L,
  ifelse(domain_freq_embedding_data$epistemic_ownership >= 5, 1L, NA_integer_)
)

domain_freq_embedding_ownership_data <- subset(
  domain_freq_embedding_data,
  !is.na(epistemic_ownership_binary) &
    !is.na(z_embedding_cosine_similarity_to_ai)
)

model_dfreq_ownership_embedding <- glmer(
  epistemic_ownership_binary ~ z_embedding_cosine_similarity_to_ai +
    z_self_reported_freq +
    z_hours_per_week +
    (1 | task_id),
  data = domain_freq_embedding_ownership_data,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_dfreq_ownership_embedding)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: epistemic_ownership_binary ~ z_embedding_cosine_similarity_to_ai +
##     z_self_reported_freq + z_hours_per_week + (1 | task_id)
##    Data: domain_freq_embedding_ownership_data
## Control: glmerControl(optimizer = "bobyqa")
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##    165.3    180.1    -77.6     155.3       138
```

```
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9137 -0.5415 -0.4500  0.8065  2.4598
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 143, groups:  task_id, 18
##
## Fixed effects:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                        -0.78720    0.19851  -3.966 7.32e-05 ***
## z_embedding_cosine_similarity_to_ai -0.91190    0.20034  -4.552 5.32e-06 ***
## z_self_reported_freq                 0.10533    0.19966   0.528    0.598
## z_hours_per_week                     0.07703    0.19955   0.386    0.699
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) z_____ z_sl__
## z_mbddn____  0.115
## z_slf_rprt_ -0.054  0.040
## z_hrs_pr_wk -0.016 -0.065 -0.162
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
fixed_effects_table(model_dfreq_ownership_embedding, exponentiate = TRUE)
```

```
##                                  term    estimate std_error      p_value
## 1                         (Intercept) -0.78720411 0.1985098 7.322144e-05
## 2 z_embedding_cosine_similarity_to_ai -0.91189777 0.2003436 5.322189e-06
## 3                z_self_reported_freq  0.10532841 0.1996556 5.978114e-01
## 4                    z_hours_per_week  0.07702609 0.1995498 6.994972e-01
##   odds_ratio
## 1  0.4551155
## 2  0.4017610
## 3  1.1110754
## 4  1.0800703
```

Domain-specific AI use frequency did not predict binary epistemic ownership whether AI uptake was measured by Jaccard similarity (OR = 1.21, p = .438) or by embedding similarity (OR = 1.11, p = .598). **In both models, similarity to the AI response was the main predictor of lower own-thinking ownership: Jaccard similarity, OR = 0.33, p < .001; embedding similarity, OR = 0.40, p < .001.**

Analysis 8: AI Engagement And Attention

This exploratory analysis asks whether participants who spent more time with the AI response, or reported reading it more carefully, showed greater uptake of the AI response or different epistemic ownership ratings. Because timing variables are usually skewed, millisecond variables are converted to seconds and log-transformed with `log1p()`.

```r
engagement_data <- subset(
  embedding_similarity_data,
  ai_used == TRUE & !is.na(task_id)
)

engagement_data$task_id <- factor(engagement_data$task_id)

engagement_data$ai_reading_depth <- factor(
  engagement_data$ai_reading_depth,
  levels = c("not_read", "glanced", "skimmed", "read_carefully"),
  ordered = TRUE
)

# Binary recodings.
engagement_data$epistemic_ownership_binary <- ifelse(
  engagement_data$epistemic_ownership <= 3, 0L,
  ifelse(engagement_data$epistemic_ownership >= 5, 1L, NA_integer_)
)
engagement_data$read_carefully <- as.integer(
  engagement_data$ai_reading_depth == "read_carefully"
)

engagement_data$response_writing_time_ms <-
  engagement_data$response_submitted_ms - engagement_data$response_started_ms

engagement_data$response_writing_time_ms[
  engagement_data$response_writing_time_ms < 0
] <- NA_real_

engagement_data$log_time_before_ai_sec <- log1p(
  engagement_data$time_before_ai_ms / 1000
)

engagement_data$log_ai_visible_duration_sec <- log1p(
  engagement_data$ai_visible_duration_ms / 1000
)

engagement_data$log_response_writing_time_sec <- log1p(
  engagement_data$response_writing_time_ms / 1000
)

engagement_data$log_task_duration_sec <- log1p(
  engagement_data$task_duration_ms / 1000
)

engagement_data$z_log_time_before_ai_sec <- z(
  engagement_data$log_time_before_ai_sec
)

engagement_data$z_log_ai_visible_duration_sec <- z(
  engagement_data$log_ai_visible_duration_sec
)
```

```r
engagement_data$z_log_response_writing_time_sec <- z(
  engagement_data$log_response_writing_time_sec
)

engagement_data$z_log_task_duration_sec <- z(
  engagement_data$log_task_duration_sec
)

table(engagement_data$ai_reading_depth, useNA = "ifany")
```

```
##
##       not_read        glanced        skimmed read_carefully
##              9             14             45             90
```

```r
summary(engagement_data$ai_visible_duration_ms)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   14993   70970  149338  217684  255126 4152051
```

```r
summary(engagement_data$response_writing_time_ms)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##     473    3595   46510   80431  134905  340073       1
```

Analysis 8A: Does Self-Reported Reading Depth Match Viewing Time?

This check asks whether people who reported reading the AI response carefully also had the AI response visible for longer.

```r
model_reading_depth_visible_time <- lmer(
  log_ai_visible_duration_sec ~ read_carefully +
    (1 | task_id),
  data = subset(
    engagement_data,
    !is.na(log_ai_visible_duration_sec) & !is.na(read_carefully)
  )
)

summary(model_reading_depth_visible_time)
```

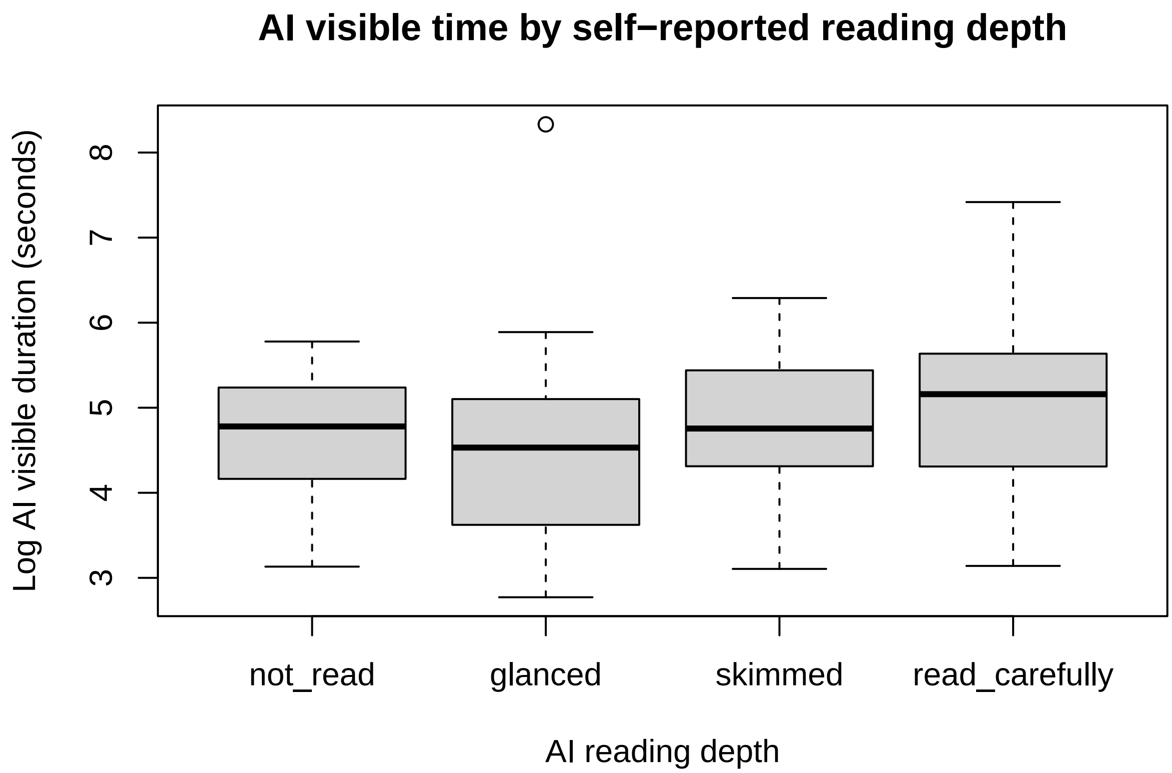
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log_ai_visible_duration_sec ~ read_carefully + (1 | task_id)
##    Data: subset(engagement_data, !is.na(log_ai_visible_duration_sec) &
##     !is.na(read_carefully))
##
## REML criterion at convergence: 424.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -2.0736 -0.6348  0.0786  0.6408  3.8822
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.09308  0.3051
##  Residual             0.77752  0.8818
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                 Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)       4.6816     0.1311  40.7406  35.711   <2e-16 ***
## read_carefully    0.3499     0.1465 153.6784   2.388   0.0182 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## read_crflly -0.637
```

```
fixed_effects_table(model_reading_depth_visible_time)
```

```
##             term  estimate std_error      p_value
## 1    (Intercept) 4.6816073 0.1310980 2.276878e-32
## 2 read_carefully 0.3499444 0.1465427 1.815458e-02
```

```
boxplot(
  log_ai_visible_duration_sec ~ ai_reading_depth,
  data = engagement_data,
  xlab = "AI reading depth",
  ylab = "Log AI visible duration (seconds)",
  main = "AI visible time by self-reported reading depth"
)
```



People who reported reading carefully had the AI response visible significantly longer than other AI users, b = 0.350, p = .018. This supports the self-reported reading-depth item as behaviorally meaningful, although it is still an imperfect proxy for attention.

Analysis 8B: Does Engagement Predict Lexical Similarity?

Outcome: `jaccard_similarity_to_ai`
Model type: linear mixed regression
Predictors: `ai_reading_depth`, `ai_visible_duration_ms`, `time_before_ai_ms`
Random intercept: `task_id`
Sample: AI users

```
engagement_jaccard_data <- subset(
  engagement_data,
  !is.na(jaccard_similarity_to_ai) &
    !is.na(read_carefully) &
    !is.na(z_log_ai_visible_duration_sec) &
    !is.na(z_log_time_before_ai_sec)
)

model_engagement_jaccard <- lmer(
  jaccard_similarity_to_ai ~ read_carefully +
    z_log_ai_visible_duration_sec +
    z_log_time_before_ai_sec +
    (1 | task_id),
```

```r
  data = engagement_jaccard_data
)

summary(model_engagement_jaccard)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## jaccard_similarity_to_ai ~ read_carefully + z_log_ai_visible_duration_sec +
##     z_log_time_before_ai_sec + (1 | task_id)
##    Data: engagement_jaccard_data
##
## REML criterion at convergence: 155.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.67357 -0.71625 -0.06888  0.80542  2.72179
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.01634  0.1278
##  Residual             0.13129  0.3623
## Number of obs: 158, groups:  task_id, 18
##
## Fixed effects:
##                               Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                    0.57931    0.05497  39.49285  10.539 4.86e-13
## read_carefully                -0.11486    0.06219 151.70802  -1.847   0.0667
## z_log_ai_visible_duration_sec  -0.15075    0.03143 153.17424  -4.796 3.80e-06
## z_log_time_before_ai_sec       -0.04092    0.03116 153.92847  -1.313   0.1912
##
## (Intercept)                   ***
## read_carefully                .
## z_log_ai_visible_duration_sec ***
## z_log_time_before_ai_sec
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rd_crf z_lg__v__
## read_crflly -0.646
## z_lg__vsb__  0.120 -0.156
## z_lg_tm_b__  0.092 -0.163 -0.168
```

```r
fixed_effects_table(model_engagement_jaccard)
```

```
##                            term   estimate  std_error      p_value
## 1                   (Intercept)  0.5793135 0.05496926 4.862625e-13
## 2                read_carefully -0.1148629 0.06218993 6.669956e-02
## 3 z_log_ai_visible_duration_sec -0.1507500 0.03143097 3.800506e-06
## 4      z_log_time_before_ai_sec -0.0409166 0.03116472 1.911658e-01
```

Reading carefully was marginally associated with lower lexical similarity, b = -0.115, p = .067. **Longer AI visible duration significantly predicted lower Jaccard similarity, b = -0.151, p < .001.** This negative association should be interpreted cautiously: it is likely a reverse-causality or strategy-difference pattern. Participants who copied the AI response verbatim (Jaccard = 1.0, n = 55) had no reason to keep the AI visible, whereas participants who spent longer viewing the response may have selected, edited, or paraphrased more. Time before viewing AI was not significant, b = -0.041, p = .191.

Analysis 8C: Does Engagement Predict Semantic Similarity?

Outcome: `embedding_cosine_similarity_to_ai`
Model type: linear mixed regression
Predictors: `ai_reading_depth`, `ai_visible_duration_ms`, `time_before_ai_ms`
Random intercept: `task_id`
Sample: AI users with embedding similarity scores

```r
engagement_embedding_data <- subset(
  engagement_data,
  !is.na(embedding_cosine_similarity_to_ai) &
    !is.na(read_carefully) &
    !is.na(z_log_ai_visible_duration_sec) &
    !is.na(z_log_time_before_ai_sec)
)

model_engagement_embedding <- lmer(
  embedding_cosine_similarity_to_ai ~ read_carefully +
    z_log_ai_visible_duration_sec +
    z_log_time_before_ai_sec +
    (1 | task_id),
  data = engagement_embedding_data
)

summary(model_engagement_embedding)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## embedding_cosine_similarity_to_ai ~ read_carefully + z_log_ai_visible_duration_sec +
##     z_log_time_before_ai_sec + (1 | task_id)
##    Data: engagement_embedding_data
##
## REML criterion at convergence: 5.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.6177 -0.4728  0.2252  0.6311  1.9004
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  task_id  (Intercept) 0.007374 0.08587
##  Residual             0.048927 0.22119
## Number of obs: 157, groups:  task_id, 18
##
## Fixed effects:
##                                Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                     0.80509    0.03467  36.55770  23.219  < 2e-16
```

```
## read_carefully                 -0.04677    0.03831 149.84274  -1.221  0.22405
## z_log_ai_visible_duration_sec  -0.05927    0.01937 152.83035  -3.059  0.00262
## z_log_time_before_ai_sec       -0.04566    0.01923 152.68718  -2.374  0.01883
##
## (Intercept)                   ***
## read_carefully
## z_log_ai_visible_duration_sec **
## z_log_time_before_ai_sec      *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rd_crf z_lg__v__
## read_crflly -0.626
## z_lg__vsb__  0.120 -0.164
## z_lg_tm_b__  0.089 -0.171 -0.164
```

```r
fixed_effects_table(model_engagement_embedding)
```

```
##                            term    estimate  std_error      p_value
## 1                   (Intercept)  0.80509151 0.03467334 1.767199e-23
## 2                read_carefully -0.04676631 0.03830563 2.240517e-01
## 3 z_log_ai_visible_duration_sec -0.05926574 0.01937441 2.623346e-03
## 4      z_log_time_before_ai_sec -0.04565783 0.01923084 1.883013e-02
```

Reading carefully did not significantly predict semantic similarity, b = -0.047, p = .224. **Longer AI visible duration predicted lower embedding similarity, b = -0.059, p = .0026**, and longer time before viewing AI also predicted lower embedding similarity, b = -0.046, p = .019. These timing effects should be interpreted as strategy differences rather than evidence that engagement itself causally reduces uptake.

Analysis 8D: Does Engagement Predict Epistemic Ownership?

Outcome: `epistemic_ownership_binary` (0 = AI's thinking, 1 = own thinking)
Model type: mixed logistic regression
Predictors: `read_carefully`, `z_log_ai_visible_duration_sec`, `z_log_time_before_ai_sec`
Random intercept: `task_id`
Sample: AI users

```r
engagement_ownership_data <- subset(
  engagement_data,
  !is.na(epistemic_ownership_binary) &
    !is.na(read_carefully) &
    !is.na(z_log_ai_visible_duration_sec) &
    !is.na(z_log_time_before_ai_sec)
)

model_engagement_ownership <- glmer(
  epistemic_ownership_binary ~ read_carefully +
    z_log_ai_visible_duration_sec +
    z_log_time_before_ai_sec +
    (1 | task_id),
  data = engagement_ownership_data,
```

```r
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(model_engagement_ownership)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## epistemic_ownership_binary ~ read_carefully + z_log_ai_visible_duration_sec +
##     z_log_time_before_ai_sec + (1 | task_id)
##    Data: engagement_ownership_data
## Control: glmerControl(optimizer = "bobyqa")
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##    187.6    202.4    -88.8    177.6       139
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.2342 -0.7408 -0.5713  1.1989  2.1889
##
## Random effects:
##  Groups  Name        Variance Std.Dev.
##  task_id (Intercept) 0        0
## Number of obs: 144, groups:  task_id, 18
##
## Fixed effects:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -0.92435    0.29101  -3.176  0.00149 **
## read_carefully                 0.36822    0.37854   0.973  0.33068
## z_log_ai_visible_duration_sec  0.36582    0.18611   1.966  0.04934 *
## z_log_time_before_ai_sec      -0.03064    0.17971  -0.170  0.86464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rd_crf z_lg__v__
## read_crflly -0.782
## z_lg__vsb__  0.036 -0.124
## z_lg_tm_b__  0.109 -0.158 -0.113
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
fixed_effects_table(model_engagement_ownership, exponentiate = TRUE)
```

```
##                            term    estimate std_error     p_value odds_ratio
## 1                   (Intercept) -0.92434782 0.2910085 0.001491358  0.3967901
## 2                read_carefully  0.36822359 0.3785408 0.330680152  1.4451651
## 3 z_log_ai_visible_duration_sec  0.36581990 0.1861065 0.049339222  1.4416956
## 4      z_log_time_before_ai_sec -0.03063548 0.1797061 0.864636253  0.9698290
```

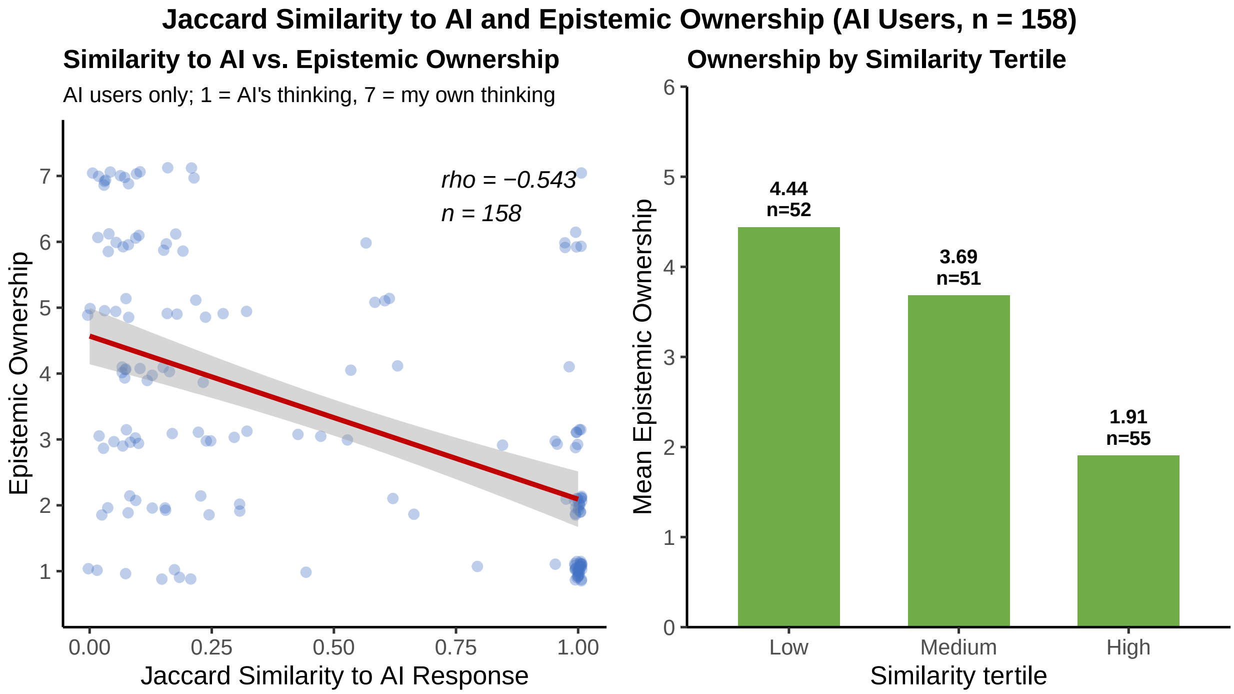
Reading carefully did not predict binary epistemic ownership, OR = 1.45, p = .331. **Longer AI visible duration predicted higher odds of primarily own-thinking ownership, OR = 1.44, p = .049.** Time before viewing AI was not significant, OR = 0.97, p = .865. Together with Analyses 8B-8C, this suggests that longer viewing time may mark a more selective or editing-oriented strategy: participants who kept the AI visible longer produced less AI-like responses and were more likely to report the final answer as primarily their own thinking.

Jaccard Similarity vs. Epistemic Ownership

```r
# Scatter: Jaccard similarity vs. epistemic ownership (continuous scale)
# Uses AI users from model_data; n is computed below.
scatter_data <- subset(
  model_data,
  ai_used == TRUE &
    !is.na(jaccard_similarity_to_ai) &
    !is.na(epistemic_ownership)
)

rho_jac_own <- cor(
  scatter_data$jaccard_similarity_to_ai,
  scatter_data$epistemic_ownership,
  method = "spearman",
  use = "complete.obs"
)

n_scatter <- sum(complete.cases(
  scatter_data[, c("jaccard_similarity_to_ai", "epistemic_ownership")]
))

# Tertile splits using rank to avoid duplicate-break error from tied 0s/1s
n_s <- sum(!is.na(scatter_data$jaccard_similarity_to_ai))
scatter_data$sim_rank <- rank(scatter_data$jaccard_similarity_to_ai,
                              ties.method = "average", na.last = "keep")
scatter_data$similarity_tertile <- cut(
  scatter_data$sim_rank,
  breaks = c(0, n_s / 3, 2 * n_s / 3, n_s + 0.1),
  labels = c("Low", "Medium", "High"),
  include.lowest = TRUE
)
tertile_means <- aggregate(
  epistemic_ownership ~ similarity_tertile,
  data = scatter_data,
  FUN = mean, na.rm = TRUE
)
tertile_ns <- table(scatter_data$similarity_tertile)

p_scatter <- ggplot(scatter_data,
    aes(x = jaccard_similarity_to_ai, y = epistemic_ownership)) +
  geom_jitter(height = 0.15, width = 0.008,
              alpha = 0.35, size = 1.8, color = "#4472C4") +
  geom_smooth(method = "lm", color = "#C00000",
              se = TRUE, linewidth = 1.1) +
```

```r
  annotate("text", x = 0.72, y = 6.7,
           label = sprintf("rho = %.3f\nn = %d", rho_jac_own, n_scatter),
           hjust = 0, size = 3.8, fontface = "italic") +
  scale_y_continuous(breaks = 1:7, limits = c(0.5, 7.5)) +
  labs(
    title    = "Similarity to AI vs. Epistemic Ownership",
    subtitle = "AI users only; 1 = AI's thinking, 7 = my own thinking",
    x        = "Jaccard Similarity to AI Response",
    y        = "Epistemic Ownership"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10)
  )

p_tertile <- ggplot(tertile_means,
    aes(x = similarity_tertile, y = epistemic_ownership)) +
  geom_col(fill = "#70AD47", width = 0.6) +
  geom_text(
    aes(label = sprintf("%.2f\nn=%d", epistemic_ownership,
                        as.integer(tertile_ns[as.character(similarity_tertile)]))),
    vjust = -0.3, size = 3.2, fontface = "bold", lineheight = 0.9
  ) +
  scale_y_continuous(limits = c(0, 6), expand = c(0, 0)) +
  labs(
    title = "Ownership by Similarity Tertile",
    x     = "Similarity tertile",
    y     = "Mean Epistemic Ownership"
  ) +
  theme_classic(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 12))

grid.arrange(p_scatter, p_tertile, ncol = 2,
  top = grid::textGrob(
    sprintf("Jaccard Similarity to AI and Epistemic Ownership (AI Users, n = %d)", n_scatter),
    gp = grid::gpar(fontface = "bold", fontsize = 13)
  )
)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Exploratory Correlation Matrix

This focused matrix gives a broad view of associations among mental-model variables, domain-specific AI-use frequency, engagement measures, similarity measures, ownership, and confidence, **among AI users only**. Spearman correlations are used because several variables are ordinal or skewed. This table is exploratory and should be used to identify patterns worth following up, not as a set of separate confirmatory tests.

```
correlation_data <- engagement_data

if (!"self_reported_freq_in_assigned_category" %in% names(correlation_data)) {
  correlation_data$self_reported_freq_in_assigned_category <- NA_real_
}

correlation_data$z_self_reported_freq <- z(
  correlation_data$self_reported_freq_in_assigned_category
)

correlation_data$z_user_decides <- z(
  correlation_data$exp_ai_does_work_vs_user_decides
)

correlation_data$z_independent <- z(
  -correlation_data$exp_independent_vs_reliant
)

correlation_data$z_instrument_vs_point_of_view <- z(
  correlation_data$exp_instrument_vs_point_of_view
)

correlation_data$z_existing_info_vs_creating_new <- z(
```

```r
  correlation_data$exp_existing_info_vs_creating_new
)

correlation_vars <- c(
  user_decides = "z_user_decides",
  independent_not_reliant = "z_independent",
  instrument_vs_point_of_view = "z_instrument_vs_point_of_view",
  existing_info_vs_creating_new = "z_existing_info_vs_creating_new",
  domain_freq = "z_self_reported_freq",
  read_carefully = "read_carefully",
  log_ai_visible_time = "log_ai_visible_duration_sec",
  log_time_before_ai = "log_time_before_ai_sec",
  log_response_writing_time = "log_response_writing_time_sec",
  jaccard_similarity = "jaccard_similarity_to_ai",
  embedding_similarity = "embedding_cosine_similarity_to_ai",
  epistemic_ownership = "epistemic_ownership",
  confidence = "confidence_level"
)

correlation_matrix_data <- correlation_data[, correlation_vars]
names(correlation_matrix_data) <- names(correlation_vars)

spearman_correlations <- round(
  cor(
    correlation_matrix_data,
    use = "pairwise.complete.obs",
    method = "spearman"
  ),
  2
)

spearman_p_values <- matrix(
  NA_real_,
  nrow = length(correlation_vars),
  ncol = length(correlation_vars),
  dimnames = list(names(correlation_vars), names(correlation_vars))
)

for (i in seq_along(correlation_vars)) {
  for (j in seq_along(correlation_vars)) {
    x <- correlation_matrix_data[[i]]
    y <- correlation_matrix_data[[j]]
    complete_rows <- complete.cases(x, y)
    if (sum(complete_rows) >= 3) {
      spearman_p_values[i, j] <- suppressWarnings(
        cor.test(
          x[complete_rows],
          y[complete_rows],
          method = "spearman",
          exact = FALSE
        )$p.value
      )
    }
```

```r
  }
}

spearman_p_values <- round(spearman_p_values, 3)

spearman_correlations
```

```
##                              user_decides independent_not_reliant
## user_decides                         1.00                    0.22
## independent_not_reliant              0.22                    1.00
## instrument_vs_point_of_view          0.06                   -0.18
## existing_info_vs_creating_new        0.08                   -0.10
## domain_freq                          0.01                   -0.10
## read_carefully                       0.02                    0.10
## log_ai_visible_time                  0.12                    0.22
## log_time_before_ai                   0.01                    0.04
## log_response_writing_time            0.12                    0.16
## jaccard_similarity                  -0.12                   -0.09
## embedding_similarity                -0.14                   -0.04
## epistemic_ownership                  0.21                    0.04
## confidence                           0.00                   -0.03
##                              instrument_vs_point_of_view
## user_decides                                        0.06
## independent_not_reliant                            -0.18
## instrument_vs_point_of_view                         1.00
## existing_info_vs_creating_new                       0.42
## domain_freq                                         0.19
## read_carefully                                      0.17
## log_ai_visible_time                                -0.02
## log_time_before_ai                                  0.03
## log_response_writing_time                          -0.08
## jaccard_similarity                                  0.05
## embedding_similarity                               -0.01
## epistemic_ownership                                 0.01
## confidence                                          0.11
##                              existing_info_vs_creating_new domain_freq
## user_decides                                          0.08        0.01
## independent_not_reliant                              -0.10       -0.10
## instrument_vs_point_of_view                           0.42        0.19
## existing_info_vs_creating_new                         1.00        0.33
## domain_freq                                           0.33        1.00
## read_carefully                                        0.13        0.08
## log_ai_visible_time                                  -0.08       -0.03
## log_time_before_ai                                   -0.12       -0.05
## log_response_writing_time                            -0.18        0.02
## jaccard_similarity                                    0.07       -0.06
## embedding_similarity                                  0.05       -0.11
## epistemic_ownership                                   0.13        0.07
## confidence                                            0.24        0.24
##                              read_carefully log_ai_visible_time
## user_decides                           0.02                0.12
## independent_not_reliant                0.10                0.22
## instrument_vs_point_of_view            0.17               -0.02
```

```
## existing_info_vs_creating_new          0.13              -0.08
## domain_freq                            0.08              -0.03
## read_carefully                         1.00               0.18
## log_ai_visible_time                    0.18               1.00
## log_time_before_ai                     0.19               0.13
## log_response_writing_time              0.23               0.63
## jaccard_similarity                    -0.20              -0.36
## embedding_similarity                  -0.22              -0.37
## epistemic_ownership                    0.15               0.18
## confidence                             0.34               0.16
##                              log_time_before_ai log_response_writing_time
## user_decides                               0.01                      0.12
## independent_not_reliant                    0.04                      0.16
## instrument_vs_point_of_view                0.03                     -0.08
## existing_info_vs_creating_new             -0.12                     -0.18
## domain_freq                               -0.05                      0.02
## read_carefully                             0.19                      0.23
## log_ai_visible_time                        0.13                      0.63
## log_time_before_ai                         1.00                      0.21
## log_response_writing_time                  0.21                      1.00
## jaccard_similarity                        -0.18                     -0.59
## embedding_similarity                      -0.25                     -0.58
## epistemic_ownership                        0.10                      0.39
## confidence                                 0.17                      0.13
##                              jaccard_similarity embedding_similarity
## user_decides                              -0.12                -0.14
## independent_not_reliant                   -0.09                -0.04
## instrument_vs_point_of_view                0.05                -0.01
## existing_info_vs_creating_new              0.07                 0.05
## domain_freq                               -0.06                -0.11
## read_carefully                            -0.20                -0.22
## log_ai_visible_time                       -0.36                -0.37
## log_time_before_ai                        -0.18                -0.25
## log_response_writing_time                 -0.59                -0.58
## jaccard_similarity                         1.00                 0.94
## embedding_similarity                       0.94                 1.00
## epistemic_ownership                       -0.54                -0.54
## confidence                                -0.11                -0.16
##                              epistemic_ownership confidence
## user_decides                                0.21       0.00
## independent_not_reliant                     0.04      -0.03
## instrument_vs_point_of_view                 0.01       0.11
## existing_info_vs_creating_new               0.13       0.24
## domain_freq                                 0.07       0.24
## read_carefully                              0.15       0.34
## log_ai_visible_time                         0.18       0.16
## log_time_before_ai                          0.10       0.17
## log_response_writing_time                   0.39       0.13
## jaccard_similarity                         -0.54      -0.11
## embedding_similarity                       -0.54      -0.16
## epistemic_ownership                         1.00       0.19
## confidence                                  0.19       1.00
```

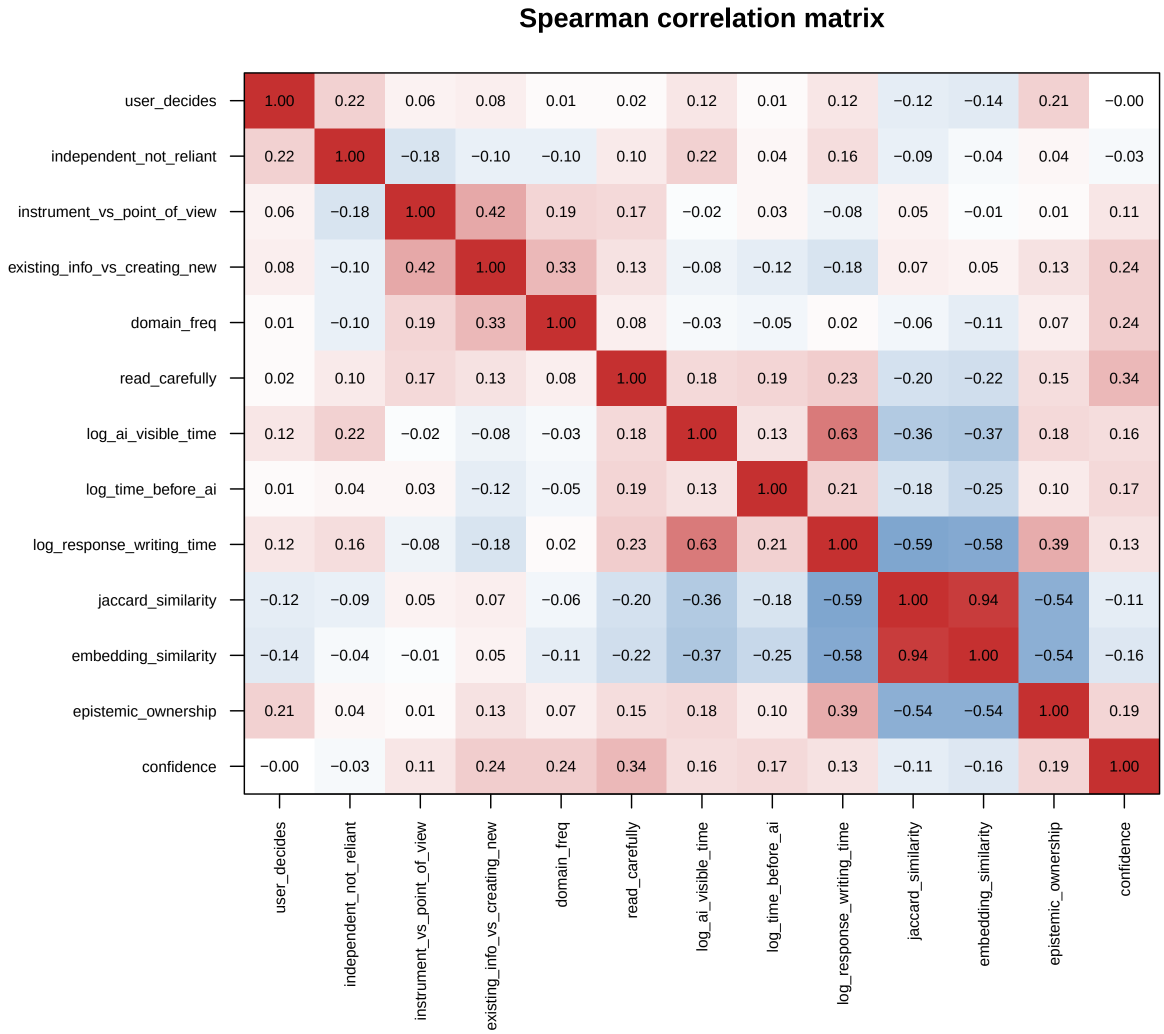
```
spearman_p_values
```

```
##                              user_decides independent_not_reliant
## user_decides                        0.000                   0.005
## independent_not_reliant             0.005                   0.000
## instrument_vs_point_of_view         0.432                   0.027
## existing_info_vs_creating_new       0.322                   0.226
## domain_freq                         0.857                   0.196
## read_carefully                      0.793                   0.234
## log_ai_visible_time                 0.128                   0.005
## log_time_before_ai                  0.875                   0.659
## log_response_writing_time           0.147                   0.045
## jaccard_similarity                  0.138                   0.255
## embedding_similarity                0.077                   0.620
## epistemic_ownership                 0.007                   0.661
## confidence                          0.952                   0.671
##                              instrument_vs_point_of_view
## user_decides                                       0.432
## independent_not_reliant                            0.027
## instrument_vs_point_of_view                        0.000
## existing_info_vs_creating_new                      0.000
## domain_freq                                        0.015
## read_carefully                                     0.037
## log_ai_visible_time                                0.804
## log_time_before_ai                                 0.687
## log_response_writing_time                          0.303
## jaccard_similarity                                 0.554
## embedding_similarity                               0.880
## epistemic_ownership                                0.940
## confidence                                         0.183
##                              existing_info_vs_creating_new domain_freq
## user_decides                                         0.322       0.857
## independent_not_reliant                              0.226       0.196
## instrument_vs_point_of_view                          0.000       0.015
## existing_info_vs_creating_new                        0.000       0.000
## domain_freq                                          0.000       0.000
## read_carefully                                       0.105       0.308
## log_ai_visible_time                                  0.325       0.686
## log_time_before_ai                                   0.135       0.537
## log_response_writing_time                            0.027       0.795
## jaccard_similarity                                   0.377       0.485
## embedding_similarity                                 0.529       0.189
## epistemic_ownership                                  0.103       0.412
## confidence                                           0.003       0.002
##                              read_carefully log_ai_visible_time
## user_decides                          0.793               0.128
## independent_not_reliant               0.234               0.005
## instrument_vs_point_of_view           0.037               0.804
## existing_info_vs_creating_new         0.105               0.325
## domain_freq                           0.308               0.686
## read_carefully                        0.000               0.021
## log_ai_visible_time                   0.021               0.000
## log_time_before_ai                    0.015               0.107
```

```
## log_response_writing_time             0.004             0.000
## jaccard_similarity                    0.013             0.000
## embedding_similarity                  0.005             0.000
## epistemic_ownership                   0.054             0.021
## confidence                            0.000             0.051
##                               log_time_before_ai log_response_writing_time
## user_decides                               0.875                     0.147
## independent_not_reliant                    0.659                     0.045
## instrument_vs_point_of_view                0.687                     0.303
## existing_info_vs_creating_new              0.135                     0.027
## domain_freq                                0.537                     0.795
## read_carefully                             0.015                     0.004
## log_ai_visible_time                        0.107                     0.000
## log_time_before_ai                         0.000                     0.008
## log_response_writing_time                  0.008                     0.000
## jaccard_similarity                         0.027                     0.000
## embedding_similarity                       0.001                     0.000
## epistemic_ownership                        0.219                     0.000
## confidence                                 0.036                     0.104
##                               jaccard_similarity embedding_similarity
## user_decides                               0.138                0.077
## independent_not_reliant                    0.255                0.620
## instrument_vs_point_of_view                0.554                0.880
## existing_info_vs_creating_new              0.377                0.529
## domain_freq                                0.485                0.189
## read_carefully                             0.013                0.005
## log_ai_visible_time                        0.000                0.000
## log_time_before_ai                         0.027                0.001
## log_response_writing_time                  0.000                0.000
## jaccard_similarity                         0.000                0.000
## embedding_similarity                       0.000                0.000
## epistemic_ownership                        0.000                0.000
## confidence                                 0.178                0.046
##                               epistemic_ownership confidence
## user_decides                                0.007      0.952
## independent_not_reliant                     0.661      0.671
## instrument_vs_point_of_view                 0.940      0.183
## existing_info_vs_creating_new               0.103      0.003
## domain_freq                                 0.412      0.002
## read_carefully                              0.054      0.000
## log_ai_visible_time                         0.021      0.051
## log_time_before_ai                          0.219      0.036
## log_response_writing_time                   0.000      0.104
## jaccard_similarity                          0.000      0.178
## embedding_similarity                        0.000      0.046
## epistemic_ownership                         0.000      0.017
## confidence                                  0.017      0.000
```

```
cor_heatmap <- spearman_correlations

heat_colors <- colorRampPalette(
  c("#2b6cb0", "white", "#c53030")
)(101)
```

```r
par(mar = c(10, 10, 4, 2))

image(
  x = seq_len(ncol(cor_heatmap)),
  y = seq_len(nrow(cor_heatmap)),
  z = t(cor_heatmap[nrow(cor_heatmap):1, ]),
  col = heat_colors,
  zlim = c(-1, 1),
  axes = FALSE,
  xlab = "",
  ylab = "",
  main = "Spearman correlation matrix"
)

axis(
  1,
  at = seq_len(ncol(cor_heatmap)),
  labels = colnames(cor_heatmap),
  las = 2,
  cex.axis = 0.7
)

axis(
  2,
  at = seq_len(nrow(cor_heatmap)),
  labels = rev(rownames(cor_heatmap)),
  las = 2,
  cex.axis = 0.7
)

for (i in seq_len(nrow(cor_heatmap))) {
  for (j in seq_len(ncol(cor_heatmap))) {
    value <- cor_heatmap[nrow(cor_heatmap) - i + 1, j]
    text(
      x = j,
      y = i,
      labels = sprintf("%.2f", value),
      cex = 0.65
    )
  }
}

box()
```




```r
par(mar = c(5, 4, 4, 2) + 0.1)
```

The strongest pattern is that Jaccard and embedding similarity are highly correlated, rho = .94, p < .001, so the lexical and semantic uptake measures capture very similar information among AI users.

Both uptake measures are strongly negatively related to epistemic ownership among AI users: Jaccard similarity, rho = -.54, p < .001; embedding similarity, rho = -.54, p < .001. In plain terms, participants whose final responses were more AI-like tended to report less own-thinking ownership and more perceived AI influence.

AI visible time and response writing time are positively correlated, rho = .63, p < .001, and both are negatively related to AI similarity. AI visible time was negatively related to Jaccard similarity, rho = -.36, p < .001, and embedding similarity, rho = -.37, p < .001. Response writing time showed an even stronger negative association with Jaccard similarity, rho = -.59, p < .001, and embedding similarity, rho = -.58, p < .001. This supports the idea that fast full-copy users and slower selective/editing users are behaviorally different groups.

Reading carefully was positively related to confidence, rho = .34, p < .001, but only marginally related to epistemic ownership, rho = .15, p = .054. Reading carefully was also negatively related to Jaccard similarity, rho = -.20, p = .013, and embedding similarity, rho = -.22, p = .005, again suggesting that more attentive engagement may reflect selective use rather than direct copying.

Instrument vs. point-of-view was moderately related to existing info vs. creating new, rho = .42, $p < .001$, suggesting these two mental-model beliefs tend to cluster together.

Existing info vs. creating new was not significantly related to epistemic ownership among AI users, rho = .13, p = .103, but it was positively related to confidence, rho = .24, p = .003. This suggests that seeing AI as helping create something new may be more connected to confidence than to ownership once people have already used AI.

User decides was positively related to epistemic ownership, rho = .21, p = .007. People who believed AI should defer to their judgment tended to report more own-thinking ownership.

Independent/not reliant was not significantly related to epistemic ownership, rho = .04, p = .661, reinforcing that this predictor is empirically distinct from `user_decides` and does not consistently predict ownership.